



# 13

CHAPTER

# PLUMBING

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## LEARNING OBJECTIVES

*By the end of this chapter you should be able to:*

- describe functional flow
- list three general things that can go wrong with supply plumbing
- list seven different materials used for supply piping
- list two common problems with service piping
- describe 11 problems with supply piping
- identify common types of water heaters
- list the common problems with various types of water heaters
- list eight materials commonly used for drain, waste and vent purposes
- list the common problems with each DWV component
- describe the implication of nonperformance of these components
- describe the inspection strategy for identifying problems
- identify each of the major plumbing fixtures, their function, and how they should be connected to the plumbing system
- list the common problems with each fixture, their implications, and the inspection strategy used to identify them

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**INTRODUCTION**

In this chapter, you will learn how plumbing systems work, and how to identify the common components that make up the supply and waste plumbing system. We'll also talk about water heaters, plumbing fixtures and faucets.

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## 13.1 SUPPLY PLUMBING

Supply plumbing is how water gets into the house. Although we can't see much of them during inspections, we'll talk about the **service pipes** or **water entry pipes** that bring the water to the house. We'll also look at the piping inside the house on the supply side. We will talk about the things that cause supply plumbing problems, such as leaks and low water pressure.

### 13.1.1 How Supply Plumbing Works

#### *Potable Water*

We need clean water at several locations throughout the house. We capture the water in piping to keep it where we want it and then push it through the pipes. If we do it right, we'll always have water available when we open a faucet. We call the clean water **potable**, which means we can use it for drinking, cooking and washing.

#### *Public or Private Water*

##### **The Source**

The source of water for a house is either a municipal supply system, or a private system such as a lake, river, or well. An investigation of the sources is beyond the scope of a professional home inspection, so we will not discuss them here. Either the municipal or private water supply system is assumed to provide us with a continuous supply of clean water at a relatively constant pressure. Since the municipal system is more common, let's use that as our example.

There are three things that can go wrong with supply plumbing:

- **Leaks:** Sometimes we get the water when or where we don't want it. We call these **leaks**. They range from an annoying dripping faucet to a split pipe that floods the house.
- **Not enough water:** People are often confused by this condition. Some say they don't have enough **pressure**, others say they don't have enough **flow** and still others say they don't have enough **volume**. These are different ways of saying the same thing. We'll call these flow/pressure problems.
- **Dirty water:** The water quality may be poor because of a problem with the municipal supply system or a well. The water may also be dirty because it's accidentally mixed with the waste water in the house. All of these situations are dangerous. Some are easy to find and others are just about impossible.

You should determine the flow/pressure performance of the supply plumbing system. Our test of **functional flow** is usually performed at the bathroom that is highest and most remote from the water supply into the house. This is the most rigorous test. Many inspectors get more than two fixtures flowing simultaneously, although you should create a situation that is typical for people living in the home.

A home with eight teenagers is more likely to have a number of fixtures flowing simultaneously than a home with an elderly couple, for example. You can ask your clients some lifestyle questions to get a sense of their needs.

### 13.1.2 Service Piping (Water Entry Piping)

This is the piping that leads from the city main to the house. In modern construction, the piping is typically 3/4-inch for a single family home. In some areas, it is common to go to 1-inch pipe if there are more than three bathrooms in the house. The largest pipe size commonly used for single-family homes is 1 1/2-inch. Some older houses had 3/8-inch diameter piping. Half-inch diameter piping was common in many areas up into the 1960s.

*Piping Materials*

Water service pipes may be—

- copper
- galvanized steel
- lead
- PVC plastic
- CPVC plastic
- polyethylene plastic
- polybutylene plastic
- cast iron
- asbestos cement
- brass

*Piping Identification*

Most of the water service piping is not visible. In some cases, it can be seen at its point of entry into the building. In warm climates, it may be seen coming out of the ground and into the building above grade from the outside. In some cases, the service piping is visible at the meter outside the house (warm climates only). In cold climates, the meter is typically inside the house.

*Piping in Trenches*

Water service piping must be buried at least 12 inches. In cold climates, the piping must be at least 6 inches below the depth of frost.

*Protecting Copper Pipe*

Where copper piping passes through concrete it is usually wrapped to protect it from corrosion.

Problems with water service pipes include low pressure and leaking.

**Low Pressure**

Low pressure is often noted inside the building, even though the piping inside the building seems to be a good size and in good condition. The culprit is often old or undersized water service piping, or a combination of both.

**IMPLICATIONS**

Reduced water pressure and flow are inconvenient for homeowners at best. In severe circumstances, the plumbing system may not be capable of providing showers while any other fixtures are operating.

**STRATEGY**

Where the water pressure in a large house seems poor for no apparent reason, you may want to check the water meter and recommend possible replacement with a larger commercial type water meter that will not restrict the flow. When you perform your functional flow test, you will get a sense of the strength of the water flow and pressure. It's not always possible to determine where the losses are greatest, and when you can't pinpoint the problem, make sure you include the water service pipe as a possible problem area.

## Leaking

A leaking water service pipe may be undetected for some time.

Leaking water service pipes can cause flooding and soil erosion. In milder cases, the leak may be undetected and water pressure and flow may suffer as a result.

Wet spots at the front of the property that are otherwise unexplained suggest a leaking water service pipe. Where a leaking water service pipe is suspected, the water can sometimes be heard flowing through the pipe. Some inspectors will ensure that there is no water being used in the house and then put the tip of a screwdriver against the pipe where it comes into the house. They will then put the handle of the screwdriver against their ear, listening for water flowing through the pipe. In some cases, this can reveal a leaking water service pipe.

Where the water service pipe enters the building below grade, a leaking service pipe may result in a wet spot around the pipe penetration point or at the wall or floor near the point of entry into the house. Where you see dampness in subgrade areas close to a water service entry point, keep in mind the possibility of a leaking pipe outside the house.

*Wet at Point of Entry*

## 13.1.3 Shutoff Valves

Every plumbing system should have a single shutoff valve that allows you to quickly turn off all the water in the house. An important and valuable part of a home inspection is locating this valve and showing the client where it is and what it does. Good shutoff valves are accessible and operable.

Many shutoff valves are **stop and waste** valves (Figure 13.1). This means that there is a bleed or drain valve on the downstream (house) side of the valve. The purpose of this stop and waste valve is to drain all the water out of the supply piping, for example, if the house is to be winterized. On some old valves, the bleed valve operates automatically when the valve is closed. In more modern stop and waste valves, there is a cap on the bleed valve that allows you to decide whether or not water is going to bleed out of the pipe downstream. **Petcock** is another term for the bleed valve on a main shutoff valve.

Let's look at some common problems with shutoff valves.

- missing or cannot be located
- leaking
- inoperable/inaccessible/buried
- damaged handle
- partly closed
- exposed to mechanical damage
- rusted

### Missing or Cannot Be Located

Sometimes there is no shutoff valve.

The implication of shutoff valve problems is an inability to shut off all the water quickly in the event of an emergency, such as a flood.

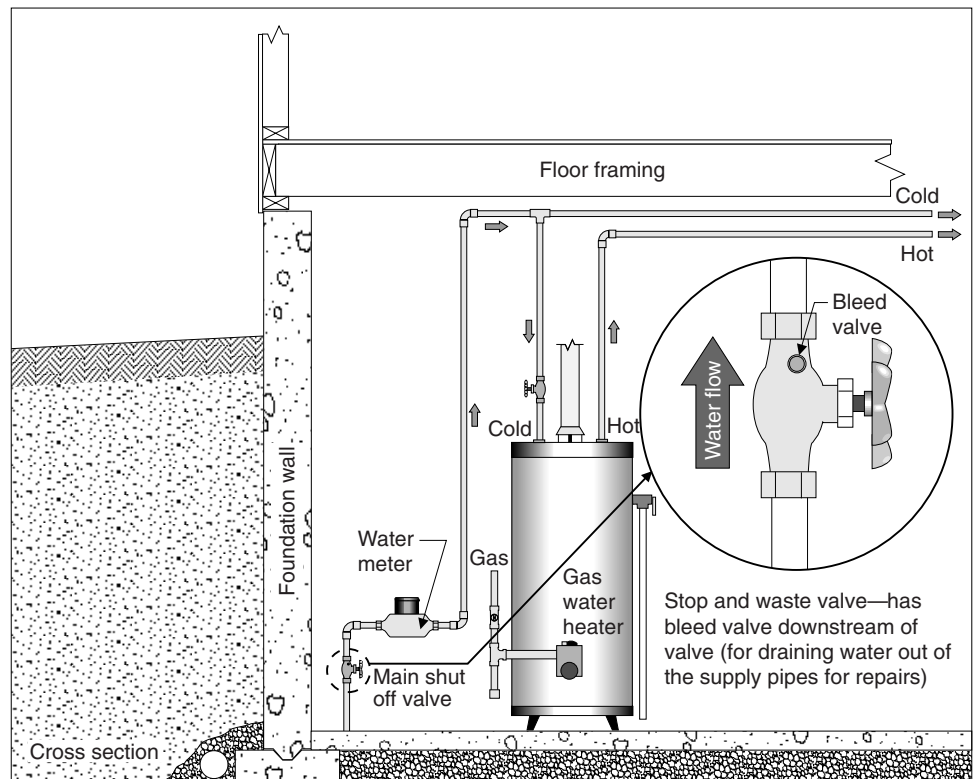
Trace the water supply piping back to its point of entry into the house, looking for a single main shutoff valve. The valve is usually upstream of the meter and

### IMPLICATIONS

### STRATEGY

### IMPLICATION

### STRATEGY

**FIGURE 13.1** Main Shutoff Valve—Stop and Waste

upstream of a pressure regulator. If you can't find a shutoff inside the house, check outside for the valve. If you can't find it, write it up as not being located, rather than missing, and recommend further investigation.

### Leaking

Sometimes the shutoff valve is leaking.

The implications of a leaking shutoff valve are water damage and perhaps rust forming on the valve, rendering it inoperable over time. If the leak is severe, the water flow and pressure in the house may suffer.

### Inoperable/Inaccessible/Buried

Again, people won't be able to shut off the water in an emergency.

Use common sense to determine whether the valve can be turned off easily. Would a homeowner have trouble turning this valve off? If so, recommend improvement.

### Damaged Handle

A damaged valve handle is common and can be dangerous.

Not only will it be difficult to close the valve, it is very easy to cut your hand on a valve with a damaged handle.

Make sure the handle is intact.

#### IMPLICATIONS

#### IMPLICATIONS

#### STRATEGY

#### IMPLICATIONS

#### STRATEGY

**IMPLICATIONS****STRATEGY****Exposed to Mechanical Damage**

The location of the shutoff valve sometimes makes it prone to mechanical damage.

If the valve is damaged, there may either be leakage or the valve may become inoperable.

Watch for valves that are susceptible to mechanical damage. On outdoor, above-grade valves, think about lawnmowers and other landscaping equipment, for example, that may damage the valves.

**Rusted**

Most of these valves are brass and will not rust. Some, however, have steel handles and other ferric (containing iron) parts.

The valve may be inoperable.

Report rust on shutoff valves. If the valve is badly rusted, you should not operate it.

**IMPLICATIONS****STRATEGY**

### 13.1.4 Supply Piping Distribution System in the House

*Function*

The house supply piping carries water from the service piping to the various points of use, including the water heater, fixtures, appliances such as dishwashers and clothes washers, and hose bibbs.

*Materials*

Several materials may be used for supply piping. These include:

- Copper
- Galvanized steel
- Brass
- Stainless steel
- Polybutylene (PB)
- Polyethylene (PE)
- Cross-linked polyethylene (PEX)
- CPVC

*Pipe Sizes*

Not all of these materials are permitted in all areas.

Common supply pipe sizes are  $\frac{1}{2}$ -inch and  $\frac{3}{4}$ -inch. On very large homes, 1-inch supply pipes are occasionally found. The longer the run, the larger the pipe diameter should be.

Now let's look at some of the common supply piping problems in houses.

1. Leaking
2. Poor pressure/flow
3. Rust
4. Split/damaged/crimped
5. Poor support
6. Cross connections
7. Excess pressure

8. Excessive noise
9. Combustible piping
10. Suspect connections on polybutylene
11. Non-standard material

### Leaking

Leaking supply piping can be very dramatic. Because the piping is pressurized, a leak can do a lot of damage quickly. If there is no functional floor drain, a leaking supply pipe can flood a house.

The big risk here is water damage to the home.

As you look at the visible supply piping in the house you may find leaks. Dramatic leaks are usually found by homeowners and corrected right away. It's the small leaks that you are more likely to find. These may be drips at poor connections or pinholes on galvanized steel pipes that rust over and scab themselves shut temporarily. Some leaks are concealed and you won't see them during an inspection.

If the house has a water meter, you can ensure that all of the plumbing fixtures are off and check the water meter for movement. If you can't tell whether it's moving or not you can come back and check in 10 or 15 minutes and see whether the meter has moved. You can put the tip of a screwdriver on the supply piping and put your ear against the handle of the screwdriver, listening for the sound of water moving through the pipe.

In most cases, supply pipe leaks will be constant. Leaks at fixtures and drain and waste pipes will be intermittent.

| IMPLICATION |
|-------------|
| STRATEGY    |

*Check the Water Meter*

*Listen with Screw Driver*

*Leaks Are Constant*

### Poor Pressure or Flow

This is one of the most common problems, particularly on older homes.

The implications are less water at each fixture than is desirable. This is most noticeable when more than one fixture is flowing. In severe cases, when fixtures on lower floors are operated, water stops coming out of the fixtures on upper floors.

Your first and most important job is to point out the low pressure and flow conditions. It's nice to be able to pinpoint the cause, but you won't always be able to do this.

You may be wise to talk to the municipal water departments about any localized supply problems they are having. Undersized mains, old neighborhoods, and neighborhoods that have been substantially expanded can all suffer water supply problems.

| IMPLICATIONS |
|--------------|
| STRATEGY     |

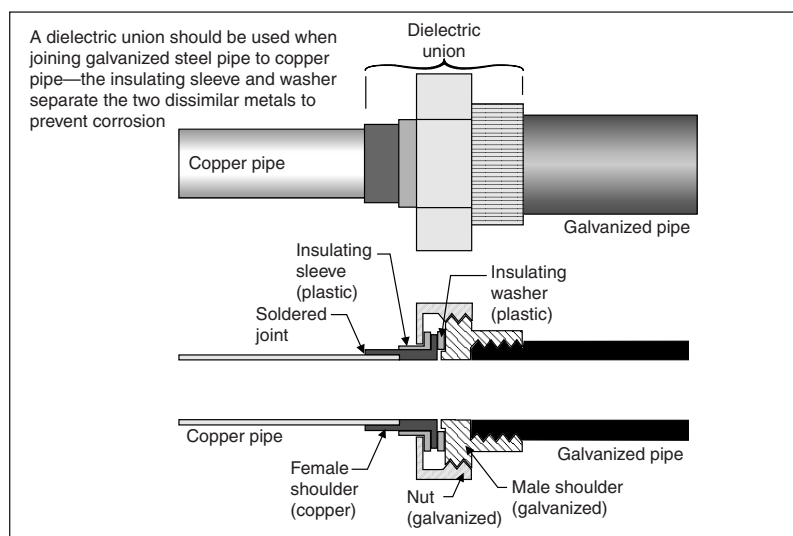
### Rust

Rust is found on metallic piping only. Galvanized and stainless steel piping are prone to rust. Copper piping does not rust or corrode under normal use, unless the water is acidic.

The implication of pipe rusting may be pipe failure.

If the piping is galvanized steel, rusting is probably a legitimate indication that the pipe is near the end of its life. You have to be careful when looking at rust. Is the pipe rusting from the inside out or is it surface rusting on the exterior only? Galvanized steel piping typically rusts through from the inside, showing first at its threaded connections. Pinhole leaks sometimes scab over and temporarily stop leaking. Surface rust that is the result of condensation or corrosive chemicals in the atmosphere tends to be uniform around the pipe surface and may be more cosmetic than a functional concern.

| IMPLICATION |
|-------------|
| STRATEGY    |

**FIGURE 13.2** Dielectric Union

If you come across rusted stainless steel, it will probably be near the end of its life.

#### *Dissimilar Metals*

Corrosion on copper piping may or may not indicate a serious problem. Where copper piping is directly connected to galvanized steel piping, corrosion usually is the result of **galvanic action**. The steel tends to be the sacrificial metal and it will corrode first and most dramatically. This kind of corrosion also occurs where copper pipes contact sheet metal ductwork, or where copper pipes are supported by iron or steel hangers, for example.

#### *Dielectric Connectors*

Special connectors are made to reduce the galvanic action and corrosion of the steel piping by the copper (Figure 13.2). These fittings separate the two materials physically and reduce the flow of electrons between them. In many cases, this is a short-term issue since the galvanized steel piping is probably close to the end of its life anyway.

### **Split/Damaged/Crimped**

Piping that is damaged won't be able to carry water effectively.

The implications of split, damaged or crimped pipes are leakage and reduced water pressure and flow.

Wherever piping is visible, make sure it is intact. Watch for crimping and other mechanical damage. If the pipe is split, flooding will usually occur. However, if the pipe is at an outside hose bibb where the water supply has been shut off for the winter, there won't be any leakage. When your client turns on the supply to the hose bibb in the spring, they will find the split pipe and may well ask you to replace it for them.

Plastic pipe can be damaged by sharp hangers. The expansion and contraction of plastic piping can cause wear at hangers and may weaken connections over time.

Some piping is particularly vulnerable to mechanical damage. You can use common sense to look at supply piping and see whether it is likely to be damaged.

#### *Vulnerable Locations*

#### **IMPLICATIONS**

#### **STRATEGY**



**TABLE 13.1** Support Requirements

| Piping Material       | Maximum Spacing of Supports<br>for Horizontal Piping |
|-----------------------|------------------------------------------------------|
| copper pipe           | 6 to 10 feet                                         |
| copper tubing         | 6 to 10 feet                                         |
| galvanized steel      | 10 to 12 feet                                        |
| plastic (PE, PEX, PB) | 32 inches to 4 feet                                  |

**Poor Support***Vertical*

All piping needs to be adequately supported. Vertical pipes should be supported at every other floor level as a general rule and at least every 24 feet. Copper piping should be supported at each floor level and at least every 10 feet.

*Horizontal*

Table 13.1 summarizes typical support requirements for horizontal piping runs. Individual codes will vary, so once again, check with your local building authorities. Flexible plastic piping, such as polybutylene, needs very good pipe support, particularly on the hot water piping, because the pipe tends to expand a lot when you heat it up.

Hangers for copper or brass pipe should be copper or brass or should be wrapped so that they are electrically insulated from the copper pipe. This prevents corrosion of the hangers if they are a dissimilar metal.

**IMPLICATIONS**

The ultimate implication of improper support is leaking pipes. Failures usually occur at joints or connections.

**STRATEGY**

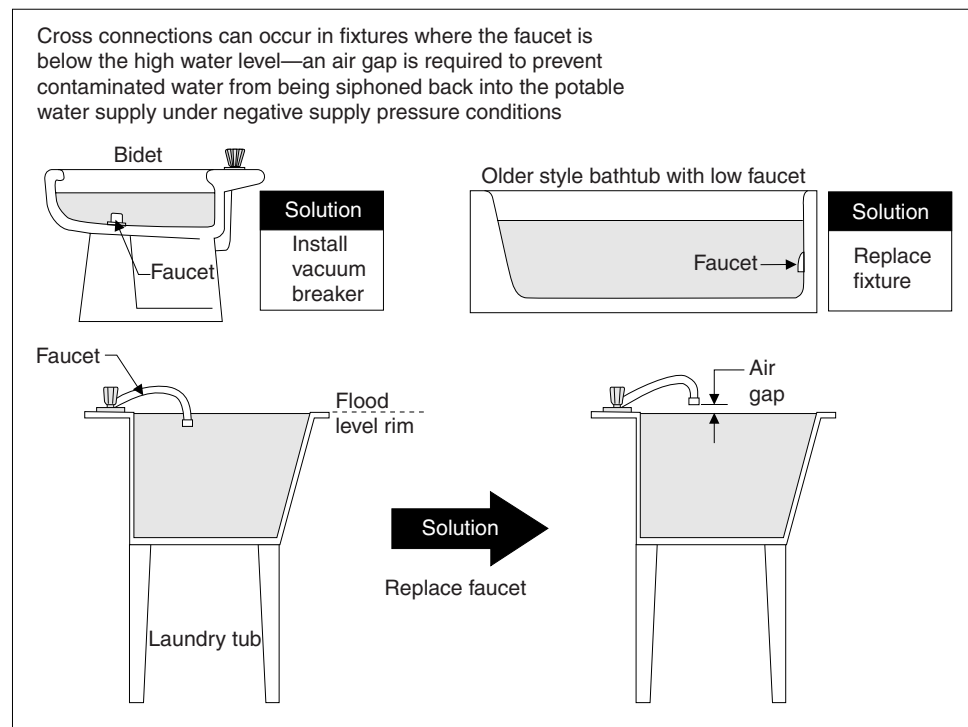
You won't be able to see the vertical supports for piping in most cases. You will be able to see horizontal piping supports in unfinished basements and subgrade areas (crawlspaces). Where piping is run in attics in warmer climates, you will also be able to see pipe supports there. Rather than applying a bunch of rules for various piping types, it is often more practical simply to look for deflection of the piping. Generally speaking, this is more of an issue with plastic piping than metallic piping.

Watch for plastic piping damage at the hangers, especially if the hangers are metal.

**Cross Connections**

**Cross connections** are a possibility wherever the supply plumbing and solid or liquid waste could come together (Figure 13.3). A cross connection is a serious health issue. There is a risk of illness or death if the potable water for drinking, cooking and washing is contaminated with waste. The Standards require us to look for cross connections. Let's look at some of the places we might find them:

1. Fixtures where the faucet spout is below the overflow or flood rim of the fixture. This occurs on older bathtubs and laundry tubs, for example.
2. Fixtures where hand-held showers, extendable faucets or vegetable sprayers can be left in tubs, basins or sinks.
3. Garden hoses left attached to a hose bibb.
4. A toilet tank where the fill water enters at the bottom.
5. A dishwasher where a solenoid valve separates the clean water from the gray water (a term used to describe water that is no longer clean but does not include human waste).
6. A clothes washing machine with a discharge pipe in a laundry tub.

**FIGURE 13.3** Cross Connections

7. A water softener.
8. A humidifier connected to the supply plumbing system.
9. A lawn or fire sprinkler system.
10. A bidet where water enters the bowl at the bottom.
11. A hot water boiler (boilers are connected to the supply plumbing piping so that make-up water can be provided to the boiler).
12. A swimming pool fill outlet that is located below the flood level of the pool.
13. A trap primer without an air gap.

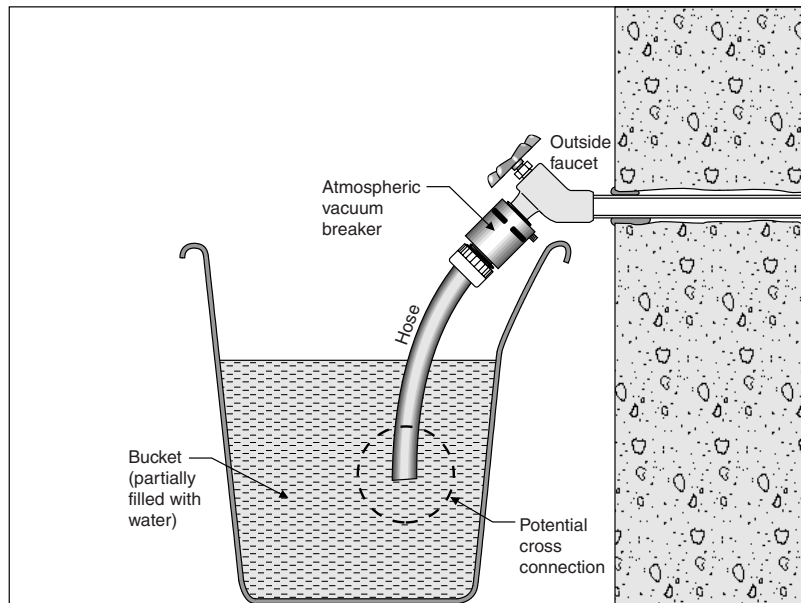
One of the simplest and most effective ways to prevent cross connections is with a 1-inch vertical air gap. If the supply device (a faucet, for instance) is located above the flood rim of a basin, the water in the basin can't get back into the faucet. Another way to prevent cross connections is with an anti-siphon device, such as a vacuum breaker. This device opens on low pressure, and discharges waste water rather than allowing it to flow backwards through the breaker.

#### *Backflow and Back Siphonage*

**Backflow** is a flow occurring in a reverse direction from that intended. A vacuum isn't necessary for backflow. The pressure on the potable water side only has to be very slightly lower than the pressure on the waste water side for backflow to occur.

**Back siphonage** is also a reverse flow of soiled water into potable water piping, but it doesn't need positive pressure on the downstream (dirty) side. Back siphonage is a specific kind of backflow caused when the potable water piping system is at a lower than atmospheric pressure. A partial vacuum is needed for back siphonage.

Backflow and back siphonage can be prevented by several different arrangements or devices, of which vacuum breakers are the most common (Figures 13.4 and 13.5).

**FIGURE 13.4** Atmospheric Vacuum Breaker**IMPLICATIONS**

As we discussed at the outset, the health implications are serious if cross connections exist.

**STRATEGY**

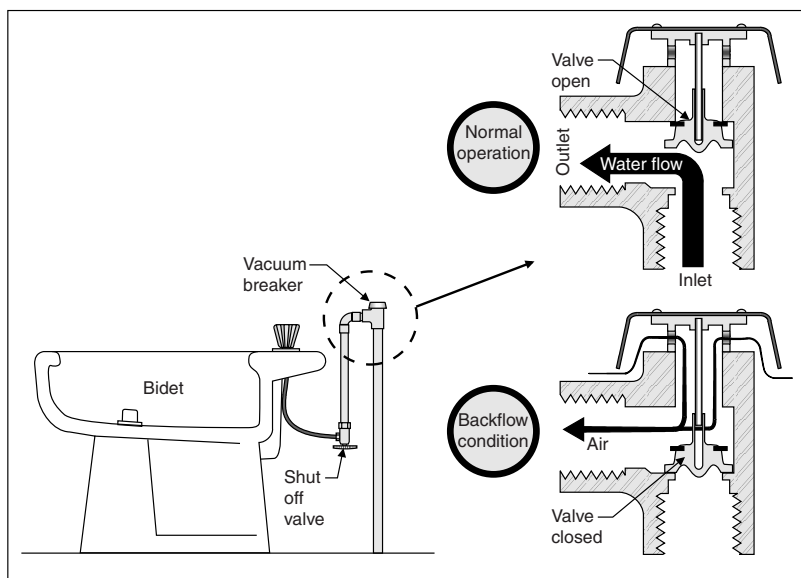
You'll need to be aware of cross connections and what methods your local authorities require to protect against them.

**Excess Pressure**

Where the water pressure exceeds 80 or 100 psi (depending on your jurisdiction) pressure regulators may be required to limit the house supply pressure to 80 psi.

**IMPLICATIONS**

The implications of excess pressure are leaking valves and faucets, as well as solenoid valve failures.

**FIGURE 13.5** Pressure Type Vacuum Breaker

**STRATEGY**

If you use a pressure gauge to check the static pressure on the supply side (beyond the Standards), you can pick up the excess pressure readily. If you don't use a gauge, you may detect the high pressure simply by the force of the water coming out of the faucets.

**Excessive Noise**

Supply plumbing systems can be noisy. Water hammer is perhaps the most serious of the noise problems.

*Water Hammer*

**Water hammer** is a loud banging sound in the pipes. It occurs when valves are closed quickly and the water, which is moving with considerable velocity and momentum, has to stop very quickly. The water bounces off the valve and actually creates a vacuum as it bounces away from the valve. The vacuum then violently attracts the water back to the low pressure area and the water again bounces off the closed valve. If the water has enough momentum, it will again bounce off the valve and create a smaller vacuum. The effect of this reverberation sounds a little bit like someone pounding on metal piping with a hammer. Water velocities can be high (up to 3,000 miles per hour) and the forces developed with water hammer can exceed 600 psi.

Noise is always an irritant. In some cases, it may also suggest premature failure of joints and early leakage.

**IMPLICATIONS**

The implications of water hammer can include deterioration of valves, loosening of piping supports and leaking at pipe joints. Metal piping is more susceptible to water hammer than plastic piping. The plastic piping tends to have enough flexibility to absorb some of the energy.

The implications of valve chatter and pipes with rough edges are less serious. Valves often chatter because the washers or seats are worn. The damage caused by pipes rubbing against other materials is not usually an issue over the short term. Over the long term, the abrasive effect can wear through the pipe and cause sudden failure and flooding. In most cases this occurs in concealed wall and floor spaces and is difficult to predict. It is disruptive to interior finishes to locate and correct pipe noises that result from pipes rubbing against framing members, for example.

**STRATEGY**

Where you hear noises from pipes when the system supply piping is tested, you'll need to determine under what circumstances the noise occurs. If the loud banging noise is heard when valves are closed, it's a water hammer problem. If a vibrating noise is heard and it's more noticeable when valves are partly open, it's very likely valve chatter. If the noise increases with the amount of water flowing through the pipe, it may be rough edges on pipe ends.

If the problem is intermittent and occurs after the water has been flowing for a while and again a few minutes after the water has been shut off, it may be the expansion and contraction of metal piping rubbing against framing members or other materials. This typically occurs on the hot water piping.

With the exception of water hammer, these problems are not serious.

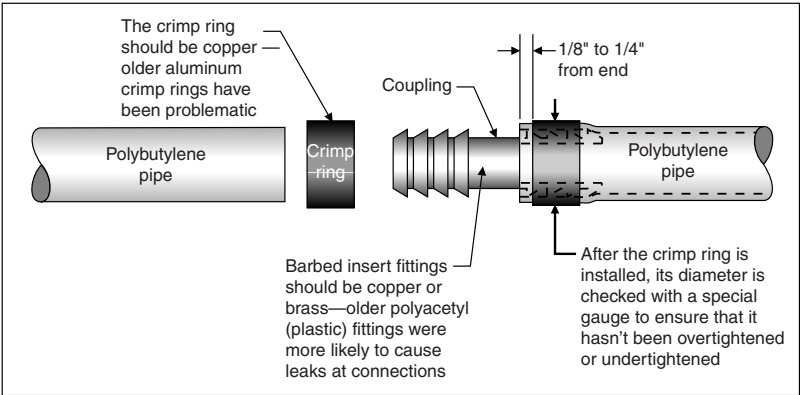
**Combustible Piping**

Plastic piping is combustible. Most jurisdictions allow it to be installed in single-family dwellings. However, plastic piping is not usually allowed to pass from one dwelling to another. The rationale is that a fire might be transmitted through the combustible plastic piping from one home to the next.

**IMPLICATIONS**

As discussed, this is a fire hazard.

FIGURE 13.6 Polybutylene Pipe—Crimp Fitting



STRATEGY

Heat Sources and Plastic Pipe

Where you find plastic piping in homes that are attached to other dwellings, look to see whether the plastic piping passes from one unit to the next. If this appears to be the case, recommend further investigation. At least some sections of the piping may have to be replaced.

Combustible piping should be kept away from heat sources, including exhaust flues for combustion appliances. The required clearances depend on the appliances. Most appliances have rated requirements for clearances from combustibles.

Suspect Connections on Polybutylene Piping

There have been so many problems with polybutylene piping that you should check the exposed connections carefully and let your client know about the potential for leaks at the connections.

The implication is primarily leakage.

IMPLICATIONS

STRATEGY

Look at exposed connections for evidence of leakage. Polybutylene connections should be at least 18 inches from water heaters. The first 18 inches of piping from the water heater should be metal. Polybutylene piping should not be installed immediately downstream of instantaneous-type water heaters because of the high temperatures.

Insert Fittings on Early Systems were Problems

The barbed insert fittings for polybutylene piping were originally made of polyacetyl (a plastic). Brand names of these fittings included Celcon and Delrin. These systems were used from about 1980 to 1986. Since then, copper and brass fittings have been used.

No Aluminum Crimp Rings

Some of the original crimp rings were aluminum. Some of the aluminum was gold-colored. More modern installations require annealed copper crimp rings. The original crimping tool, which was a combination tool, has also been discarded.

Crimp Rings and Bending the Pipe

Crimp rings should be 1/8-inch to 1/4-inch away from the end of the tube (Figure 13.6). The crimp ring should be perpendicular to the pipe run. The crimp ring should not show evidence of having been crimped more than once. Bends in polybutylene piping should be on a radius that is at least 12 times the tube diameter. For example, the radius of a turn for 1/2-inch piping should be 6 inches minimum.

Protection and Support

Piping that passes through metal wall studs should be protected with sleeves or bushings. Protective sleeves should be used where polybutylene piping runs through floors or concrete slabs.

*Metal Hose Bibbs*

Polybutylene piping should not be used to make outdoor hose bibb connections. A metal pipe should be used for the hose bibb.

*No Circulating Loops*

Some authorities say that houses with recirculating loops for hot water piping should not use polybutylene on these loops.

*Leak Evidence and Fitting Clues*

You should be watching for evidence of leaks, which includes water or a whitish deposit on the pipe surface, usually near fittings. You may also find that there are more fittings than you would expect. This suggests repairs and possible chronic problems. This is especially true if you see different kinds of fittings in the same house.

*Heat Sources Summary*

Watch for piping close to recessed lights, exhaust flues, or other heat sources.

Even if you don't see any problems with polybutylene piping, you may want to make your client aware of the issues surrounding polybutylene piping. The bottom line is that crimp rings are questionable, even when well done, and several authorities recommend the use of compression fittings with polybutylene pipe.

### Non-Standard Materials

We have listed materials that are approved for water distribution systems. These are the only materials that you should find.

#### IMPLICATIONS

Leakage, premature failure, poor water pressure and possible health issues, including cross connections, may result from the use of non-standard materials.

#### STRATEGY

Watch for radiator hose, garden hose, corrugated piping, PVC piping used for hot water supply, and any other piping arrangements that appear unconventional. Where you are unsure as to the suitability of materials, simply raise the question and call for further investigation by a specialist.

## 13.2 WATER HEATERS

Water heaters are designed to provide hot water at fixtures such as sinks, basins, laundry tubs, bath tubs and showers. We'll concentrate on the most common types, which are gas, oil and electric. These heaters usually have storage tanks with capacities of 30 to 60 gallons.

Most tanks are insulated steel cylinders with an enamel coating on the inner surface. They are referred to as **glass-lined** tanks. The lining helps but does not absolutely prevent corrosion.

*Other Materials*

Some water heater tanks are copper, some are aluminum and some are an alloy of copper and nickel. Some modern water tanks are plastic coated. The copper-aluminum and copper-nickel alloy tanks all last longer than steel tanks. Some of these can last up to 50 years.

*Life Expectancy*

Conventional domestic water heaters have life expectancies that vary across North America. In some areas, life estimates are as low as 8 to 12 years. In other areas, life expectancies can be 15 to 25 years.

*Temperatures of Hot Water*

**Water heaters**, or **hot water heaters** as they are often (redundantly) called, are designed to raise the water temperature in the house from approximately 50°F to 120° F. Some people call for temperatures of 140° F or even higher but for safety, economy, and conservation reasons there is a trend toward lower temperatures for hot water. Most equipment manufacturers, installers and local authorities now recommend 120°F water.

A water heater is a single pass system that always sees new, not recirculated, water. Because water heaters are constantly exposed to fresh water with lots of

*Water Heater Is Just a Bulge  
in the Pipe*

oxygen, they are prone to corrosion. As a result, water heaters are usually provided with an anti-corrosion rod or sacrificial anode. Sacrificial anodes sit in the tank water allowing the chemical reaction from the water to attack them rather than the tank.

The water heater is really just a fat part of the piping system where the water passes through. As it passes through, it gets heated. The pressure in the water heater is the same as the pressure throughout the supply plumbing system (typically 60 psi). The pressure on the cold water inlet pipe is virtually the same as the pressure on the hot water outlet pipe.

*Vacuum Relief Valves*

Some water heaters are provided with a vacuum relief valve on the cold water inlet side. This prevents cross connections by preventing the hot water from backing into the cold water line in the event the cold water pressure is low. The vacuum relief valve also protects against possible collapse of the storage tank by preventing a vacuum inside the tank.

## 13.2.1 Gas- and Oil-Fired Water Heaters

*Burner in Center*

On conventional propane-, gas- or oil-fired water heaters, heat is generated by a burner at the bottom of the tank (Figures 13.7 and 13.8). The tank is like a very tall doughnut, with a hole up through the central part of the tank. The exhaust products from the burner go up through this core and heat is transferred through the tank walls from the core into the water. The exhaust products go into a vent connector above the core and into a chimney or vent system.

*No Separate Heat Exchanger*

Domestic water heaters transfer heat directly from the combustion products through the tank wall and floor into the water. The tank is both the storage compartment and the heat exchanger.

*Dip Tube*

Cold water is introduced through a **dip tube** near the bottom of the tank (Figure 13.9). The cold water supply pipe usually attaches to the tank at the top. The dip tube carries the cold water down through the tank, discharging it near the bottom of the tank. The hot water is drawn off from the top of the tank.

*Thermostat Control*

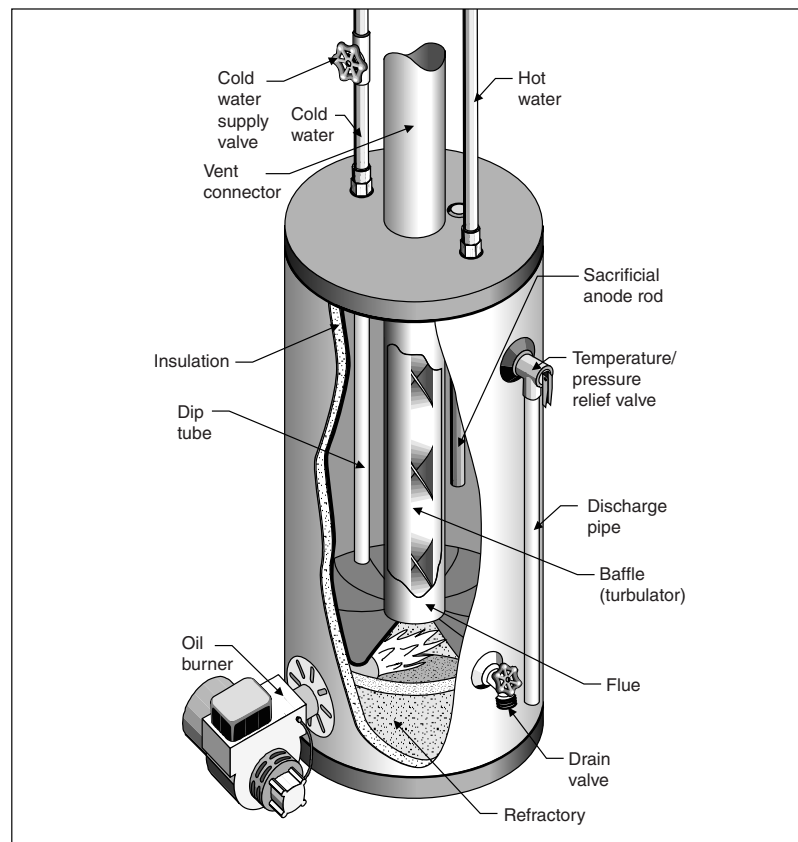
At rest, the tank is filled with water heated to the set temperature (e.g., 120°F). As water begins to be drawn off, the 120°F water leaving the tank is replaced by 50°F water. This cool water activates the thermostat which tells the burner to come on. The burner comes on and heats the water. Depending on how quickly the water is flowing out, the temperature in the tank may rise toward the set temperature, or may continue to drop because cold water comes in faster than the burner can heat it up.

*Size Issues*

The effective size of a water heater is a function of—

- the volume of hot water it can keep on hand (capacity)
- the size of the burner (recovery rate)

The larger the burner, the faster the **recovery rate** of the heater. The recovery rate is the time it takes to heat the tankfull of water from 50°F to 140°F, for example. (Current conventions measure recovery rates over a 90°F temperature rise, even though most of us don't heat water to 140°F anymore.) Generally speaking, the faster the recovery rate, the smaller the tank can be. Oil-fired water heaters typically have larger burners than gas-fired water heaters. Oil water heaters have a faster recovery rate, so their tanks are often smaller than gas water heater tanks.

**FIGURE 13.7** Oil-Fired Water Heater

### 13.2.2 Electric Water Heaters

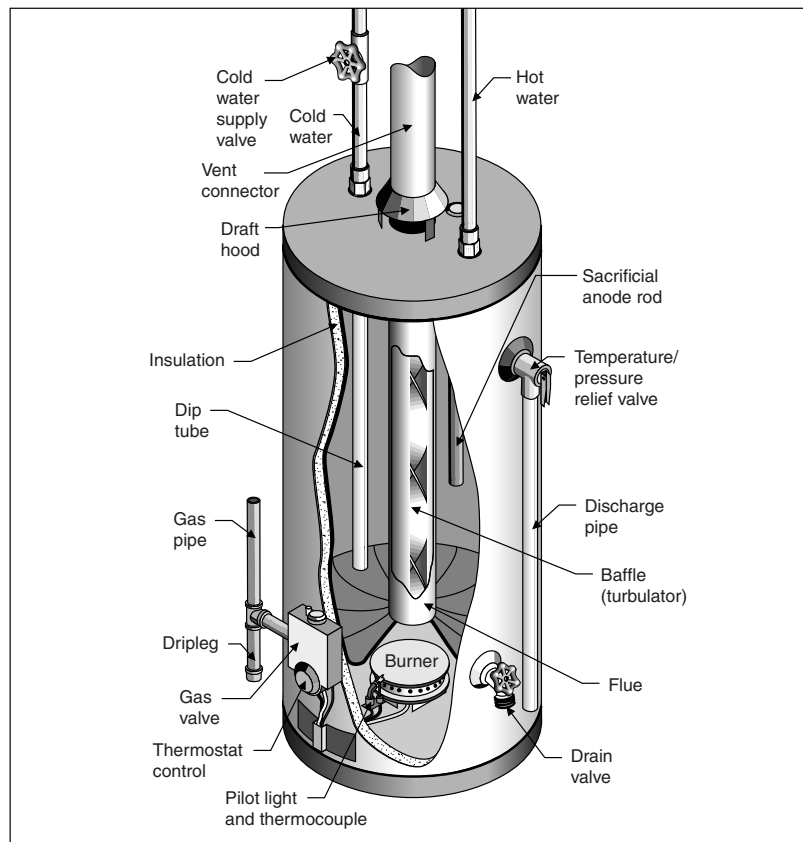
Electric water heaters work on a different principle (Figure 13.10). There is no core in the center, since there is no burner, and no heat exchanger or venting system is required. Electric water heaters typically employ one or two immersion elements that are similar to the elements in a kettle.

#### *Thermostat Control*

The elements are thermostatically controlled. The thermostats are not normally accessible without removing an access panel, and often some insulation. If the electric power to the water heater is on, **there is a risk in removing these access panels and accessing the thermostats.** There are live electrical connections adjacent to the thermostats. Unless you are comfortable and skilled working with electricity, you should leave these access panels closed. The Standards don't ask us to remove these panels during an inspection.

These elements provide the heat and, for the most part, the water configuration is the same as on a gas or oil system. Cold water is introduced near the bottom of the tank and hot water is drawn off near the top. The cold pipe may enter the top of the tank or the side of the tank near the bottom.



**FIGURE 13.8** Gas Water Heater

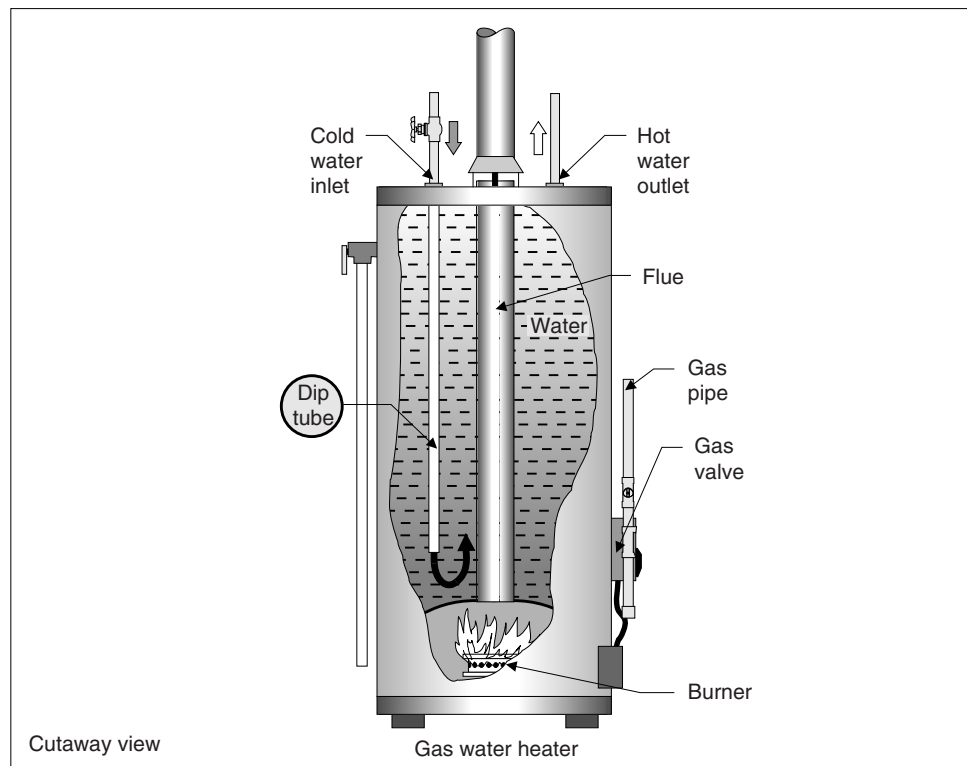
### 13.2.3 Conventional Tank-Type Water Heaters

#### *Inspection Easy*

Inspections of water heaters are very straightforward in that you are looking for hot water and the absence of leaks. Water heaters have many of the same components as and are subject to many of the conditions we discussed with respect to furnaces in Chapters 8–10. We will not repeat that information here, but encourage you to review the components of those systems, their conditions and strategies for inspecting them.

Let's look at some problems we can find with conventional tank-type water heaters.

1. Inoperative (no hot water)
2. Inadequate capacity/recovery rate
3. Leaking
4. Rust
5. Gas burner cover or roll out shield missing/damaged/rusted
6. Damaged tank
7. Inadequate clearance from combustibles
8. Insulation obstructing combustion air or draft hood (gas or oil)
9. Poor location
10. Low water pressure and flow

**FIGURE 13.9** Dip Tube in a Water Heater

11. Noisy water heater
12. Temperature/pressure relief valve problems
13. Baffle collapsed or missing (gas or oil)
14. Tank not stable/wobbly
15. Hot/cold piping reversed
16. No isolating valve
17. Drain valve problems
18. Propane water heater in low areas

**IMPLICATIONS****Inoperative (No Hot Water)**

Most people won't do without hot water for any length of time. If there is no hot water in the house, people can't wash comfortably. Dishwashers and clothes washing machines will not work efficiently.

**STRATEGY**

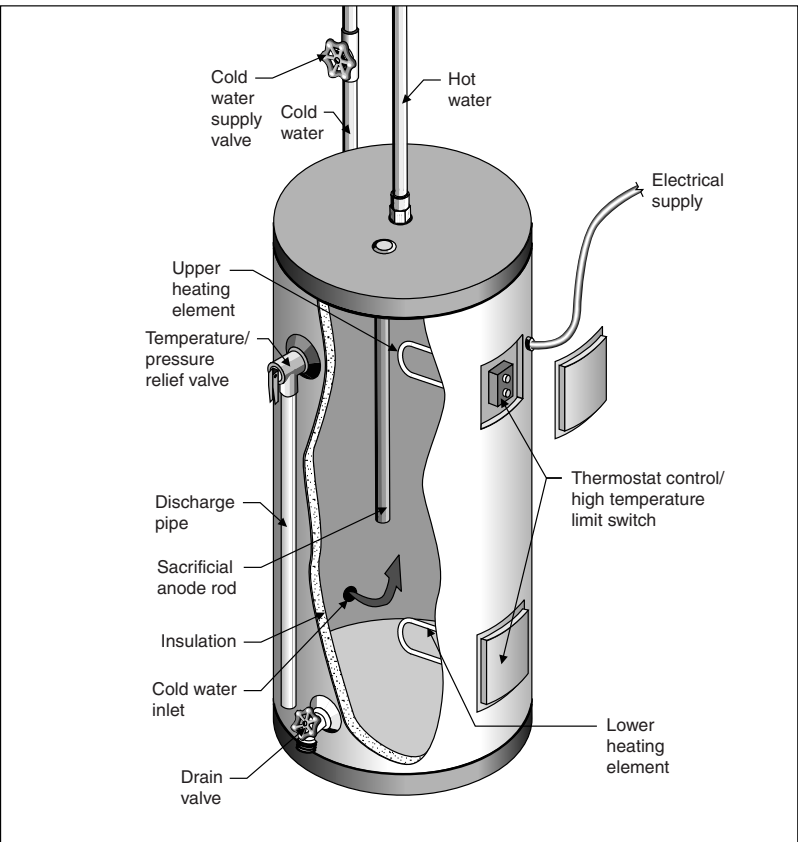
As you perform your flow tests on the plumbing fixtures, you'll notice if there is no hot water. You may have been able to predict this based on your inspection of the water heater.

**Inadequate Capacity or Recovery Rate**

The amount of hot water required by a family is very much an issue of lifestyle. There are lots of ways to estimate it, but what is fine for one family of four won't be anywhere near enough for another family of four.

*How Much Is Enough*

**FIGURE 13.10** Electric Water Heater



*General Rule*

**IMPLICATIONS**

**STRATEGY**

Here's a very rough guideline. For a family of four, a 40-gallon gas water heater might be considered typical. A 60-gallon electric water heater and a 30-gallon oil water heater might be considered equivalent.

An undersized water heater is frustrating and, although the tank may be filled with hot water during 23 hours of the day, if there isn't enough water to allow everyone to have a shower in the morning, it's going to be considered inadequate.

Most home inspectors don't spend a lot of time interviewing their clients and trying to anticipate their lifestyles. We report on water heater sizes and give people an indication as to whether that's typical. If a family uses more hot water than average, a larger or second water heater will be necessary. Water heaters are not terribly expensive, and an undersized heater should not be a deterrent from buying an otherwise satisfactory house. Further, if the tank size is typical, a high-usage family will probably have to add capacity in whatever home they buy.

**Leaking**

A leaking water heater may be simply a nuisance. The water may run over to a floor drain and disappear without any great effect. Serious leaks in water heaters can cause flooding. Obviously hot water leaking out of the plumbing system will reduce the efficiencies of water heaters. Leaks also reduce the pressure and flow available at fixtures throughout the house.

**IMPLICATIONS**

**STRATEGY**

Look for leaks around the water heater and at the top and bottom. Be careful with a gas-fired water heater. When the burner starts up, there will often be condensation in the combustion chamber running down through the center of the water heater. You may hear or see water droplets at the bottom of the heater. Don't mistake this for leakage. The condensation will typically stop within five minutes of the burner starting.

**Rust**

The tank itself or its feet may be rusted.

Rusted tanks are prone to leakage. Rusted feet may make the tank unstable.

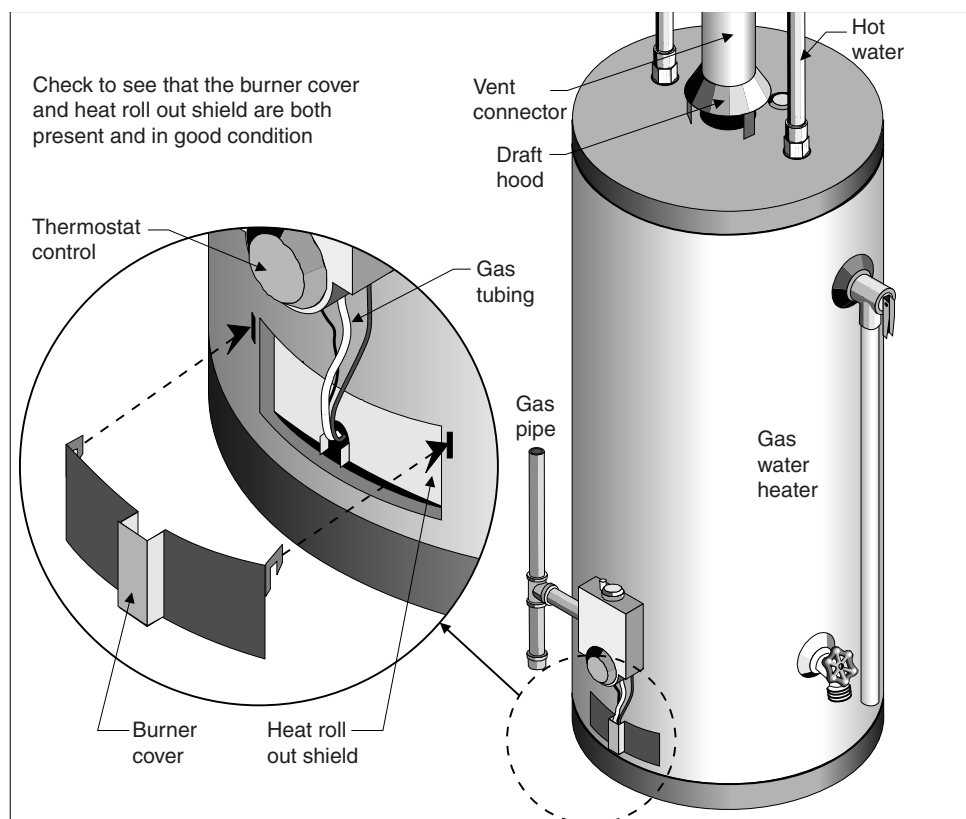
**IMPLICATIONS****STRATEGY**

When looking for rust on water heaters, also check the inlet pipes and isolating valves immediately above or beside the tank. Rust in these areas is also common. Pay particular attention to the bottom part of the tank and to the feet. If the tank is against a wall or close to it, a mirror and flashlight may be helpful in getting a look at the inaccessible areas.

### **Gas Burner Cover or Roll Out Shield Missing/Damaged/Rusted**

On gas water heaters, the burner is typically concealed behind a door and a **heat roll out shield** (Figure 13.11). Open the door and move the shield during your inspection to get a look at the burner in operation and at rest.

**FIGURE 13.11** Gas Burner Cover and Roll Out Shield



**IMPLICATIONS**

A missing burner cover or shield will allow the burner to get dirty. It will also allow flames to roll out of the burner in a backdrafting situation. This can be a fire hazard and may damage the controls.

If the burner cover or shield are damaged, they won't be able to do their job. If one of these is rusted, it probably is not easy to operate, and may not be able to perform its function.

**STRATEGY**

When you remove the cover and move the shield, make sure that both are intact and in good condition. If you see rust on either of these, you might suspect rust on the burner and combustion chamber. You should look more closely.

The rust may also be an indication of leakage. Look carefully for evidence of leakage at the tank or fittings.

**Damaged Tank**

Mechanical damage of a tank is often in the form of a dent. Sometimes the tank is split along a seam. Surprisingly, the tanks are often watertight, even when they show this kind of damage.

**IMPLICATIONS**

If the tank leaks, obviously this is a serious implication. If the tank has been dented noticeably, its capacity will be somewhat reduced and a glass lining may be ineffective on the inside. A shorter life expectancy for the tank may result. However, if there is no evidence of leakage and the system is delivering adequate hot water during your inspection, this is not usually something that you'll recommend changing.

**STRATEGY**

Note any damage to the tank and look for evidence of leakage or premature rusting as a result. Make sure that the tank is still stable on its base and that the draft hood, if it's a combustion type, is intact.

**Inadequate Clearance from Combustibles**

Most water heaters do not have large requirements for combustible clearances. Clearances of less than 6 inches are common. Some gas units only need 1 inch clearance. There does, however, have to be adequate working space (30 inches wide and deep commonly) around the control side of the heater to service the burner and replace the heater. Doors into closets or rooms with water heaters should be wide enough to allow for removal and replacement.

*Combustible Floors*

Most gas and oil water heaters are designed for installation on combustible floors, but not carpeting. You're wise to check with the manufacturers of the common systems in your area.

**IMPLICATIONS**

This is a fire hazard or a service and replacement problem. If someone has built walls around a water heater, the replacement costs are going to be much higher than average.

**STRATEGY**

Look on the data plate for the required combustible clearances and make sure they are met.

**Insulation Obstructing Combustion Air or Draft Hood (Gas or Oil)****IMPLICATIONS**

The combustion air inlet for the burner must not be obstructed with insulation. This often happens if the insulation drops down after it's been wrapped around the tank. Similarly, care has to be taken that insulation is not placed around the top of the tank, obstructing the draft hood on a gas water heater. If this happens, combustion may be inefficient at best and the burner may go out. Another possible implication is backdrafting or combustion gas spillage.

**STRATEGY**

Check both the bottom and top of gas fired water heaters for adequate combustion and dilution air. On an oil burner, make sure that the burner itself is not wrapped with insulation and that the combustion air inlet is not obstructed.

**Poor Location**

*Bedrooms, Closets, Bathrooms*

Gas or oil-fired water heaters should not be located in sleeping areas, including bedrooms or bedroom closets. We also don't want gas or oil-fired water heaters in clothes closets or bathrooms. We're worried about backdrafting appliances killing people in their sleep and about inadequate combustible clearances. Some areas allow direct vent water heaters in these areas because the chance of backdrafting is very small.

*Garages*

We don't want gas or oil-fired water heaters that are less than 18 inches above the floor in garages. Actually it's the burner and/or pilot that has to be at least 18 inches above the floor. We don't want gasoline vapors to be ignited by a pilot or burner on a water heater. Mechanical protection from vehicles should also be provided for water heaters in garages. Concrete-filled steel pipes set into the floor are one common way to accomplish this.

*Outdoors*

If the water heater is located outside, it should be on a pad above grade level. We don't want to rust the feet of the water heater.

*Drain Pans above Living Space*

If a water heater is located above a living space (in warm climates only) a drain pan is usually recommended below the water heater in case of leakage. Pan walls are usually at least 1½ inches high and the pan area is at least larger than the water heater. The pan has to be piped to a suitable location.

**IMPLICATIONS**

There are life safety implications where water heaters are in bedrooms, bathrooms and closets. Fire and explosion are the implications where a water heater is not high enough off a garage floor. Damage to the water heater and possibly fire and explosion are issues if the water heater is not mechanically protected. Water damage to the living space is a possibility if there is no drip pan under a water heater above a living space.

**STRATEGY**

Check for the issues we've talked about and recommend improvements needed.

**Low Water Pressure and Flow**

Low pressure and flow on just the hot water plumbing is fairly common. Where the piping is galvanized steel, you can expect the rust to be heavier on the hot water piping and the low water pressure on the hot side is not a surprise. The hot water piping arrangement is usually longer than the cold water, so slightly lower pressure may be expected.

The low water pressure and flow reduces the usability of the house plumbing.

If the water pressure is low for no apparent reason, you might recommend flushing of the water heater through the drain valve. Attaching a hose to the drain and directing water to a floor drain or outside the building is fairly straightforward. Most people recommend shutting off the unit and opening the drain valve until the water runs clear.

Some large homes have water service pipes of 1- to 1½-inch diameter. The pressure and flow in these houses can be affected by residential water heaters which typically have ¾-inch connections.

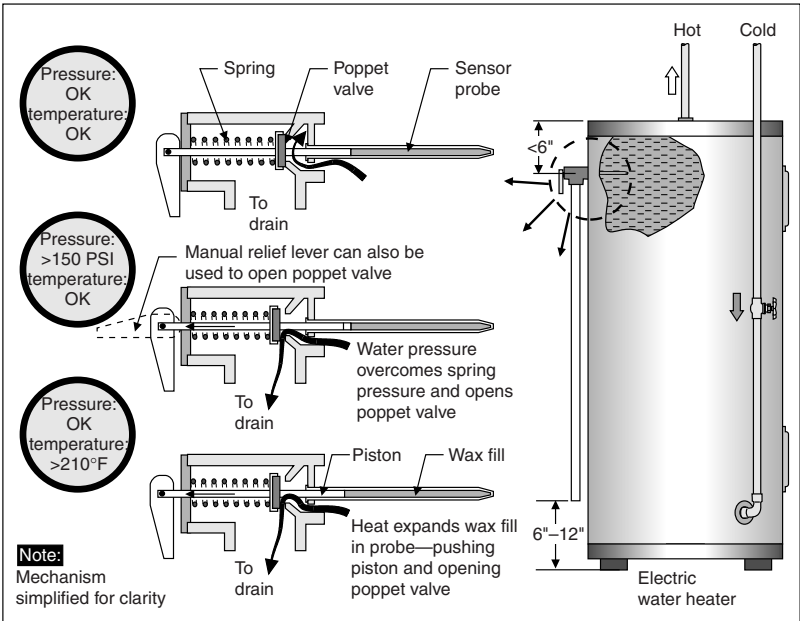
**IMPLICATIONS****STRATEGY**

*Poor Water Pressure and Flow on Larger Homes*

**Noisy Water Heater**

Water heaters are sometimes noisy because the burner is a problem. Popping sounds at the burner on start up or shut down may be an issue with the gas pressure and/or

FIGURE 13.12 Temperature/Pressure Relief Valve



burner adjustment. It may also be condensation dripping onto the burner. This should be checked by a specialist.

Sometimes the water heater rumbles. The rumbling sound appears to come from the tank.

Noisy burners may deliver less hot water than they should and be inefficient. Rumbling sounds indicate sediment in the tank and inefficient operation. Lower recovery rates and sometimes low water pressure and flow result.

If the burner is noisy, recommend servicing. Where a rumbling sound is heard at the water heater, again recommend that the tank be flushed.

|              |
|--------------|
| IMPLICATIONS |
| STRATEGY     |

Temperature/Pressure Relief Valve Problems

Like hot water boilers, all water heaters must have some way of protecting against high temperatures and high pressures in the tank.

Temperature/pressure relief valves (TPR valves) do two jobs (Figure 13.12). They will open and dump water out of the tank if the temperature exceeds 210°F (just below boiling). They will also open if the pressure in the tank exceeds 150 psi (the maximum normal operating pressure for a water heater). The temperature/pressure relief valve will have a BTU rating stamped on it. Its rating must be the same as or larger than the BTU rating of the water heater.

A temperature/pressure relief valve has to have its probe in the top 6 inches of the tank where it sees the hottest water. Location is more important for temperature than pressure, since the pressure will be uniform throughout the tank.

The temperature/pressure relief valve (TPR valve) has to have a discharge tube attached to it. These tubes used to be metal (typically steel or copper). Now most are plastic, typically CPVC. The discharge tubes cannot have any shutoff valves and have to extend down to a safe location, less than 6 inches above the floor level. This minimizes the chance of spraying anyone with scalding hot water.

Temperature/Pressure Relief Valves

Location

Discharge Tubes

*Pipe Diameter and Ends*

The discharge tubes have to be at least the same diameter as the relief valve. The tube isn't allowed to have any threads or cap on the end because we don't want it to be closed off. The old metal tubes are often rusted on the interior and almost closed off.

*Other Ways of Accomplishing the Same Thing*

Some systems have an internal high temperature shutoff. Many electric water heaters have an energy cutout (ECO), for example. In an electric water heater, the energy cutout may be inside the tank and not visible, although its presence is usually indicated on the data plate. If there is an ECO, the water heater should have a pressure relief valve only, instead of a temperature/pressure relief valve.

*Gas Shut Off on High Temperature*

Some gas water heaters use temperature cutouts, such as the Watts 210 valve. This is a valve in the gas supply line to the water heater. There is a temperature probe in the top 6 inches of the water heater. The valve will shut off the gas supply to the water heater if the water temperature exceeds 210°F.

*Oil Burner Temperature Control*

Oil-fired water heaters may also have an internal high-temperature limit control that will shut off the burner.

*Pressure Relief Valve*

If the temperature control is provided, we only need a **pressure relief valve**, not a temperature/pressure relief valve. If we have only a pressure relief valve, its location is not so critical because the pressure will build up uniformly in the tank and in the immediate vicinity. The pressure can be sensed anywhere on the tank (not just within the top 6 inches) and in some jurisdictions it can even be on the piping near the tank. There cannot be any valves between the pressure relief valve and the tank itself.

Let's list some of the specific problems you might find on a temperature pressure relief system:

1. Missing relief valve.
2. Undersized relief valve (BTU rating less than water heater).
3. Missing temperature protection (pressure relief valve only, no ECO).
4. Poor location for temperature sensor of temperature/pressure relief valve (must be in top 6 inches of tank).
5. Shut off valve between the pressure relief valve and the tank (not allowed).
6. No discharge tube (dangerous—may scald people).
7. Discharge tube too short or too small diameter.
8. Discharge tube damaged or split.
9. Discharge tube dripping or leaking.
10. Discharge tube capped, threaded or rusted closed.

**IMPLICATIONS**

Temperature/pressure relief devices are safety items. If any of these conditions exist, safety may be compromised.

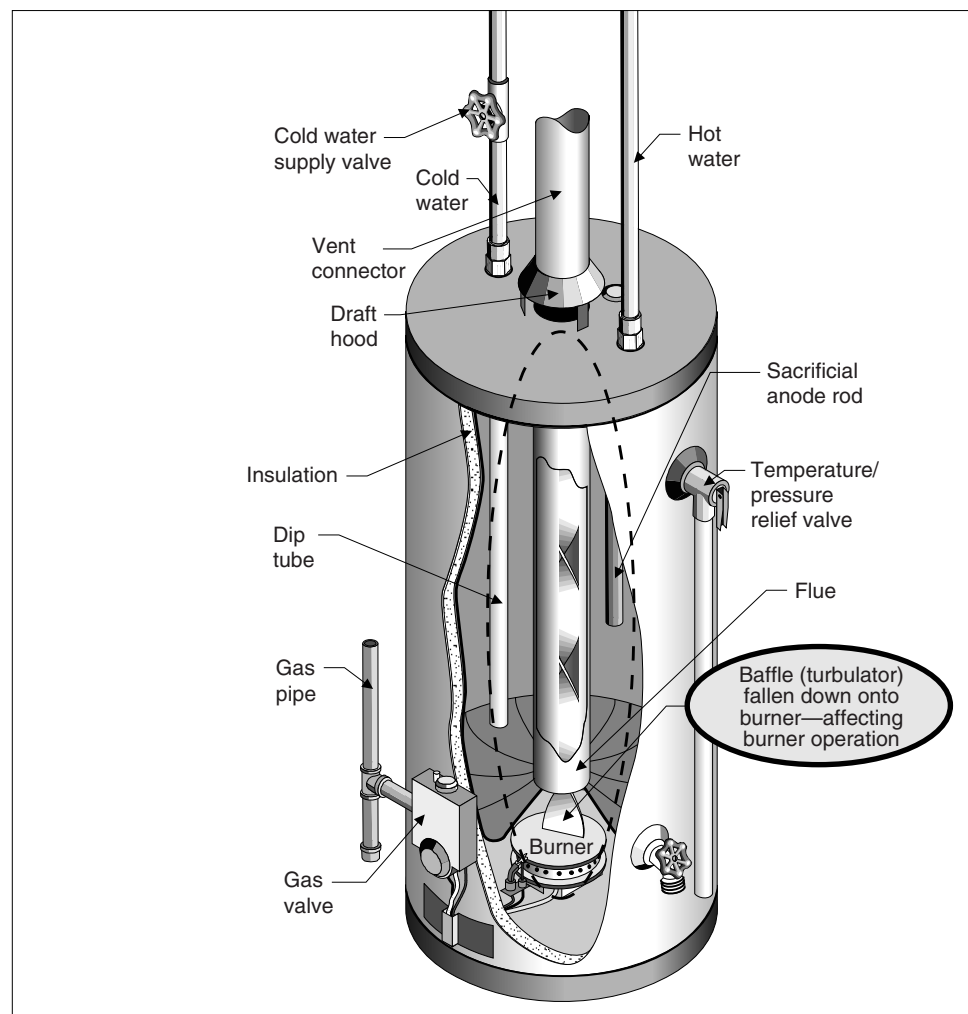
**STRATEGY**

Check temperature/pressure relief valves, high temperature limits and pressure relief valves for all of the problems we've listed above. The life safety issues here are significant.

**Baffle Collapsed or Missing (Gas or Oil)***Turbulator*

The baffle, or **turbulator** as it is sometimes called, is a stainless steel (typically) helix installed in the flue passage to slow down the gas or oil exhaust products as they move up through the water heater and out toward the vent connector (Figure 13.13). The longer the exhaust products linger in the water heater, the more heat will be transferred into the water.



**FIGURE 13.13** Baffle Collapsed or Missing**IMPLICATIONS**

The baffle may fall onto the burner, affecting burner operation and reducing the efficiency of the system. A missing or displaced baffle will allow exhaust products to move too quickly through the tanks. This increases fuel consumption.

**STRATEGY**

When looking into the burner compartment on a gas-fired water heater, make sure that the baffle has not fallen down onto the burner. You may be able to look around the draft hood at the top and see the top of the baffle. Where it is not intact, it may have fallen or it may be missing.

In an oil-fired water heater, you may find that the baffle has fallen down into the firepot or refractory. This can sometimes be seen looking in through the viewing port.

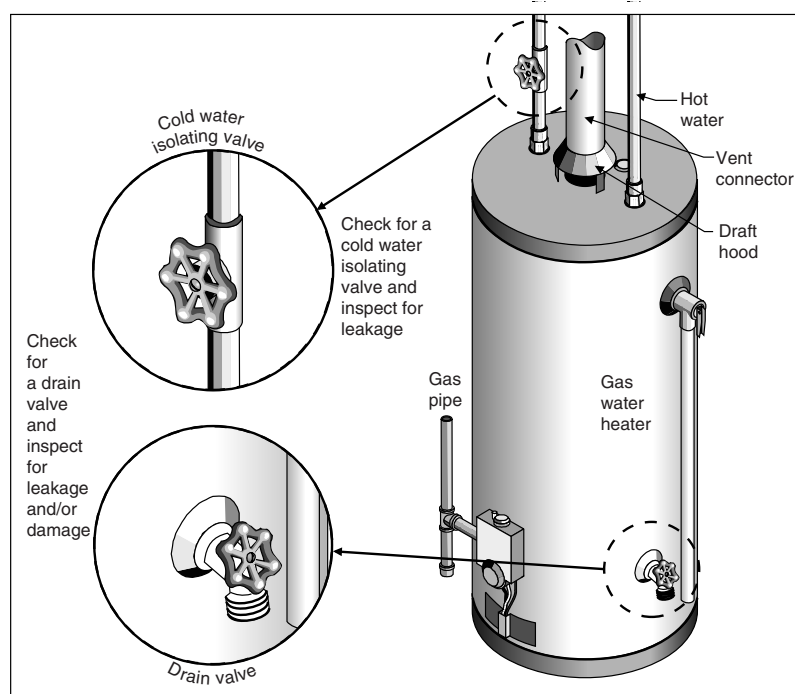
**Tank Not Stable/Wobbly**

Tanks need to be well supported and secure on the floor.

The implications of an unstable tank are leaks, either as water connections are strained or as the tank falls.

When inspecting a water heater, push and pull with reasonable pressure on the top of the tank to make sure it is not wobbly or loose.

**IMPLICATIONS****STRATEGY**

**FIGURE 13.14** Isolating and Drain Valves

### No Isolating Valve

There should be an isolating valve on the cold water inlet in case the tank ruptures and floods (Figure 13.14).

If there is no isolating valve, the system can't be turned off in an emergency situation. Additional water damage may occur.

Check that there is an isolating valve on the cold water piping adjacent to the water heater.

### Drain Valve Problems

Water heaters should have a drain valve at the bottom. The valve may be missing, damaged or leaking.

If there is no drain valve, it will be difficult to get the sediment out of the bottom of the tank. This affects the efficiency and amount of hot water available. It can also result in noisy tanks.

If the valve is damaged, it may not be operable. If the valve is leaking, obviously water damage is a possibility.

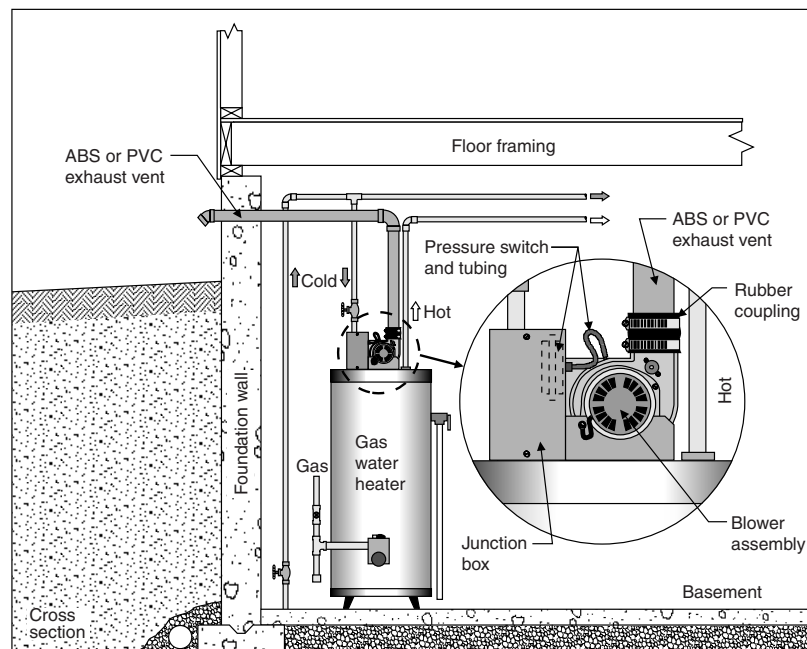
Make sure that there is a drain valve and that it appears intact. We do not recommend that you open the drain valve. Hot water will be discharged immediately. The other risk is that the valve will not shut off when you close it. Again, you don't want to create a leak.

#### IMPLICATIONS

#### STRATEGY

#### IMPLICATIONS

#### STRATEGY

**FIGURE 13.15** Power-Vented Water Heater

### 13.2.4 Fan-Assisted Gas Water Heaters

Fan-assisted or power-vented gas water heaters have become popular in some areas. These are typically used with modern heating systems that don't use chimneys. Where the heating system is sidewall-vented or electric, no chimney is needed. It becomes impractical to install or maintain a large chimney for a domestic water heater. If an existing chimney remains in service, it is often too large for the water heater once the furnace or boiler is replaced with a sidewall-venting unit. Oversized chimneys are prone to condensation and deterioration.

#### *Get Rid of the Chimney*

A common approach is to replace the water heater with a sidewall-vented unit. **Power-vented** or **fan-assisted** water heaters can be vented through the sidewall using ABS or PVC piping, typically (Figure 13.15).

#### *Added Complexity*

Power-vented water heaters have an exhaust fan that requires electricity, controls and a safety device. These water heaters are more complex, more expensive and more prone to failures.

#### *Automatic Ignition Systems*

Most power-vented water heaters have either an intermittent pilot or hot surface ignition for the burner. These units do not typically have a standing pilot.

#### *Making Sure the Blower Is On*

These water heaters won't operate if the blower is not on. We run the risk of dumping exhaust products into the house and overheating the exhaust piping if the blower is not working. A pressure switch checks to see that the blower is working before the burner is allowed to come on.

#### *Venting Issues*

The vents on power-vented water heaters are smaller diameter, typically, than conventional water heater vents. They are plastic vents that do not require any combustible clearances. The vents usually terminate outdoors and have the same rules for termination location as other sidewall-vented appliances.

#### *Slope and Supports for Vents*

Since this is not a condensing appliance, the manufacturers may not require any slope on the vent piping. These water heaters do not have a condensate management system, so we don't necessarily want to slope the vent back down to the water heater

to collect condensate. Most manufacturers will allow a slope back down toward the appliance, but they don't want to have to handle a lot of condensation. The piping should be supported every 4 feet horizontally, as is typical for plastic piping.

Inspection of power-vented water heaters is similar to conventional water heaters with a few added conditions to watch for. Problems with power-vented systems include the following:

1. Inoperative
2. Blower intake obstructed
3. Poor venting arrangement
4. Poor blower wiring

### **Inoperative**

There will be no hot water.

We're simply testing to make sure that hot water is delivered. If you operate hot water faucets and get only cold water, go down to the water heater. If it's a power-vented heater, you should hear the blower operating even before you remove the burner cover to check for burner operation. If the blower is not operating, the burner should not be on.

### **Blower Intake Obstructed**

These systems require considerable amounts of dilution air as discussed.

The pressure switch may not be satisfied and the system will not operate. We don't want the water heater to operate if it can't get enough dilution (cooling) air.

Look to see that the blower intake area is free and open.

### **Poor Venting Arrangement**

Venting issues can be similar to those on high efficiency furnaces. Look for such things as poor termination points, inadequate support, excessive elbows and inappropriate vent size. You may have to refer to an installation manual for details on specific systems. Where the venting system is suspect and you can't verify a problem, recommend further investigation.

### **Poor Blower Wiring**

Any electrically operated device needs to have safe wiring. You should be looking for a grounded system with wiring of an appropriate size. Fuses or breakers for the wiring should be appropriate. The wires should be of a suitable type and adequately secured and protected. Connections should be suitably made inside junction boxes.

Many power-vented water heaters simply use a cord and plug that is put into a 120-volt convenience receptacle. The cord should not be in the path of the exhaust products and should not contact the exhaust piping. In some areas, a dedicated circuit has to be provided for the water heater fan.

Electrical problems are safety issues.

Follow the electrical supply system to the water heater and to the fan itself. Look for the problems we've listed above.

#### **IMPLICATIONS**

#### **STRATEGY**

#### **IMPLICATION**

#### **STRATEGY**

#### **IMPLICATION**

#### **STRATEGY**

## 13.3 DRAIN, WASTE, AND VENT PIPING

*Waste Includes Solids, Liquids, and Gases*

In the discussion of supply plumbing we learned how to get clean water into the house and control it. Now we have to get rid of the waste that includes the solids, liquids and gases. Once again, the concept is very simple; the practice is more complicated.

*Flows by Gravity*

We get rid of our waste by letting it fall or flow through a series of nearly horizontal and vertical pipes, down through and out of the house. Unlike the supply system, we do not need any pumps or pressure to move the waste, at least not in most cases.

*Private or Municipal Sewer System*

Once the waste leaves the house, it is no longer the concern of the home inspector. It goes either to a municipal sewage treatment facility (via city sewers), or an on-site facility such as a septic system, for example. There are other options including holding tanks, seepage tanks and cesspools, but we won't address these here, since they are outside the scope of a Standards inspection.

*Don't Guess Where Sewage Goes*

One word of caution—if you are going to identify the sewage handling system as public or private, we recommend that you do some research. It can cost you several thousand dollars if you are wrong on this one. If you mistakenly tell someone that they are on a public sewage system, you have a serious problem.

The components of the drain, waste and vent plumbing system that we will look at are: drain piping, traps, floor drains, venting systems, sewage ejector pumps, sump pumps, and laundry tub pumps.

*Drain, Waste, and Vent Pipes*

There are different names for pipes on the waste side of plumbing systems, based on their functions.

- **Soil pipes** carry liquids and solids. The pipe from a toilet is a soil pipe.
- **Waste pipes** carry only liquids according to some authorities and liquids and solids according to others. The terms **drain** and **waste** are often used interchangeably.
- **Drain pipes** are intended to carry liquids and solids (some say liquids only).
- **Vent pipes** carry air only (no liquids or solids).

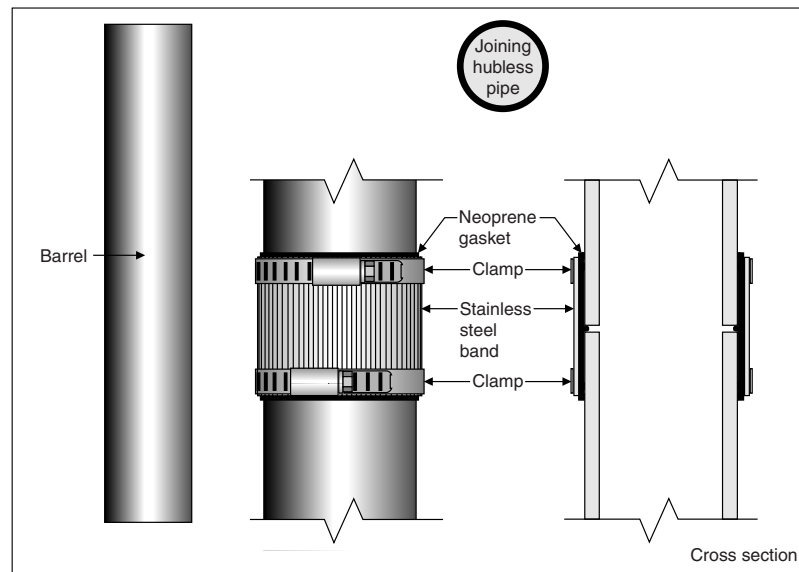
Most pipe materials are described as **DWV** (Drain, Waste and Vent), which means they are suitable for solids, liquids, and gases.

### 13.3.1 Drain Piping

*Materials*

Drain, waste, and vent piping may include the following materials:

- Copper
- ABS
- PVC
- Cast iron
- Lead (drain and waste only)
- Galvanized steel (typically drain and vent only)
- Brass (typically fittings)
- Clay (underground drain and waste only)

**FIGURE 13.16** Connecting Cast Iron Waste Pipe—Hubless

### Pipe Supports

#### Vertical Supports

Generally speaking, vertical supports should be provided at every other floor level. Some areas call for supports at every floor level, particularly with heavy pipe like cast iron. Most areas call for the support of copper piping every 10 feet vertically. Plastic piping needs guides at the mid story height between supports in some jurisdictions. We don't usually see this during a home inspection, so we won't be able to comment.

#### Sway Bracing

On cast iron waste piping with flexible hangers that are longer than 18 inches, sway bracing is often required to stabilize the pipe in areas where seismic activity is an issue.

#### Additional Support for Cast Iron Piping

Where cast iron piping has hubless connections (with a sleeve, gasket or shield and clamps) additional supports may be required (Figure 13.16). You should check to see what your area calls for here. Understand, however, that most new work will not include cast iron piping, and that unless supports have been damaged or removed, the age of the system usually proves the success of the installation.

#### Fittings Cannot Reduce Pipe Diameter

It is critical that the interior of waste pipes be smooth and that size changes always go from smaller to larger. The fittings are not permitted to reduce the interior diameter on waste piping. We don't want to create surfaces or edges that hang up solids.

#### Bell and Spigot Connections

Bell and spigot connections were common on cast iron pipes, although they have largely been replaced with hubless connectors. The pipes have a male and female end. The pipes are fit together and the base is packed with oakum or jute. This material is sealed in place with molten lead to a depth of at least 1 inch. The oakum or jute is sometimes called **caulking**.

#### Lead Piping Replacement

Lead piping hasn't been used commonly since the 1950s. It is prone to leakage in most installations after 40 years or so. Galvanized steel supply piping often needs to be replaced after 40 years because of leaks or low pressure. Many home inspectors recommend replacement of lead waste piping whenever galvanized steel supply

*Galvanized Steel Pipes*

piping is being replaced. It's common to find both materials at bathrooms, for example. Since changing the supply piping is disruptive, it's a good idea to replace the lead waste piping at the same time.

Galvanized steel pipes are strong and, when used for vent piping, often have indefinite life expectancies. However, when they are used for drain pipes, they will rust on the inside. The rusting creates an uneven surface which can lead to clogging before the pipes rust through. Galvanized steel pipes very often rust through at the threads first. Galvanized steel drain pipes were commonly used under sinks and laundry tubs.

*Copper Piping*

Copper drain, waste and vent pipes are considered by many to be the best material. They were popular from the 1950s to the early 1970s.

*ABS and PVC*

Lightweight plastic ABS and PVC replaced copper drain, waste and vent systems in the 1970s in many areas because they are inexpensive and easy to work with. The chief criticism of these pipes is that they are noisy. In some areas, plastic stacks in walls and ceilings are wrapped with insulation to minimize the noise. This helps, but does not completely eliminate the problem.

*Combustibles and Plastic Piping*

Plastic piping cannot typically penetrate floors, ceilings or walls that separate two dwellings. Usually, there has to be a fire separation between adjacent dwellings. A plastic pipe would compromise this separation.

*Supports for Plastic Piping*

Supports for plastic pipes shouldn't rub on the piping. The pipe can wear through as a result of abrasion. The pipe tends to expand and contract as it heats up and cools down and is susceptible to being worn through over time. Insulating the pipe to protect it from hangers is one solution, although this is rarely done. Metal hangers are worse than plastic hangers.

*Connecting ABS and PVC*

The solvents used to join ABS piping sections together are not the same as the solvents used to connect PVC piping. The solvents are not interchangeable. Where ABS and PVC piping are to be joined together, a special combination solvent can be used.

*Sunlight Attacks ABS*

In some areas, authorities do not allow plastic piping (especially ABS) outside where it may be exposed to direct sunlight, which will break down the plastic over time. PVC may have UV (ultraviolet) stabilizers that make it more durable outside. Where ABS or PVC stacks extend above roofs, they are supposed to be painted in some jurisdictions, to protect them from the UV rays. Connections are weak spots on drain, waste and vent systems and you should check all connections, paying particular attention to those between dissimilar materials.

We've talked about the general function of drain, waste and vent piping and the common materials used. Let's look at some of the problems we find in the field, starting with drain and waste piping.

## Conditions

Common problems with drain, waste and vent piping include the following:

1. Leaks
2. Rust
3. Freezing
4. Split/damaged/crimped
5. Clogged
6. Poor support
7. Cross connections
8. Undersized/overloaded

9. Poor slope
10. Cleanouts missing or inaccessible
11. Exposed to mechanical damage
12. Pipe size reduced downstream
13. Galvanized steel buried in soil
14. Combustible piping
15. Noisy
16. Non-standard materials and patches
17. Missing air gap for dishwashers
18. Clothes washer drainage improper
19. Condensate drain connections improper
20. Poor manifolding

### Leaks

Leaks in drain and waste piping systems can be more difficult to find than in supply systems because the pipes are usually idle. Supply pipes are always full of pressurized water. Waste pipes intermittently see water at atmospheric pressure. The implications of a leak can be more serious because of the possible impact of sewage getting into the home. It's one of the reasons that waste pipes cannot run above food preparation areas in some jurisdictions.

Sewage leaking into the house can make people sick. Leaks can also cause damage to finishes and to the structure.

You can only verify active leaks in waste plumbing systems by operating the fixtures. In many cases, you will see evidence of leakage but won't know whether it is active until there is water running through the pipes. The more water you run, the more likely you are to find the leaks.

We talked earlier about the importance of looking at connections. Wherever the pipe changes direction or materials, we have a potential leak spot.

Where pipes pass through unheated spaces in cold climates, we have the possibility of leakage as a result of freezing. Sometimes the piping is insulated. Touching the insulation, or removing part of it, especially near the bottom of any exposed sections, may reveal a slight leak. Traps in unheated areas are vulnerable to freezing.

Watch for rusting on metallic piping. Cast iron piping will often leak at small pinholes. The pinhole leaks will often scab over with rust and stop leaking. We strongly recommend that you don't pick or poke at these pinholes. The pipe may be very thin and could collapse if you handle it roughly.

Check the top side of horizontal cast iron piping systems. It's not unusual to find a small split running parallel to the length of the pipe.

### Rust

Rust on copper pipe is typically green. It may be the result of exposure to acids or a corrosive atmosphere. It may also be caused by contact with dissimilar metals.

The implications are ultimately leakage.

Most rusting is on cast iron or galvanized steel piping. The majority of it is on the inside of the pipe, and you won't be able to see the rust until it comes through the pipe surface. This can be pinholes, blisters or splits. As we've talked about before, the rust on threaded piping often appears at the threaded connections first.

#### IMPLICATIONS

#### STRATEGY

*Focus on Connections*

*Freezing Pipes*

*Rust*

*Horizontal Splits in Cast Iron Pipe*

*Rusted Copper*

#### IMPLICATIONS

#### STRATEGY



### Galvanized Steel Piping Close to Grade

#### Surface Rust

Galvanized steel or iron piping should be at least 6 inches above grade. It may have been buried in the soil on original construction or be a result of renovations or earth being moved carelessly in crawlspaces. Where you see earth in contact with galvanized steel piping, look closely for evidence of rusting on the piping exterior.

Rusting on the surface of piping can be caused by condensation or by leaks running along the outer pipe surface. This may be less serious than rust coming through from the inside, but it is often tough to know how extensive the damage is during an inspection.

### Freezing

Freezing drain and waste piping is only an issue where the piping is filled with water in cold climates and where the piping is exposed to outdoor temperatures. These conditions may exist in garages and crawlspaces, for example.

Clogging, splitting, and leaking of the piping are obvious implications.

In many cases, pipes pass through unheated spaces without insulation. Because the warm water passes through quickly and infrequently, there may be no freeze up problems. However, if there is a faucet that drips, a trickle of water running down the inside of the pipe may be cooled enough to freeze. If uncorrected, the ice can build up to the point of obstructing the drain. This is more common on horizontal pipes but can also happen on vertical pipes.

Where waste pipes pass through unheated areas, they should be insulated and/or protected from freezing with electric heat tracing wires. Heat tracing is more effective because, although the insulation slows heat loss, it doesn't stop it. If things stay cold enough long enough, insulated pipes may freeze.

#### Insulation and Heat Tracing

#### IMPLICATIONS

#### STRATEGY

### Split/Damaged/Crimped Pipe

The implications of split, damaged or crimped piping are leaks and clogging.

Follow the lengths of pipes looking for irregularities such as dented or crimped spots, splits or areas of obvious mechanical damage. Watch for abrasion of plastic piping against metal or masonry, for example. Metal hangers on plastic pipes can wear through the piping.

#### IMPLICATIONS

#### STRATEGY

### Clogged

You won't always be able to identify clogged piping and where you can find that it is clogged, you will often have difficulty finding the clog.

This is a health issue since waste will back up in the fixtures.

We normally find clogged drain pipes by flowing the fixtures. Good plumbing fixtures will drain water out of them as quickly as it can be introduced with both faucets fully open. Many fixtures cannot meet this standard, but that does not mean that they are defective or that the drains are clogged.

You have to use your judgment to determine what is a reasonable amount of time for draining. Many full basins can drain in roughly 15 seconds. If the basin takes more than 30 seconds to drain, there may be an obstruction.

In addition to the drainage piping system, the fixture drain stopper, strainer, outlet pipe or trap may be clogged. Don't be fooled by pop-up drain stoppers that don't pop up high enough to drain freely. Also watch for hair, soap scum, and other junk hung up on strainers. Generally speaking, you would have to observe backup at more than one fixture before describing the waste piping as being clogged.

#### 15 Seconds to Drain a Basin

#### IMPLICATIONS

#### STRATEGY

### Poor Support

Generally speaking, you can't see vertical supports except at sub-grade areas and in attics. Supports for horizontal pipes can often be seen in basements or crawlspaces.

Improper slope, clogging and opening at joints are all implications of poor support.

Your first step should be to look at pipes for evidence of movement. If a low spot is visible in a horizontal run, the supports are probably not adequate. Sometimes building settlement or heaving may leave piping unsupported, with the hangers in mid air below the pipe. Watch for pipes that are suspended above their hangers.

Watch for metal piping with metal hangers that are dissimilar. Copper piping, for example, will eventually rust steel hangers.

If your area is susceptible to seismic activity, sway bracing may also be necessary for near horizontal pipes.

#### IMPLICATIONS

#### STRATEGY

*Corroded Hangers*

*Sway Bracing*

### Cross Connections

Cross connections include the supply and waste piping coming together. Examples are given in the discussion of supply piping earlier in this chapter.

Contamination of a potable water supply is the implication of a cross connection.

Look for the common cross connection areas and recommend rearrangement.

#### IMPLICATIONS

#### STRATEGY

### Undersized

Drain and waste pipes have to be large enough to carry away the waste material. The minimum sizes of traps and trap arms for some common fixtures are listed in Table 13.2.

Where drains carry the discharge from multiple fixtures, the drain pipe sizing is determined from **unit loading** tables (unit loading numbers for each fixture are set out in plumbing codes). The drain size depends on several variables including the orientation (vertical or horizontal), the length of the drainage pipe and the length of the vent piping.

Slow drain performance, clogging and possible backup are the implications.

It's beyond the scope of a home inspection to size every drain and vent pipe in the house. For the most part, they're not visible anyway. You should, however, be able to look at individual fixture drains, including their traps, and see if they are suitably sized for the fixture. The chart above is helpful in this regard.

Look under each fixture where visible and ensure that the size of the fixture outlet pipe (tail piece), trap and trap arm (fixture drain) are at least as large as that indicated on the chart.

#### IMPLICATIONS

#### STRATEGY

### Poor Slope

Nearly horizontal drain pipes should generally slope at  $\frac{1}{4}$  to  $\frac{1}{2}$  inch per foot. Large pipes (more than 4-inch) can slope at about  $\frac{1}{8}$  inch per foot.

A pipe that has too low a slope will be prone to clogging. A pipe with too steep a slope may be prone to clogging as the liquids flow by quickly and the solids get hung up in the pipe.

Most home inspectors don't measure pipe slope. You should be able to sight along drain pipes and see a positive slope. Where you can't see a slope or you can see an obvious low spot, recommend further investigation.

#### IMPLICATIONS

#### STRATEGY

**TABLE 13.2** Minimum Trap and Trap Arm Sizes

| Fixture                                 | Minimum Size of Traps and<br>and Trap Arm                                  |
|-----------------------------------------|----------------------------------------------------------------------------|
| single wash basin                       | 1 <sup>1</sup> / <sub>4</sub> inch                                         |
| bidets                                  | 1 <sup>1</sup> / <sub>4</sub> inch                                         |
| dishwashers                             | 1 <sup>1</sup> / <sub>2</sub> inch                                         |
| kitchen sink with or without dishwasher | 1 <sup>1</sup> / <sub>2</sub> inch                                         |
| laundry tub                             | 1 <sup>1</sup> / <sub>2</sub> inch                                         |
| bathtubs                                | 1 <sup>1</sup> / <sub>2</sub> inch                                         |
| floor drains                            | 2 inches                                                                   |
| clothes washers                         | 2 inches                                                                   |
| showers                                 | 1 <sup>1</sup> / <sub>2</sub> –2 inches (depending on<br>shower flow rate) |

**Cleanouts Missing or Inaccessible**

There is typically one cleanout for every horizontal drain. The cleanouts can be above grade, in a basement or in a crawlspace. In some cases, the cleanout cap is flush with the basement floor. In some cases, the cleanout is near the base of a stack. It's good practice to have 12 to 18 inches of clear space at the cleanout to gain access depending on the size of the pipe.

When problems occur, service can be much more expensive if the cleanout is missing or inaccessible.

Make sure that there is an accessible cleanout provided.

**IMPLICATIONS****STRATEGY****Exposed to Mechanical Damage**

In some cases, the pipe is not damaged, but its location makes it vulnerable.

Mechanical damage causing leakage and/or obstruction of the piping is the implication.

Watch for piping that may be damaged by opening doors or by vehicles (in garages), or is otherwise susceptible to being bumped. You'll have to use some common sense here.

**IMPLICATIONS****STRATEGY****Pipe Size Reduced Downstream**

As waste flows it should never go from a large pipe to a smaller one. The pipe size should always increase downstream and as waste from other fixtures joins in.

A poor arrangement is prone to clogging.

As you follow the waste plumbing system, look for spots where liquid or solid wastes will get hung up. A pipe size reduction is one such area. Watch for traps that are smaller than the fixture tailpiece or trap arms that are smaller than the traps.

**IMPLICATION****STRATEGY****Galvanized Steel Pipe Buried In Soil**

The implication is premature failure of the pipe as a result of rusting.

Check for good clearance (6 inches) of these pipes from the soil. Where the pipe is at or below soil level, recommend rearrangement and further investigation of the pipe.

**IMPLICATIONS****STRATEGY**

### Combustible Piping

PVC or ABS plastic piping is combustible. It's generally only allowed in single-family homes. There is usually a fire separation between adjacent dwellings or dwellings above each other. A combustible waste pipe that passes from one dwelling to the next compromises the fire separation.

This is a fire safety concern.

If you are in a multi-family building, watch for PVC or ABS piping passing from one dwelling to the next.

### Noisy

This is a nuisance for homeowners. They can insulate the pipe to reduce the noise somewhat, but this is disruptive and expensive.

Where the waste piping is ABS or PVC plastic, let your client know that some noise should be expected. You can flush a toilet in an upper floor and let your client listen in the room below, so they will have a sense of what to expect.

### Non-Standard Materials and Patches

Non-standard materials to watch for include rubber hose (radiator hose), garden hose, corrugated pipe and polyethylene piping. Waste piping materials are not flexible.

Patches are not generally successful on waste piping.

Leaks and related health problems can result from using improper materials.

Watch for flexible hoses used as piping and patches on piping material including duct tape. Damaged waste piping should generally be replaced rather than repaired. While repairs are common on cast iron plumbing stacks, they should be considered temporary measures.

### Poor Manifolding of Drain Piping

There are several rules for how drain pipes should be interconnected. It's beyond our scope to look at all of these, but you can use some common sense to see whether connections are arranged to encourage the flow of waste in the proper direction.

One example of a poor arrangement is a straight T connection into a waste pipe from fixtures that are back to back or side by side. We don't want the waste from one fixture to flow past the stack and deposit itself in the drain from the other fixture.

Horizontal drain lines joining a vertical stack should connect through:

- 45-degree Y connections
- 60-degree Y connections
- combination Y and one-eighth bends
- sanitary Ts or
- sanitary tapped T branches, for example.

There are other permitted fittings.

When one horizontal drain line connects with another, the connection should be made through a 45-degree Y, combination Y and one-eighth bend, or other similar fitting. Again, the idea is to make the change of direction gradual and to encourage the waste to flow in the desired direction.

Vertical into horizontal connections should be made with 45-degree branches or similar fittings (Figure 13.17).

If the vertical pipe is truly vertical, a 60-degree branch may be used.

| IMPLICATION |
|-------------|
| STRATEGY    |

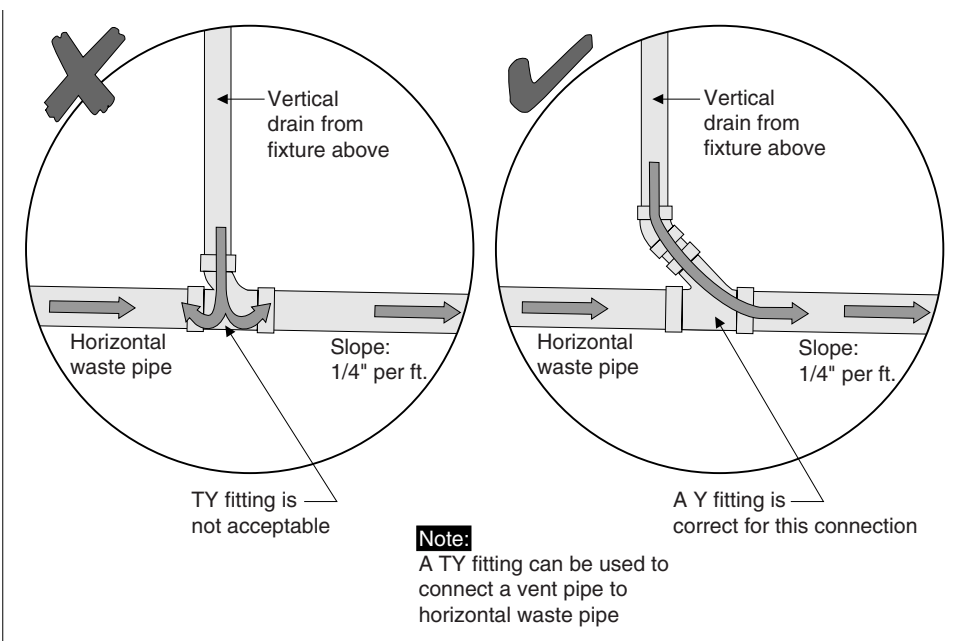
| IMPLICATIONS |
|--------------|
| STRATEGY     |

| IMPLICATIONS |
|--------------|
| STRATEGY     |

*Horizontal Drain to Horizontal Drain*

*Vertical into Horizontal Drains*

**FIGURE 13.17** Connecting Vertical Drain Pipes to Horizontal Drain Pipes



|                     |
|---------------------|
| <b>IMPLICATIONS</b> |
| <b>STRATEGY</b>     |

Poor connections may result in clogging of pipes.

Look at drain connections from a common sense standpoint.

Watch for connections that will not promote flow of waste in the intended direction. Waste should never be asked to make a sharp 90° turn and should never be asked to flow even slightly uphill.

### 13.3.2 Traps

Traps are designed to prevent sewer odors from entering the home through the plumbing fixtures. The seal in the trap is provided by the waste water. Every time we use a fixture, we flush out the water that’s forming the trap seal and replace it with new water.

Traps are carefully engineered systems. They are designed to be self-scouring so that they don’t collect debris, yet retain water to form a seal.

Even though traps are designed to flow freely, they sometimes get plugged. Traps should be provided with slip joints (removable threaded connections) that can be removed, or there should be a cleanout plug at the bottom of the trap (Figure 13.18). Slip joints have a disadvantage because they are slightly more vulnerable to leakage than other types of connections. However, the accessibility issue is important.

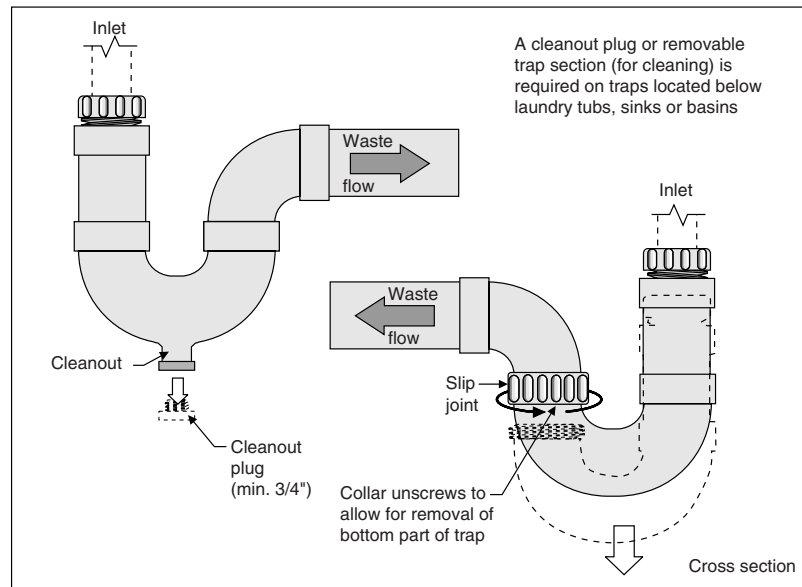
The only fixture that doesn’t need a trap is the toilet because the water in the toilet bowl acts as a trap. The water in the toilet bowl is replenished every time the toilet is used.

Running traps are traps located in the drainage piping that are not directly connected with a fixture. Many older homes had running traps in the building sewer, typically in the front lawn. These traps were designed to prevent sewer gases from entering the house. Modern homes do not have running traps since they were found to be unnecessary once effective venting systems were designed.

*Need to Be Cleaned*

*No Traps for Toilets*

*Running Traps*

**FIGURE 13.18** Trap Cleanout Required

Let's look at some of the problems we find with traps.

1. Missing
2. Wrong type
3. Leak
4. Double trapping
5. Split, rusted or damaged
6. Freezing
7. Tail piece (fixture outlet pipe) too long
8. Trap arm (fixture drain) too short or too long
9. Traps too small or too big
10. Trap primer problems
11. Non-standard shape or material
12. Clogged/no cleanout provision

### Missing

Every fixture should be provided with a trap, except for a toilet.

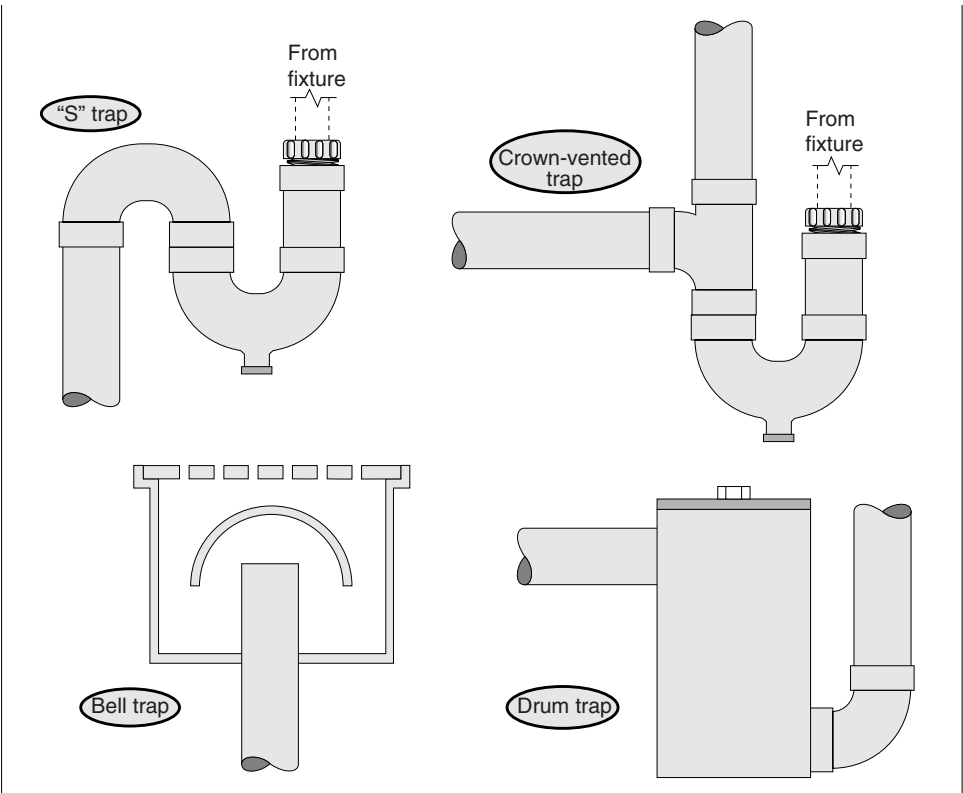
An unhealthy environment in the house will result if sewer gases find their way into the home.

Where you can see below fixtures, look for a trap. If you can't see below the fixture, look down the fixture drain. You should see standing water. You can also use your nose to search for sewer odors.

#### IMPLICATIONS

#### STRATEGY

FIGURE 13.19 Illegal Traps



**Wrong Type**

Residential traps should be **P-type**. **S-traps, drum traps, bell traps, and crown-vented traps** are usually not permitted (Figure 13.19).

The implication of an S-trap is siphoning. The implication of a crown vented trap is obstruction of the vent and, ultimately, siphoning. Bell traps are unreliable and drum traps are not self-scouring. They tend to accumulate solids.

Look for **P-traps**. Recommend that other types of traps be replaced or closely monitored for sewer odors or clogging.

**Leaks**

Leaks are common at traps.

Obviously, waste leaking into the home is a health issue and a water damage issue.

When you are operating each plumbing fixture, look below if possible, and make sure the trap is not leaking. Many inspectors fill the basin and pull the plug, rather than simply run water through. Their thinking is that they will ensure that the tail piece, trap and trap arm are completely flooded. Some leaks only show up when these components are flooded.

You can also watch for evidence of leakage including—

- staining or streaking on the trap
- rusting on cleanout plugs
- buckets or trays below the trap
- water damage on the floor, finishes or structural members below the trap

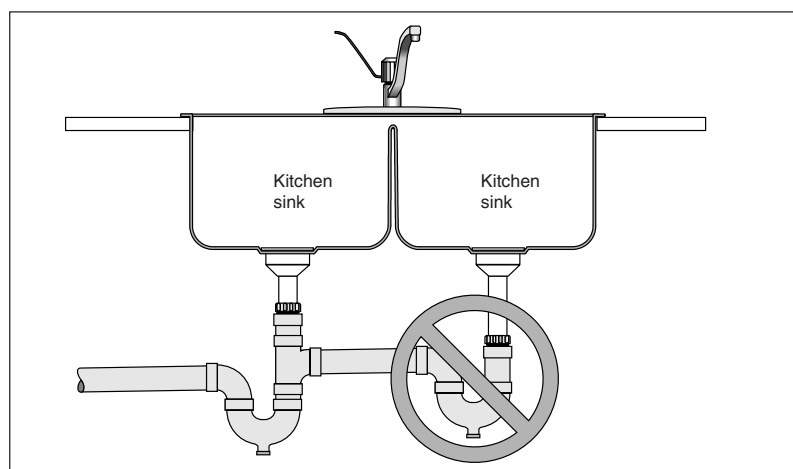
*Evidence of Leakage*

IMPLICATIONS

STRATEGY

IMPLICATIONS

STRATEGY

**FIGURE 13.20** Double Trapping Doesn't Work*A Word of Caution*

Many old tail pieces and traps are brass, often chrome- or nickel-plated. Be careful touching these components. They are sometimes so thin or so badly deteriorated, that simply putting your hand on them can cause them to leak or break.

**Double Trapping**

Double trapping sounds like a good idea but it's not (Figure 13.20).

The implication of double trapping is clogging of the trap. If we have double trapping, the velocity coming into the second trap will be reduced and solids are more likely to collect and clog the second trap.

A common spot for double trapping is under double sinks in kitchens, laundry areas or bathrooms. This can occur elsewhere in the house as well, but may be more difficult to detect.

**Split, Rusted, or Damaged**

In some cases you won't be able to run the water and find the leaks. In other cases there won't be any apparent leakage but you can see obvious weakness in the trap.

Trap leakage is the risk here.

Look at the trap for evidence of splitting, rusting or damage. A mirror and a flashlight are helpful. Looking at just the front of the trap doesn't give you the whole picture, obviously.

**Freezing**

Traps in cold areas are susceptible to freezing. In most cases, traps will be in heated areas, but traps in bathrooms above unheated garages or crawlspaces, for example, can be vulnerable.

A freezing trap will split and leakage will result.

Where plumbing fixtures are close to cold areas, think about the possibility of freezing traps. Pay particular attention to showers and bathtubs, for example, which typically have their drains and traps below the floor level. This can also be an issue on island sinks, where in some jurisdictions it's common to drop the trap below the floor level.

**IMPLICATIONS****STRATEGY****IMPLICATIONS****STRATEGY****IMPLICATIONS****STRATEGY**



### Tail Piece (Fixture Outlet Pipe) Too Long

The vertical or horizontal distance between the fixture and the trap is important. We talked earlier about maintaining adequate waste velocity through the trap.

Most codes call for this distance to be a maximum of 24 inches.

The implications are siphoning of the trap if the tail piece is vertical, or clogging of the trap if the tail piece is horizontal.

If you can get a look, check to see how far the trap is from the fixture. Depending on your jurisdiction, it should be no more than 24 or 36 inches.

#### IMPLICATIONS

#### STRATEGY

*Minimum Trap Arm Length*

*Maximum Trap Arm Length*

### Trap Arm Too Long or Too Short

The distance from the trap to the vent can't be too long or we may siphon the trap. If the trap arm is too short, waste might splash up into the vent and eventually clog it.

A trap arm should be at least two pipe diameters long. For example, if the trap is 1½ inch diameter, the trap arm should be at least 3 inches long.

Many jurisdictions use a 5 foot rule. In some areas, different rules apply based on different pipe diameters. Here is an example of one jurisdiction's rules:

| Trap Arm Diameter<br>in Inches | Maximum Trap Arm<br>Length in Feet |
|--------------------------------|------------------------------------|
| 1¼                             | 5                                  |
| 1½                             | 6                                  |
| 2                              | 8                                  |
| 3                              | 12                                 |

The object is to have the vent coming off the top of the trap arm higher than the weir of the trap. We don't want the vent pipe flooded because then it won't be able to add air to the trap arm to break the vacuum. Breaking the vacuum is important in preventing direct siphoning of the water in the trap.

Siphoning the trap or blocking the vent pipe are the implications of trap arms that are too long or too short.

While it is sometimes difficult to see much, where you can determine the length of the trap arm, check that it falls within the parameters we've outlined above (depending on your jurisdiction). You'll have to find out what rules apply in your area if you are going to call these defects out for improvement. It is more important to look for the functional implications such as siphoning traps, than to memorize rules.

#### IMPLICATIONS

#### STRATEGY

### Traps Too Small or Too Big

Most authorities specify minimum trap sizes. Most also make a general comment that the trap size shouldn't be so large that the velocity of the waste slows down to a point where solids will collect in the trap. This is hard to define. As a general rule, the trap should be the same diameter as, or one pipe size larger than, the tail piece (fixture outlet pipe) and the same size as the trap arm. Where the drains from three sinks or basins are manifolded, the trap arm is usually one size larger than the traps and the tail pieces.

The implications of both an undersized and oversized trap are clogging.

Traps should be the same size as the trap arm if we are draining a single fixture. Where the trap is a different size, watch for backup and/or siphoning at that fixture.

#### IMPLICATIONS

#### STRATEGY

**Nonstandard Shape or Material**

Homemade traps, or traps made from different materials than those that are approved, are usually not successful. Their depth of seal may be appropriate, but they won't maintain the proper velocity and self-scouring effect.

The trap is not likely to perform well and is more likely to clog or siphon.

Whenever you see a trap that does not have the characteristic P-shape, you should note it as a questionable performer. Watch also for either a cleanout plug or slip joints that allow the trap to be removed for cleaning.

**IMPLICATION****STRATEGY****Clogged**

When you are performing your drainage test, you may find that an individual fixture is clogged. Very often, the trap is the problem. While we shouldn't be troubleshooting, you may be able to determine that the trap is clogged. We do not recommend that you remove cleanouts or slip joints to inspect traps. The chances of breaking a fitting or making a mess are significant.

The fixture will not drain, at least at its normal speed.

You will be describing drainage problems where fixtures don't get rid of their water in a reasonable amount of time. You may or may not know that it's a clogged trap, but include this as a possibility. This is more likely if only one fixture is slow to drain. If all fixtures in the house are slow to drain, the problem is more likely in a main drain, for example.

**IMPLICATIONS****STRATEGY**

### 13.3.3 Floor Drains

Floor drains are needed to protect the house from water damage. Watch for them during inspections and look for these common problems:

1. Missing
2. Poor location
3. Backup
4. Grate missing, rusted or obstructed
5. No trap
6. Downspout connection upstream of trap
7. No primer or poor primer arrangement

**Missing**

Every home should be provided with a floor drain. Is it required in your area?

There is no place for water to go if there is a flood.

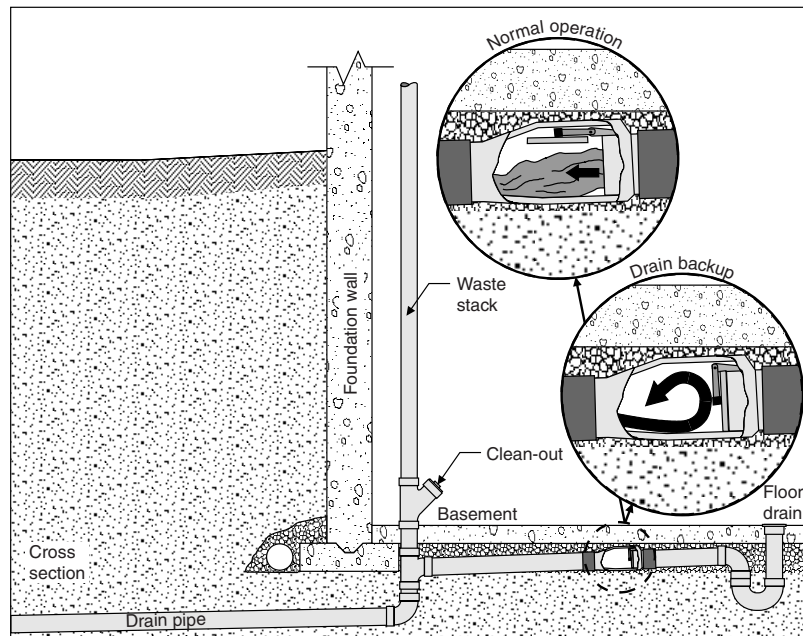
Look for a floor drain. If you can't find one, recommend that it be located or that one be provided.

**IMPLICATIONS****STRATEGY****Poor Location**

Floor drains work best if they are at the low point in the floor. Most are not, although they should be relatively close. You'll have to use some common sense to determine whether the location is acceptable or not.

A considerable amount of water may accumulate and damage the home before it finds its way to the drain if the drain is too high.

**IMPLICATIONS**

**FIGURE 13.21** Backwater Valve**STRATEGY**

Look to see that the drain is approximately at the low point of the floor. If the house is multi-level, make sure the lowest level has a floor drain.

**Backup**

Drains may back up for a number of reasons.

If the floor drain backs up, the house may flood.

**IMPLICATIONS****STRATEGY**

Look for evidence of the floor drain backing up. Watermarks around the floor close to the drain are one indication. In areas where municipal storm sewers are likely to back up, **backwater valves** are sometimes provided just downstream of the floor drain trap (Figure 13.21). This is a type of check valve designed to protect the home against sewer backup. In most cases you won't know whether one has been provided. In some cases you can see a backflow prevention device just below the floor drain grate. This may be a clue that you should be looking for evidence of prior flooding. Most people do not put these in as a preventative measure. They only get installed after there has been a problem.

**Grate Missing, Rusted, or Obstructed**

Every floor drain should have a grate with holes in it to allow water to go down.

The drain may become obstructed by materials falling into the 3-inch hole in the floor if the cover is missing. The 3-inch hole for the floor drain is also a safety hazard since someone may step on the hole and twist an ankle, for example. A rusted grate may also be a safety hazard because it may not carry the weight of a person. A rusted or obstructed grate may not let water flow down the drain.

**IMPLICATIONS****STRATEGY**

Where the grate is missing, recommend that one be added.

Ensure that there is enough drainage that a large quantity of water can escape through the floor drain. Remember, these are usually 3 inches in diameter, so that the drain can handle large amounts of flooding.

Recommend replacing rusted, damaged or missing grates and cleaning obstructed grates.

### No Trap

Every floor drain inside a building should have a trap.

Sewer odors may find their way into the house if there is no trap.

When you look down the floor drain, you should see standing water indicating that there is a trap that is filled. If you see no water, there may either be no trap, or there may be a trap that has had its water evaporate. The trap may also be cracked or broken so that it won't hold water. If you look down and see no water, you should be smelling for sewer odor and recommending further investigation.

|              |
|--------------|
| IMPLICATIONS |
| STRATEGY     |

## 13.3.4 Venting Systems

It's hard to see the venting system in a house and, while our discussion will focus on the things that we can check, you should understand that there's a lot that you won't know about the venting system.

### Function

The venting system on a house is designed to equalize the air pressure in the waste plumbing system. The venting system should allow air to escape so waste can move through the pipes. Vents should also allow air to be drawn in to prevent a vacuum from forming. We don't want vacuums to form because we don't want the water to be siphoned out of traps. Some people think of venting systems as **trap protectors**. Venting also allows sewer gases to escape outdoors rather than fight their way through the water seal and into the house.

### What You Can See

In many houses, the only part of the venting system you can see is the part that extends above the roof. You may be able to see vent piping in the attic, but you can't see the individual vents and branch vents from fixtures. You may be able to see some of the vents in the basement or crawlspace areas. Where there are plumbing fixtures in unfinished areas such as basements, you may see the vents for the fixtures there. For the most part though, the vent system is invisible.

### Testing the Venting System

As you flow water at each fixture and watch the functional drainage, you are also testing the venting system. As you watch the water drain out of the fixture, water should go down the drain with about the same noise level right to the end. If the last bit of water gurgles loudly, it's probably because air is being pulled through the water in the trap. This is siphoning the water out of the trap. This characteristic gurgling noise tells us that the venting system is not working.

### Toilets Gurgle

The toilet is the exception. It is allowed to gurgle because it is a siphoning fixture. Let's look at some of the common venting problems.

1. Missing
2. Ineffective
3. Poor vent pipe arrangements
4. Too small or too long
5. Island venting problems
6. Vent termination problems
7. Automatic air vents

|              |
|--------------|
| IMPLICATIONS |
| STRATEGY     |

**Missing**

Vents should be continuous up through the roof of the building.  
The drainage system won't work very well and traps might siphon.

When looking at the outside of the building, you should see a vent pipe projecting up through the roof. Where there is none, you can be sure the venting system is missing or incomplete.

**Ineffective**

Sometimes you can see the vents come up through the roof but when you test the plumbing system, all the fixtures siphon. In this case you should describe the venting system as ineffective.

|              |
|--------------|
| IMPLICATIONS |
| STRATEGY     |

Again, siphoning of traps will allow sewer gases into the house. Drains may be slow because they can't push the water out of the way.

Performing your flow tests on fixtures will tell you whether the venting is working effectively.

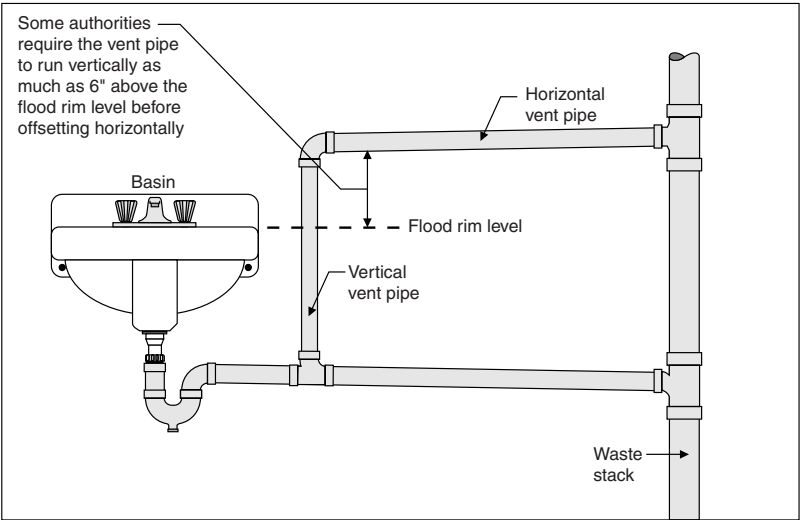
When you are on the roof, you can usually hear water flowing through the fixtures by listening at the top of the vent. You can also look for frost closure when on the roof in cold weather.

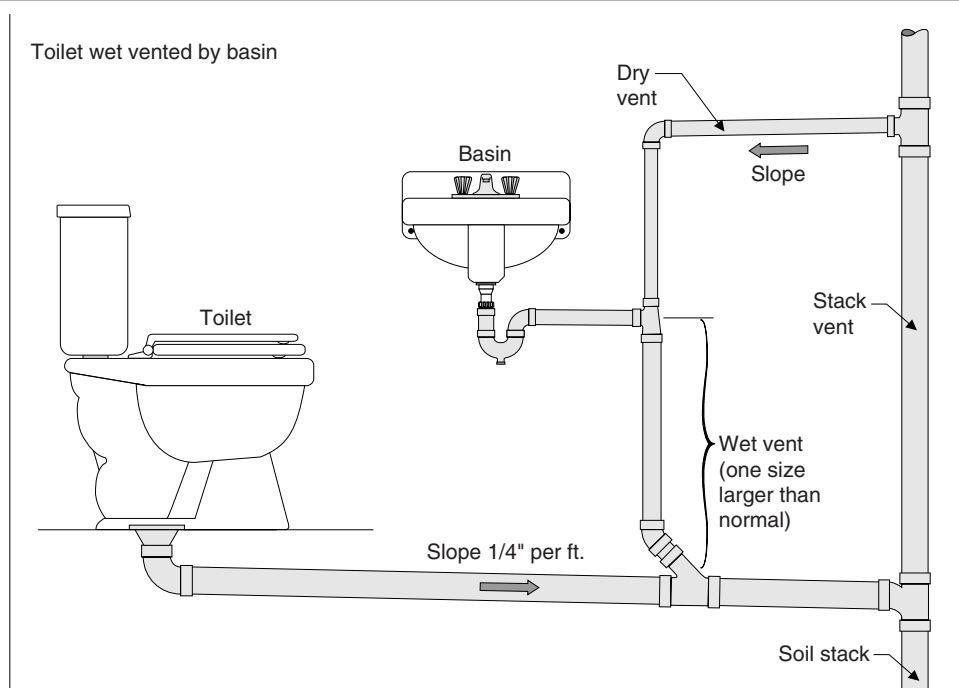
**Poor Vent Pipe Arrangements**

Many arrangements for vents are not recommended. You won't be able to see most of them during the inspection, but we'll mention a few. As vents come off trap arms, they should run vertically until they get above the flood rim level of the fixture they are venting (Figure 13.22). This isn't always practical. Sometimes the vents run horizontally. These vents are more likely to become flooded and clogged if there is a backup in the drain system.

The vent has to be on the downstream side of the trap. Occasionally it will be on the upstream side of the trap. This problem is most common on multiple sink or basin arrangements. The only place you should see something on the upstream side of the trap is when it is a cleanout or fresh air intake for a building trap. This will be on the building drain or building sewer and will not be associated with a fixture.

**FIGURE 13.22** Horizontal Vent Offset



**FIGURE 13.23** Wet Venting*Improper Wet Venting*

**Wet vents** are permitted under some circumstances. (A wet vent is a vent that also acts as a drain for another fixture; Figure 13.23). The fixtures have to be on one floor level and the drain sizes have to be increased to allow for the wet venting. (An exception in some areas allows a basement laundry tub and clothes washing machine stand-pipe to be wet vented through the drain for a first-floor kitchen sink plus dishwasher.)

It's possible that there is inappropriate wet venting or that the drain pipe serving the wet vent was not increased appropriately in size.

*Slope of Vents*

The near horizontal vent pipes should slope up slightly away from the trap they are protecting. We want the sewer gases to be able to move up through the venting system by convection. We don't want to create any low spots where condensation may collect. If condensation does develop, it should flow back down through the venting system into the drain and out to the sewer system.

The implication of these venting problems is traps losing their seals and allowing sewer gases into the home.

In most cases you won't be able to inspect the entire venting system. This makes it tough for you to remember what the important issues are on the components that you usually cannot see. If you are doing an inspection where the entire venting system is exposed, you may want to either get some outside help, or refresh yourself as to the proper venting practices in your area. Look for—

- horizontal vent pipe runs below the fixture (less than ideal)
- vent on the wrong side of trap
- wet venting over two floor levels
- non-standard materials
- poor slope of near horizontal vent pipes

**IMPLICATION****STRATEGY**

Too Small or Too Long

Vent piping can be undersized. Again, for the most part you’ll have trouble picking this up. In cold climates, the vent piping extending through the roof may have to be 3-inch diameter to avoid frost closure. In other areas, vent piping can be 1½-inch or 2-inch diameter where it passes through the roof.

The minimum vent size for any fixture is 1¼-inches. This is adequate for a 1¼- and a 1½-inch trap. Anything larger, including a toilet, requires a 1½-inch vent.

Just as drain sizes get bigger as you go further downstream, so do vent pipes. The further away from the fixture we get, the larger the vent pipe should be. Vent pipes should never reduce in size.

Maximum Length of Vents

Some jurisdictions limit the length of vent pipes. For example—

| Vent Diameter | Maximum Vent Length |
|---------------|---------------------|
| 1¼ inch       | 45 feet             |
| 1½ inch       | 60 feet             |
| 2 inch        | 120 feet            |
| 2½ inch       | 180 feet            |
| 3 inch        | 212 feet            |

Again, you are going to have a tough time determining vent sizing and length throughout the house.

|             |
|-------------|
| IMPLICATION |
| STRATEGY    |

The implication of undersized or overly long venting is ineffective venting.

Where you can see vents at fixtures, check their size. Where you can see the vent coming through the roof, make sure it is appropriate.

Island Venting Problems

Kitchen sinks in islands present a special venting problem. Because most homeowners don’t want vents running up through the middle of their kitchen, we have to find another way to handle the venting. In some jurisdictions, the trap from the kitchen sink can be below the floor, and a vent can come off the trap arm under the floor and run horizontally to a wall. While this is a less than ideal arrangement, it is not a bad compromise for an otherwise difficult situation. (Why is this less than ideal? Because if the drains back up, the vent will be flooded and may clog.) The alternative (and required) solution in many areas is as illustrated in Figure 13.24.

The Easy Way

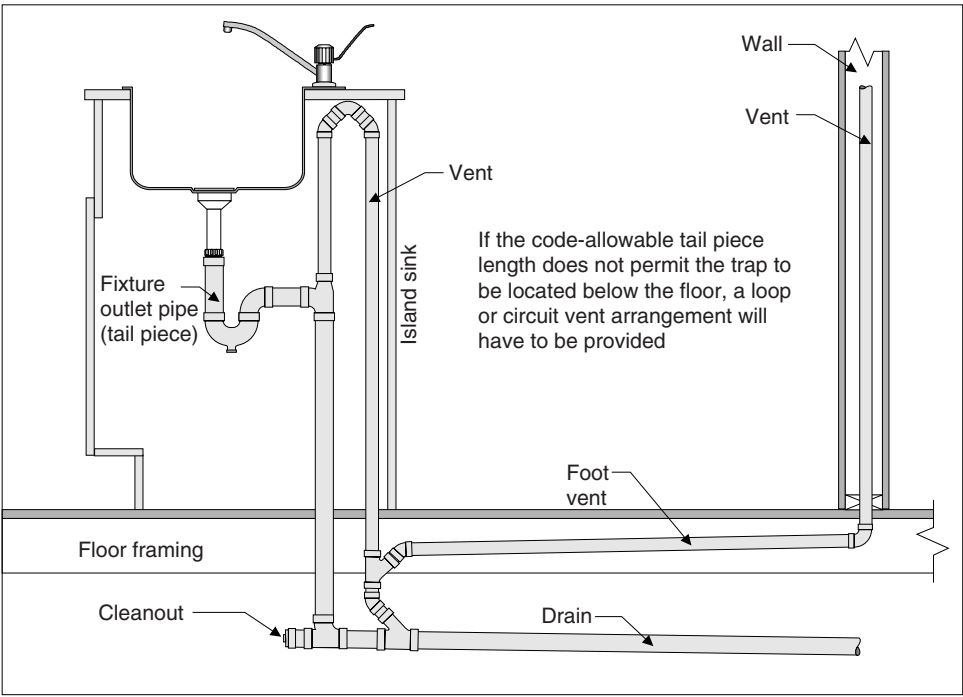
The trap comes off the sink in a conventional fashion. The trap arm is connected to a vertical drain that goes down through the floor and also a vent that goes up as high as possible under the counter before turning 180 degrees and coming down to join the horizontal drain below. A nearly horizontal vent comes off this vertical fixture vent. It’s called a foot vent and it’s 1½ inches in diameter. It slopes up slightly until it gets to a wall where it can extend up through the roof or connect to other vents.

The Hard Way

The purpose of the vertical fixture vent making a second connection to the horizontal drain is in anticipation of waste getting up into the loop vent in the event of a backup. If this happens, we want to make sure that the material can drain out and not block the vent.

The drain lines are typically 2-inch and the vent lines are typically 1½ -inch diameter in this arrangement. In some jurisdictions, island venting may be accom-

**FIGURE 13.24** Venting an Island Sink (If Below Floor Trap Is Not Permitted)



plished by using an air admittance valve. This is a one-way air valve that is installed in place of conventional venting. The AAV should be at least four inches above the drain.

**Vent Termination Problems**

Vents must be tall enough to be appropriately flashed where they extend through the roof.

We don't want vents too tall in cold climates because they will be more prone to frost closure. Generally speaking, the maximum height of vent pipes above roofs is 12 inches. Vents that are slightly taller than this are not noteworthy, but where the vents are more than 2 feet tall, for example, and you are in a cold climate, you may worry about frost closure.

Sometimes vents have to be extended to clear adjacent structures. In this case, there is a risk of frost closure. Vents are sometimes insulated to reduce this risk, although this is rare on houses. If the vent is more than 6 feet tall, consideration should be given to bracing the vent.

Vents have to be at least 2 feet above any operable door or window and 10 or 12 feet away from any door or window at the same elevation or higher. Different areas have different requirements.

Vent pipes must be at least 10 feet above ground level in most jurisdictions. Again, you may want to check what is appropriate in your area. We mentioned earlier that plastic vents exposed to sunlight have to be painted in some areas to protect against ultraviolet rays attacking the plastic.

Vents that are too short may result in flashing leaks at the roof. Vents that are too tall may result in frost closure and possible instability of the vent. Vents that are

*Flashing*

*Not Too Tall*

*Tall Vents*

*Vent Clearances  
from Windows*

*Vent Height above Grade*

**IMPLICATION**



**STRATEGY**

too close to windows or other building openings may allow sewer gases back into the home. Similarly, vents too close to property lines may allow sewer gases into neighboring homes.

You will need to find out what the standard practices are in your area. Checking the vent terminations is fairly easy. Watch for—

- flashing problems
- too small
- too short (less than 6 inches)
- too tall
- not supported if more than 6 feet tall
- frost closure
- less than 2 feet above doors or windows
- less than 10 feet from door or windows beside or above the vent
- less than 10 feet above grade

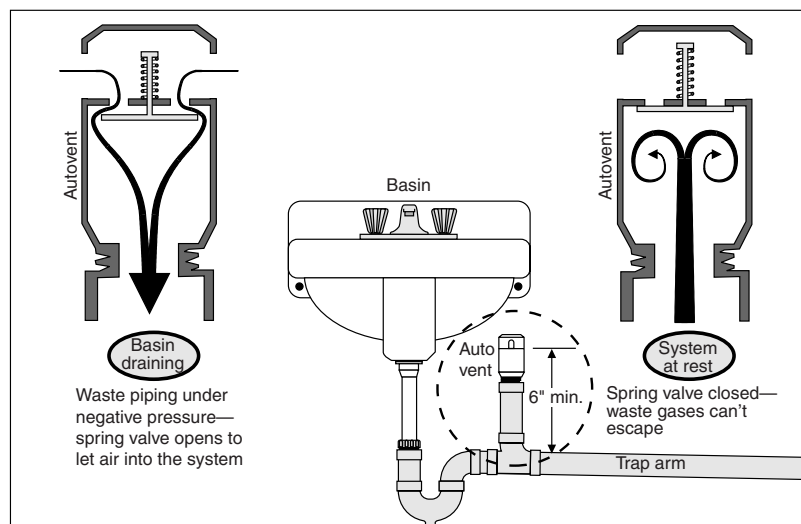
**Automatic Air Vents**

Automatic air vents are similar to air admittance valves; however, automatic air vents are illegal in most areas and you should tell your client that these devices are usually not accepted by authorities (Figure 13.25). However, they may work to some extent. These vents allow air to be drawn in when the waste plumbing system in that area is under negative pressure. This occurs when a vacuum is being formed and the system is about to siphon. These mechanical vents, when working properly, can allow house air to be drawn in and break the vacuum, preventing the siphon problem. The vents do not allow air to be pushed out through the vent, so they don't perform all of the functions of a venting system. Automatic air vents are not recommended for toilets.

**IMPLICATION**

The automatic air vents perform only some of the functions of a plumbing venting system. Since they are mechanical devices operating in a hostile environment,

**FIGURE 13.25** Automatic Air Vent



**STRATEGY***If It Works*

they are prone to failure. Many authorities won't accept their installation and require replacement with a conventional venting arrangement.

When we see an automatic air vent, we advise our client that a full venting system is probably not in place. We perform our plumbing tests and get a sense of whether the device is operating.

*If It Doesn't Work*

If no performance problems are noted, we'll advise the client that this is not an appropriate installation but it may work, at least over the short term. We will let the client know that replacement may be needed.

If there are performance issues, we will recommend replacement with an appropriate venting system right away.

### 13.3.5 Sewage Ejector Pumps

Sewage ejector pumps are a big deal. They aren't all that common, but if you miss one, neither you nor your client will be happy.

*Sewage Pumps*

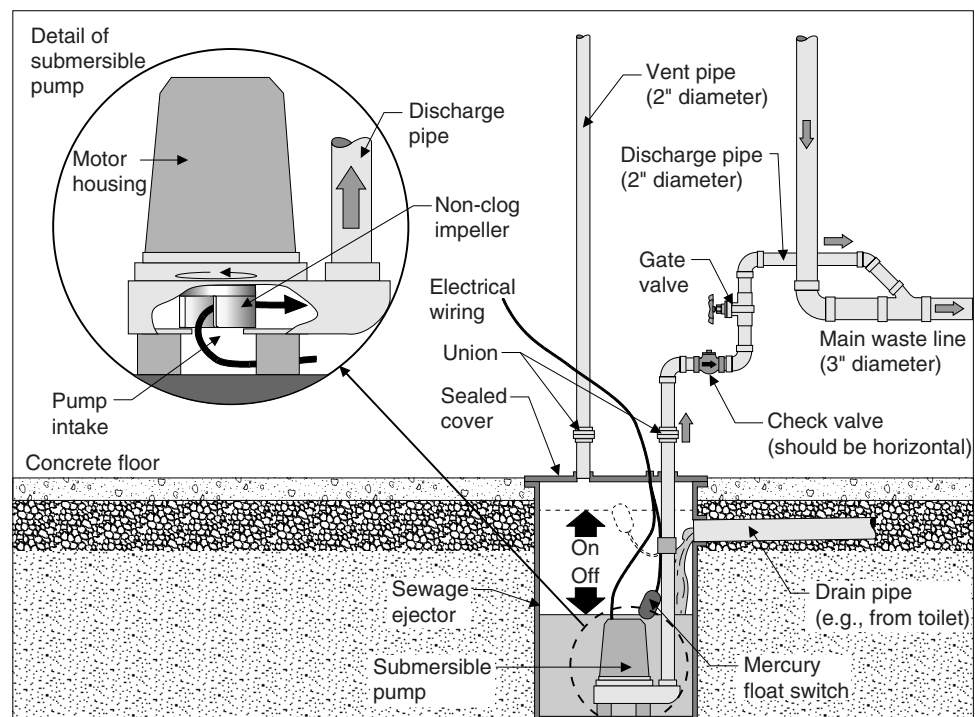
**Sewage ejector pumps**, also called **sewage pumps**, **sanitary sewer pumps**, or **solid waste pumps**, are used when there are toilets below the public or private sewer line (Figure 13.26). If waste can't flow to the sewer line by gravity, the waste has to be lifted to be discharged into the sewer.

These pumps are able to pump solids and liquids from the sump or tank up to the sewer. They consist of —

*Buried Tank*

- a buried tank or sump made of corrosion resistant material (modern ones are often fiberglass or polyethylene) with a sealed, watertight lid. The top of the tank is often flush with the floor of the lowest living space level. A diameter of 18 inches and a 30-inch depth are typical.

**FIGURE 13.26** Sewage Ejector Pump



*Electric Pump and Controls**Pump Arrangements**Electric Supply**Union, Check Valve, Gate Valve, in Discharge Line**Vent Pipe**Check Valves on Sewage Pump Lines*

- a side inlet to receive discharge from toilets and other fixtures. This is typically 3 to 4 inches in diameter.
- an electric pump and motor inside, operated by a float or diaphragm type level sensor to activate the pump. When the liquid level in the tank rises, the pump is activated. Pump motors are typically  $\frac{1}{2}$  horsepower.
- in some cases, a pedestal pump with the motor located above the tank or sump. This makes it easier to see. Most modern sewage pumps have a submersible pump with the pump and motor completely inside the tank.

These systems can be hard wired or plugged in. Some systems have separate electrics for the automatic controls. These are 120-volt systems.

A discharge pipe, typically 2 inches in diameter, comes off the top of the tank. Some jurisdictions require that the first connection be a union. This makes disassembly and servicing easier. Many installations require a check valve and then a gate valve as we move away from the tank in the discharge line. The check valve is used when the sewage ejector pump is connected to a public sewer system and protects the tank and fixtures that feed the tank from backup of the municipal sewer system. This is necessary since the pump and these fixtures are below the sewer line.

The check valve may have to be in a horizontal section of pipe to work properly. The gate valve is to isolate the tank and check valve from the discharge line for servicing.

A vent pipe comes off the top of the tank and vents like any other plumbing fixture. It is typically  $1\frac{1}{2}$  to 3 inches in diameter, with 2-inch diameter piping being the most common.

Seasonal properties such as cottages with private sewage disposal systems often have no check valve on the discharge line because any liquids in the line beyond the check valve would freeze during cold weather. With no check valve, the discharge line can run from the pump to a septic tank, for example, without our worrying about burying the line below the frost level. There is less risk of catastrophic sewer backup with a private disposal system than with a municipal sewer connection, so the check valve isn't as critical here.

Let's look at what can go wrong with these sewage pumps.

1. Inoperative
2. Short cycling or running continuously
3. Odor
4. No vent
5. Discharge pipe problems
6. Missing, rusting or inoperative union, gate valve or check valve
7. Alarm sounding
8. Electrical problems

### **Inoperative**

As you do your inspection, you will be testing the plumbing fixtures. Where you find a sewage pump, you should be flowing enough water to make the pump operate.

### **IMPLICATIONS**

There are obvious health implications to an inoperative pump. Waste will back up into the house.

**STRATEGY**

Flow enough water to ensure that the pump operates. You may have to flow several gallons. Flow all the fixtures that feed into the sump. It's unusual to have an inoperative pump if there are people living in the house. Most people will not live with plumbing fixtures they cannot use.

**IMPLICATIONS****Short Cycling or Running Continuously**

The life expectancy of the pump and motor is going to be shorter if the pump cycles frequently or if it never shuts off.

**STRATEGY**

Before you run any water, note whether the pump is running. If so, does it shut off or is it on constantly?

When you run water, watch for the pump cycling on and off several times a minute. If either of these conditions are noted, recommend servicing.

**Odor**

Sometimes you'll notice an odor in the area of the sewage pump. Sometimes it's the odor that makes you notice that there is a sewage pump.

**IMPLICATIONS**

Sewer odors are unhealthy and should not be allowed to accumulate in a house.

**STRATEGY**

Where odors are noted, recommend servicing.

**No Vent**

Every sewage pump should have a vent coming off the top of the sump and venting to the building exterior.

**IMPLICATIONS**

If no vent is provided, the pump may not work properly. It may create a vacuum in the sump and may siphon the plumbing fixtures that are connected to it. This will lead to odors getting back into the house. If the vent connection is missing at the top of the sump and the hole is open, sewer gases can enter the house directly.

**STRATEGY**

Ensure that there are two pipes coming out of the tank lid. One will be the discharge, and the other, the vent. The vent system for sewage pumps should be treated as any other venting system.

**Discharge Pipe Problems**

Discharge pipes may be leaking, crimped or poorly connected to the sewer piping or the sump.

**IMPLICATION**

Discharge pipe problems may result in sewage being discharged directly into the home. Obviously this is unsafe.

**STRATEGY**

When the pump is operating, check the performance of the discharge pipe. Look for leaks or vibration of the piping, which may suggest a blockage. Check that the pipe is well supported and does not sway excessively when in use.

**Missing, Rusting, or Inoperative Union, Check Valve, or Gate Valve**

All discharge pipes should have a check valve or backwater valve and a gate valve. Many jurisdictions also call for a union before the check valve, to facilitate servicing.

**IMPLICATIONS**

The implications of problems with these devices are harder work and shorter life for the pump and more difficult servicing. A defective check valve may allow a sewage backup to enter the house.

**STRATEGY**

Check for a union, check valve and gate valve on the discharge pipe close to the sump. You should determine whether a union is required in your area. If any of the required components are missing, recommend improvements. Check valves usually have to be installed in horizontal piping. Recommend further investigation if the check is on vertical piping (it may be all right).

**IMPLICATION****STRATEGY****Alarm Sounding**

Sewage will soon back up through the fixtures into the home.

Recommend servicing immediately. Don't operate plumbing fixtures feeding the sump. Document this limitation to your inspection.

**Electrical Problems**

A number of electrical problems may be found with the supply wiring and controls. Chapter 7 covers these.

This can be a life safety issue.

**IMPLICATION****STRATEGY**

Check that the electrical arrangements to the pump follow good electrical practice. The electrical system should be grounded and well-secured and controls and junction boxes should be intact and well supported. These pumps are typically 120 volts. They can be either plugged in or hard wired.

We strongly recommend that you use a voltage detector or similar device to ensure that pumps and their controls are safe before touching them.

**Inspector in the House: A Work in Progress**

The new main floor washroom was very attractive, and a bit of a surprise—there was no other plumbing in this part of the house, and I had seen no vent piping in the attic or above the roof in this part of the house either. But I didn't think too much of it at first. I remembered the locked basement room in the finished basement below and assumed that's why I did not see the plumbing below.

I ran the basin and flushed the toilet. Everything seemed fine, although the waste sounded a little funny. I am a naturally curious type, and wondered why. I asked about access to the locked basement room. The sellers were not home and the agent said she had never seen that room either. She was clever enough to find a key on the top of the door casing and in we went.

We were amazed at what we saw—the hot and cold supply piping were garden hoses, and the waste piping was drained into a 45 gallon drum! That explained the strange noise. We later learned that the owner's son was doing the work and had been out of town for some time. Rather than admit the work was not finished, the sellers created a temporary arrangement. The agent was assured that the work would have been properly completed before the new owners moved in. Our discovery ensured that that happened.

## 13.4 FIXTURES AND FAUCETS

So far we have looked at the supply piping, water heaters, and the drain, waste and vent system. Now we are going to look at the fixtures where the water is actually used.

### 13.4.1 Basins, Sinks, and Laundry Tubs

Each of us uses these fixtures every day, so in a sense, inspection is very easy. You're familiar with how to operate these fixtures. There are certainly some things to watch for on inspections.

Let's look at some common problems.

- leaks
- rust
- surface defects
- loose
- overflow missing or inappropriate
- cross connections
- not level
- slow drains

### Leaks

Leaks at fixtures are probably the most common problem. The leaks may be at the—

- faucets (especially vegetable sprayers and extendable faucets)
- bowl (especially if the bowl is concrete, synthetic marble or china)
- drain connection or pipe
- trap
- pop-up drain stoppers (where they go through the tailpiece)
- overflow

#### IMPLICATION

The implication of leaks is primarily water damage.

#### STRATEGY

As you flow the fixture, look above, around and below for evidence of leakage. Make sure you pull out and operate extendable faucets and vegetable sprayers. Where there are cracks in basins, they may only be surface cracks. It's best to fill the bowl and watch for leaks.

With older enameled steel sinks, it was common to weld a steel overflow onto the front of the sink. It's very common for these to rust out and cause the bowl to leak near the overflow connection, even when no water goes down the overflow (Figure 13.27).

Many of these overflow connections are now siliconed in place, rather than welded.

### Rust

Rust is an issue on metal fixtures and on metal faucets, drains and stoppers.

The implications are mostly leakage related. If the rusting in the bowl is bad, the water may not remain clean as rust flakes come off in the water when the sink is in use.

Watch for leaks at the faucets, bowls and drain connections. On enameled steel basins, check for rust around the overflow, on both the inside and outside of the bowl.

### Surface Defects

Chips, cracks and stains are common in basins and other similar fixtures.

In many cases these are simply cosmetic defects. Where enamel is chipped away from stainless steel, rusting is the longer term implication.

Where cracks appear in the surface of bowls, leakage through the crack may be the long term implication. Mold growth in the cracks is a common problem.

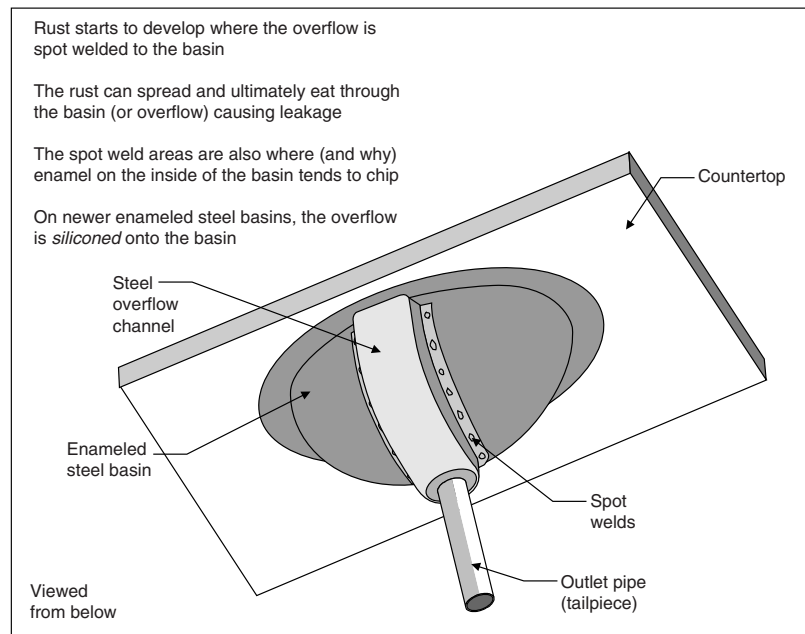
Look for surface defects before operating the fixture.

#### IMPLICATIONS

#### STRATEGY

#### IMPLICATIONS

#### STRATEGY

**FIGURE 13.27** Rusting Overflow

### Loose

Basins, sinks and laundry tubs may be supported by countertops, walls, or floors (by pedestals or legs). Wall-hung and pedestal type basins are the most likely to be loose.

#### IMPLICATIONS

Loose fixtures are likely to strain supply and drain pipes. Leakage is a likely implication. Wall-hung fixtures may fall off the wall and injure someone. Pedestal sinks can also have the tops fall off and injure people. Some pedestal sinks are one piece, although many are two piece with a separate bowl and pedestal. China and cast iron fixtures are extremely heavy and can cause serious injury as well as flooding if they fall.

#### STRATEGY

With both pedestal and wall-hung fixtures, it's best to test for looseness by lifting on the fixture, rather than pressing down on it. Sinks are not designed to be stepping stools. Putting all your weight on the edge of a pedestal sink or a wall-hung sink, for example, may result in sudden collapse.

#### *Slope on Wall Hung Fixtures*

Use some common sense in determining whether the fixture is loose. One of the clues with wall-hung fixtures is that the basin may slope down away from the wall if it's loose. Lifting up gently on the front of the basin will very often allow it to rotate back up into a level position with the back of the basin tight against the wall.

#### *Legs*

Many laundry tubs and some basins are supported on legs. The legs may be concrete, metal or plastic. Watch for rusted metal legs and spalling or cracked concrete legs. Again, if you can see deterioration to the legs, there is no need to do any rough physical testing. Your chances of breaking things are significant.

### Overflows Missing or Inappropriate

#### *No Overflows on Sinks*

Kitchen sinks should not have overflows. Any fixtures that are used for food preparation or dishwashing do not have overflows for fear of trapping food particles and creating an unhealthy situation.

*No Overflows on Laundry Tubs*

Many laundry tubs also do not have overflows. Double kitchen sinks and double laundry tubs very often have a compromise on overflows. The partition between the two sinks will often be  $\frac{1}{4}$  inch lower than the flood rim of the sink. If one of the sinks or tubs overflows, the water will run into the other sink before it spills out onto the floor. This is not completely reliable because the other sink may be in use or its drain stopper may be in place. However, this is certainly better than not having this arrangement.

Basins have overflows, although most home inspectors don't test them.

We've talked about the implications already. We don't want bits of food being stuck in an overflow in a kitchen. We want to avoid flooding in lavatories where people may be called away to the phone while washing.

Look for overflows where they should be and where they shouldn't. Where you find an inappropriate installation, recommend rearrangement.

Feel free to test overflows by filling the basin up to the height of the overflow, but be prepared for leakage.

**IMPLICATIONS****STRATEGY****Cross Connections**

Cross connections are very common on laundry tubs, and are always a possibility with vegetable sprayers or faucets that can be left lying in a sink full of dirty water.

The implications of cross connections are obviously health issues. The plumbing system could make people sick.

Look for cross connections and advise people of the risks inherent in vegetable sprayers and extendable faucets. Most extendable faucets now have integral protection against cross connections, although you typically won't know during an inspection.

**IMPLICATIONS****STRATEGY****Not Level**

The laundry tub is the most likely fixture to be out of level. Many laundry tubs have a large, flat base to the tub. If the tub is even slightly out of level, water may pool in the bottom of the tub. This is particularly true of plastic tubs.

The water pooling in the bottom of the tub is a nuisance and can be a health issue. Stagnant water is not a good thing. A small amount of water will typically evaporate. Rarely is the amount of water enough to raise a serious concern.

As you test each fixture, you should make sure that when the drain stopper is pulled, the fixture drains completely. Note any areas where water pools in the fixture.

**IMPLICATIONS****STRATEGY****Slow Drains**

Slow drains are a common problem on plumbing fixtures. We consider slow drains as:

- basins that take more than 30 seconds to empty
- sinks or laundry tubs that take more than 60 seconds to empty

If the drain is slightly slow, most people adjust their lifestyle very slightly and live with it. There are no serious implications. If the drain is so slow that water sits in the drain for extended periods, the fixture may not be usable, and there are health concerns.

As you perform your operating tests, you'll discover slow drains. Sometimes you can see a strainer or pop-up drain problem and recommend appropriate action. In other cases, you'll advise your client that if they can't live with the slow drain, further investigation is needed to find and correct the problem.

**IMPLICATIONS****STRATEGY**



## 13.4.2 Faucets

Faucets are found on sinks, basins, laundry tubs, bathtubs, showers, bidets and hose bibbs. They are used to deliver hot and/or cold water to fixtures.

The older traditional faucet is known as a **compression faucet**. A threaded stem is turned to open or close the valve. At the end of the stem is a disk with a washer on it. When the valve is closed, the washer is pressed against a seat to prevent water flow (Figure 13.28).

More modern faucets are typically **cartridge** or **ball type**. These faucets align holes in two different materials to allow water to flow and misalign the holes to stop the water flow.

*Water Hammer*

These modern **washerless faucets**, as they are often called, can be fully opened and closed quickly. This is convenient for the user, but may make a plumbing system prone to water hammer. As part of your test of each plumbing fixture, you will want to close the valves as quickly as an enthusiastic homeowner might. You should then listen for the characteristic banging sound of water hammer.

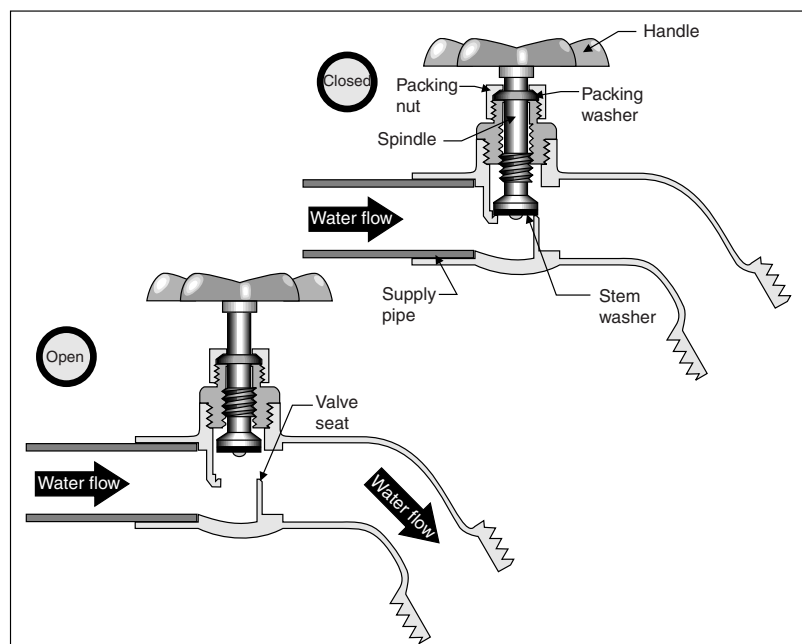
*Single Lever*

Another feature of the modern washerless faucet is that many systems now use a single lever or knob to control both the hot and cold water and the volume of water. These faucets are mixing valves that allow you to adjust the ratios of hot and cold water.

Let's look at some of the common faucet problems.

1. Leaks or drips
2. Stiff or inoperative
3. Loose
4. Cross connections
5. Noisy

**FIGURE 13.28** Compression Faucet



6. Damaged, rusted, loose or missing handles
7. Hot and cold reversed
8. Obstructed aerator
9. Shower diverter inoperative/defective

### Leaks or Drips

The implications of dripping faucets are usually not severe, since the water ends up in the fixture. Over time, you might get some discoloration in the fixture bowl. Leaks through the body of the faucet are usually more dangerous because of the water damage they can do. Damage to countertops caused by leaking faucets is a very common problem at kitchen sinks. If the faucet set or the edge of the sink is not tightly sealed to the countertop, water can get under either the lip of the sink or the faucet base and damage the countertop over time.

Some leaks are obvious. Others have to be searched out. As you operate the faucets, you should be looking around and below the faucets. This means you have to get down on your hands and knees and look under sinks. Some bathtubs have access panels to the piping on the wall opposite the tub. Look for these panels and check for leakage here.

You should operate vegetable sprayers and, if there are extendable faucets, you should pull them out and operate them. It's very common for leaks to develop at the hoses. Some extendable faucets are very subtle. Unless you are thinking about the possibility of an extendable faucet, you may just look at it and operate it as a fixed faucet. It's a good habit to get into the practice of pulling on the spout to see whether it is extendable. If you can get under the fixture, you'll usually see the hose below. This should remind you that you're dealing with an extendable faucet.

### Stiff or Inoperative

Faucets should not require superhuman strength to turn the water on and off.

It will be difficult to turn the water on and off.

You will notice stiff or inoperative faucets as you perform your normal test.

### Loose

Faucets should be fixed to the wall, to the countertop or to the fixture itself. Many people prefer the faucets fixed to the fixture itself when dealing with kitchen sinks or basins, for example. Many sinks and basins are designed so that if there is leakage around the faucet, the water will run into the sink or basin, rather than onto the countertop or floor.

Loose faucets are prone to leakage as a result of strain on fittings. Leaks at the point where the faucets connect to the supply piping system are a concern.

As you operate faucets, it's very easy to detect loose faucet systems. Understand that a loose faucet is different from a loose handle. Sometimes the handles are not well secured to the faucets. This is a different issue.

### Cross Connections

This is a health problem.

Watch for faucet spouts that are below the flood level of fixtures. Advise clients that vegetable sprayers and extendable faucets can form cross connections and should not be left in sinks or other fixtures. These faucets may be protected from backflow, but you usually won't know by looking.

#### IMPLICATIONS

#### STRATEGY

*Vegetable Sprayers*

#### IMPLICATION

#### STRATEGY

#### IMPLICATION

#### STRATEGY

#### IMPLICATION

#### STRATEGY

**IMPLICATIONS****STRATEGY****Noisy**

Faucets can be noisy in a number of ways. They can bang, chatter or whistle, for example.

The implications of chatter and whistle are valve failure. The implications of water hammer can be flooding if the piping ruptures as a result of the tremendous forces.

As you operate the fixtures, you will notice hammer, chatter and whistle. There's no need to go too far into troubleshooting. You should simply report the problem and recommend correction as necessary.

**IMPLICATIONS****STRATEGY****Damaged, Rusted, Loose, or Missing Handles**

There are many problems that you may find with faucet handles.

You may not be able to operate the valves or it may be dangerous to operate the valves. It's very easy to cut your hand on a damaged faucet handle.

This is another problem that's easy to find during your testing of plumbing fixtures.

**IMPLICATIONS****STRATEGY****Hot and Cold Reversed**

The hot water should be on the left side and the cold on the right. This is a convention throughout North America, and it applies to single lever as well as dual faucet arrangements.

People can be surprised at best or scalded at worst if they're expecting to get cold water and they get hot.

This is another easy check during your test of plumbing fixtures.

**IMPLICATIONS****STRATEGY****Obstructed Aerator**

It's very common for aerators to become obstructed and the water flow and pressure to be poor at a given fixture.

Poor water flow and pressure at that fixture are the implications.

Watch for plumbing fixtures that have considerably lower pressure than other fixtures in the house.

**IMPLICATIONS****STRATEGY****Shower Diverter Inoperative or Defective**

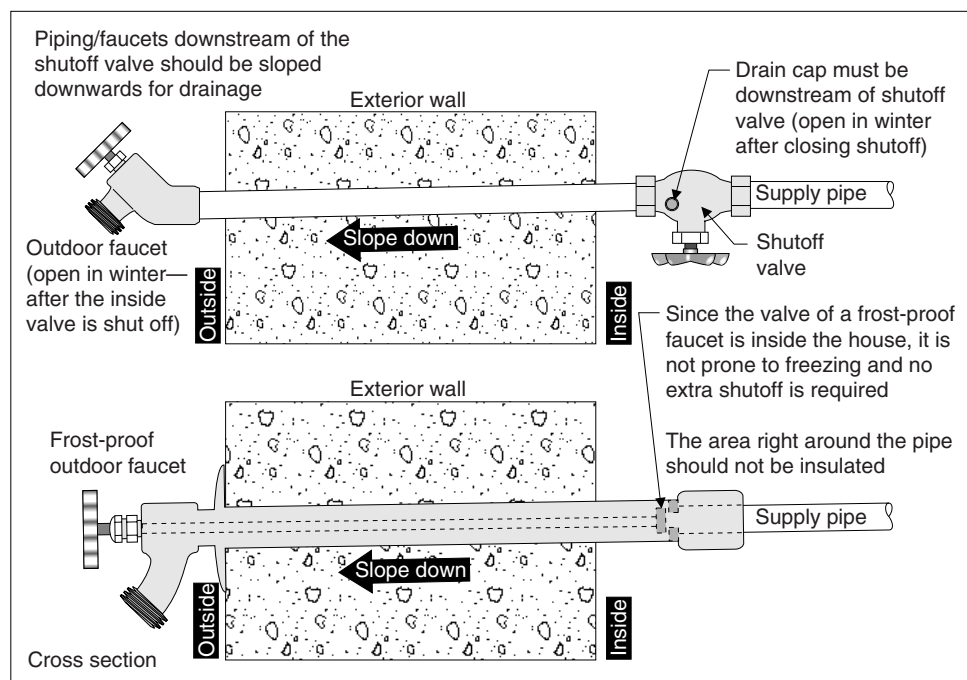
We've already talked about having to test the shower. Sometimes you can't get the shower to come on.

Obviously, people can't take showers.

As you test the shower, make sure that the diverter will stay in place. Also ensure that the flow out of the tub spout stops when the shower is operating. In some cases, the diverter doesn't close tightly, and some water will continue to come out of the tub spout while the shower is operating. This is inefficient and makes for low flow and pressure at the showerhead.

## 13.4.3 Hose Bibbs

Most houses are provided with outdoor hose bibbs (faucets) to connect hoses for watering lawns, washing cars, etc. In some areas these are called **hydrants**. The piping supply to the outside hose bibb is usually metal where it passes through the wall, even if the house has plastic supply piping.

**FIGURE 13.29** Shut Off Valves for Outside Faucets**Function**

The piping to the hose bibb is connected directly to the cold water supply piping in most cases.

As lengths of hose are added to the faucet, the water supply may be expected to deteriorate because of frictional pressure loss through the hose, so if there is a hose attached, don't criticize the flow based on the amount of water coming out of it.

**Conventional Hose Bibbs**

Conventional hose bibbs have a compression-type, hand-operated valve outdoors. In freezing climates, there is an indoor shutoff valve as well (Figure 13.29). This is usually a stop and waste valve. This means that the valve shuts off the water to the outdoor faucet and has a drain fitting that allows water to be drained from the pipe between the shutoff and the outdoor faucet. If water is left in the pipe, the pipe may split as the water freezes, even though no more supply water can flow into the pipe. The damage is often not noticed until the following spring when the stop and waste valve is reopened.

**Winterizing of Hose Bibbs**

Homeowners should be advised that when winterizing a conventional hose bibb, the indoor stop and waste valve should be closed, then the waste should be opened. We also recommend leaving the outdoor faucet open so that any water can drain out of that end. The slope of the pipe going through the wall may prevent the water draining out of the waste valve. An open waste valve inside and an open faucet outdoors assures that there will be no water left in the pipe.

**Frost-Proof Faucets**

Some outdoor hose bibbs are **frost-proof**. This is just a long stem valve that goes through the wall to the building interior. When the valve is closed, the water flow is shut off inside the house. These systems need to slope down toward outdoors, so that the water in the valve body drains out through the faucet when the valve is shut off. If the pipe slopes down toward the indoor side, water may remain in the valve body as it passes through the wall and freeze during cold weather. This usually ruins the valve, although no leakage may be noted until the valve is opened again.

*Don't Operate in Winter*

Many inspectors do not operate frost-proof hose bibbs in the winter for fear of creating an ice patch outdoors. They are also sensitive to the possibility that the valve body may be split inside the wall and by operating the valve, they may create an accumulation of water in the valve body where it passes through the wall and cause a freeze-up problem.

Common hose bibb problems include:

1. Leaks or drips
2. Stiff or inoperative
3. No backflow prevention
4. Loose
5. Pipe frozen or split
6. Poor slope on a frost-free hose bibb
7. Mechanical damage

Several of these conditions have been discussed already in the context of faucets; we'll just touch on the unique ones.

### No Backflow Prevention

In most areas hose bibbs need to have some kind of backflow prevention device. It can be integral or added to the hose bibb.

The implication is a cross connection. When a hose is attached to the bibb, the hose may be left in a bucket of dirty water, connected to a pesticide or fertilizer sprayer, or in a roof gutter filled with dirty water, for example. A problem would only arise if the water pressure in the house dropped (because someone shut off the water to work on a plumbing fixture, for example) and the hose bibb was left open. This is not an impossible combination by any stretch of the imagination.

Look for a backflow prevention device on the end of the hose bibb. If you can't see one, there may be an integral backflow prevention device built into the hose bibb (especially if it's frost-free). You should probably report the absence of a hose bibb backflow prevention device as a question, rather than a conclusion, and recommend that it be checked the first time a plumber is at the house.

#### IMPLICATIONS

#### STRATEGY

### Pipe Frozen or Split

Pipe going through the wall can be frozen and split.

The implications are water damage and the inability to use the hose bibb.

Since most of this piping isn't visible, you may have trouble determining this condition. Some of the clues to watch for include—

- a closed stop and waste valve during warm weather (you can either tell by looking or opening the hose bibb valve)
- an open valve at a conventional hose bibb during warm weather (this means a closed stop and waste inside)
- leakage to the building interior or exterior when operating a frost-proof hose bibb

Be careful operating frost-proof hose bibbs. If there is leakage, it often shows up only on the inside of the building as you flow the faucet. You risk doing water damage unless someone is watching the inside of the house as you operate these faucets.

#### IMPLICATIONS

#### STRATEGY

You may also be able to see a split pipe, more often on the inside of the wall than the outside. Conventional hose bibbs that are improperly winterized are immediate suspects for frozen or split piping. With frost-proof hose bibbs, an operational test may be possible, but remember the downsides of the test as we discussed earlier.

### Poor Slope on Frost-Free Hose Bibbs

We mentioned that the valve body of a frost-free hose bibb has to slope down to the outside to drain water out of the valve body when the valve is shut off.

A frozen and split valve is the implication. This will result in leakage, of course.

Sometimes you can see enough of the valve to get a sense of its slope. Sometimes when the valve is shut off, you can put your finger up inside the outlet and feel whether there is water sitting in the valve body. In many cases, it will be hard to know whether or not the slope is correct.

### Mechanical Damage

Hose bibbs, like any other faucets, are subject to mechanical damage. Because they are located outside and usually near grade level, their risk of mechanical damage is greater than the average faucet.

The implications are an inoperative valve or a leak.

Watch for mechanical damage when operating outdoor hose bibbs.

#### IMPLICATIONS

#### STRATEGY

#### IMPLICATIONS

#### STRATEGY

## 13.4.4 Toilets

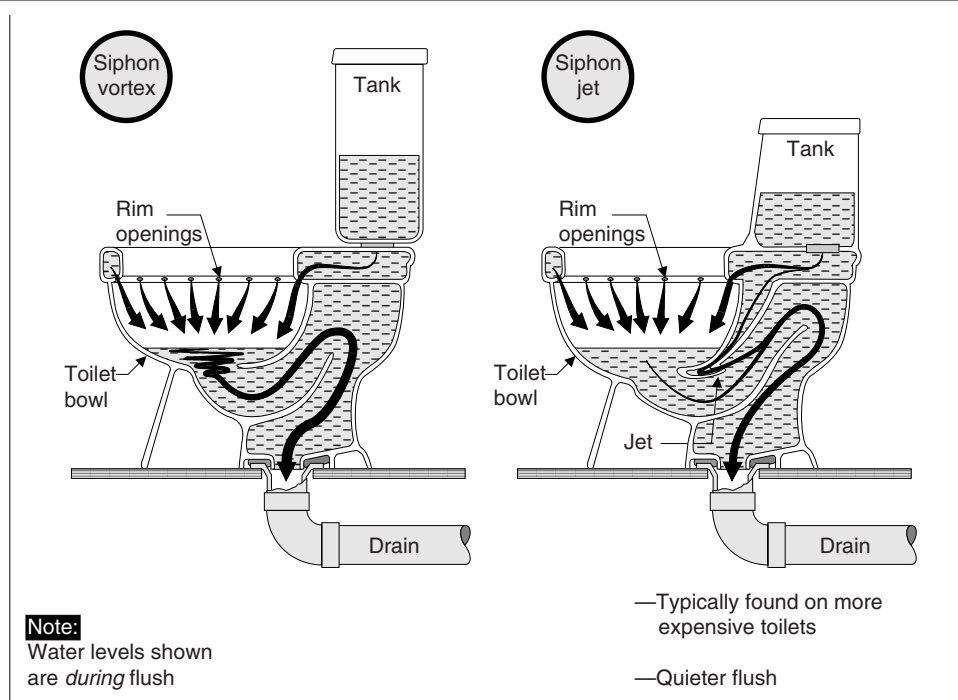
### Toilet Types

#### One- and Two-Piece Toilets

Toilets are classified by their flush mechanism. Figure 13.30 describes two types.

Two-piece toilets are very common. These have a bowl and a tank that may be bolted together on site. There are also one-piece toilets where the bowl and tank are integral.

**FIGURE 13.30** Toilet Flushing Actions



*What about Noise?*

Toilets are inherently noisy since they are siphoning fixtures. This means they draw some room air with them as they operate. The toilet itself is a trap, and is designed to siphon to some extent.

*Sweating Toilets*

Since the water supply in the house is often very cold (50°F is common with both a public and private water supply), it's not unusual to find the toilet tank and the toilet fill pipe sweating, particularly in warm, humid weather. If the house is air-conditioned, the toilets will be less inclined to sweat.

*Insulated Tanks*

Many tanks have an extruded polystyrene foam insulation available as an option that fits against the inside walls of the tank to minimize condensation. These are successful up to a point. Where condensation is a serious problem, the toilet shutoff valve will often be corroded and the escutcheon plate where the fill pipe comes up through the floor or in through the wall will often be rusted. Water damage can also be done to the flooring around the back of the toilet below the bowl and fill pipe.

*Mixed Hot and Cold Water to Toilets*

One of the unusual but successful solutions to condensation is a mixing valve that mixes some hot water in with the cold water at the toilet supply. If room temperature water is used to fill the toilet tank, there is much less chance of condensation.

*Running Toilets Usually Sweat*

Toilets that continuously run water through are prone to sweating. We don't want toilets running in any case because of the wasted water. Let's look at some of the common toilet problems.

1. Leaks
2. Loose
3. Flush mechanism inoperative
4. Obstructed
5. Lazy flush
6. Odor around toilet
7. Floor damage suspected
8. Running continuously
9. Shutoff valve missing or inoperative
10. Broken tank lid or seat
11. Surface defects
12. Crowded
13. Cross connection

**Leaks**

As with most plumbing fixtures, leaks are the big problem. Toilet leaks may occur at a number of locations, including—

- the bowl
- the tank
- supply piping
- the wax ring or gasket that makes the seal where the toilet joins the waste pipe
- the bowl/tank connection
- the bowl refill line (this tube often comes disconnected and may spray water out between the tank and lid when the toilet is refilling)

**IMPLICATIONS****STRATEGY***Check Under the Lid*

Toilet leaks can waste water and cause water damage.

Before flushing the toilet, look for evidence of leakage around and below the toilet. Make sure you don't confuse sweating (condensation) with leakage. Sweating tends to appear uniformly on the sides and bottom of the tank. It may run down the bowl or supply piping, but initiates around the perimeter of the bowl in a uniform pattern.

Before flushing the toilet, lift the tank lid off carefully (sometimes these are cracked or broken and are sitting precariously in place). Ensure that the bowl refill tube is in place. Ensure that the toilet is not running. Put the tank lid back on and flush the toilet. Watch for evidence of leakage around and below the toilet. Leaks in the supply line and tank will appear whether or not the toilet is flushed. Leaks around and below the bowl may only show up when the toilet is flushed.

**Loose**

Poorly secured toilets are common.

The implications of a loose toilet are leaks, particularly at the wax ring and at the supply piping connection.

Many modern toilets are secured to the toilet flange which is in turn secured to the floor. This is often done with two or sometimes four bolts.

Your inspection should be in two parts here. To check that the bowl is well secured to the floor, some inspectors straddle the bowl and with the insides of their legs, try to move the bowl side to side. They will also push on the front of the bowl to see if the toilet moves front to back. You don't want to use superhuman strength here. You simply want to make sure that it is secure and that the toilet won't move when people sit on it.

We're also interested in knowing whether the tank is well secured to the bowl. To test this, grasp the tank and try to move it side to side and front to back. Watch carefully to see if the connection between the tank and the bowl moves. Again, this is not a test of strength. Use reasonable pressure. If the toilet is a one-piece system, you won't need to check how well the tank is secured to the bowl, since it is integral.

If the toilet is loose, recommend resealing and watch carefully for leaks below and damaged flooring, subflooring, structural members and finishes below the toilet.

**Flush Mechanism Inoperative**

Sometimes, nothing happens when you operate the flush lever (or push the button in or lift the operating mechanism).

Obviously, the toilet won't work if you can't initiate a flush.

When you operate the toilet, this failure will become obvious. This is not usually a significant or expensive problem, but it can frustrate the average homeowner.

**Obstructed**

Nothing strikes more fear into the heart of a home inspector than flushing a toilet and watching the water level in the bowl rise rather than fall. Sometimes the bowl has an adequate capacity to hold the contents of the tank without spilling. In other cases, you won't be so lucky. You will be able to tell your client that the toilet or waste pipe is obstructed. Trying to shut off the toilet isolating valve can elevate the crisis level if it leaks.

**IMPLICATION****STRATEGY**



**IMPLICATIONS****STRATEGY**

Obviously, the toilet won't work and you run the risk of flooding the bathroom if the toilet is obstructed.

When you flush the toilet, watch the flush performance. Obviously, the water in the bowl should be washed down and out of the bowl prior to the water level rising. If the water level rises, rather than drops, the toilet is probably obstructed. It's not unusual to have the toilet and/or waste pipe partially obstructed. The water may not rise to the top of the bowl, but may not flush properly either. This brings us to our next condition.

**Lazy Flush**

Sometimes the toilet flushes, but not very enthusiastically.

**IMPLICATIONS**

The implication of a lazy flush may be an obstruction, which means that the toilet will be clogged every time solid waste is introduced. If it's a design issue, the toilet will probably have to be replaced. If it's a drain or venting issue, piping revisions in the house may be necessary.

**STRATEGY**

We typically use three or four feet of toilet paper to check the flush function. This isn't a particularly rigorous test, but if the toilet paper won't disappear with a single flush, we have a suspect toilet.

**Odor around Toilet**

An odor around a toilet usually indicates leakage of the wax ring or gasket where the toilet connects to the waste piping.

**IMPLICATIONS**

There are health implications and damage possibilities to finishes and structural members around and below the toilet.

**STRATEGY**

Your nose will help you complete this part of the inspection.

**Floor Damage Suspected**

We encourage you to watch for damaged flooring around and below a toilet.

**IMPLICATION****STRATEGY**

You should step heavily on as much of the flooring immediately adjacent to the toilet that you can. Some inspectors bounce on the flooring. If the floor is spongy, you can usually feel it. Apply your weight gradually, because if the floor is in really bad shape, you might put your foot through it.

**Running Continuously****IMPLICATIONS**

The implications of a running toilet are—

- wasted water.
- sweating (condensation). If a toilet runs continuously, the waste pipe may also sweat over its entire length. This can cause damage to finishes in areas quite remote from the toilet. This can be a difficult problem to diagnose.
- nuisance noise.

**STRATEGY**

You can often see a toilet running by looking into the bowl and/or the tank. You can often hear a toilet running. If you are not sure, you can put a screwdriver to the tank or fill line, if you can get close enough, and put the handle of the screwdriver against your ear. You can also use a flashlight and look down the overflow tube. In some cases, you'll see water flowing at the bottom.

**Shutoff Valve Missing or Inoperative**

All toilets should have an operable shutoff valve located adjacent to the toilet.

The toilet can't be shut off in an emergency.

**IMPLICATIONS****STRATEGY**

Make sure that there is a valve adjacent to the toilet. We don't recommend that you try to operate it. Some inspectors recommend that the first time clients have a plumber in the house, the plumber be asked to make sure that all isolating valves operate. A missing or inoperative shutoff valve is not a problem until it's needed in an emergency.

**Broken Tank Lids or Seats**

It's not unusual to find the tank lid cracked, broken or even missing. Similarly, seats and seat covers can be cracked, broken or missing.

The toilet may not be usable if the seat is missing or broken. A crack in a seat can pinch skin and be quite painful. Missing or broken tank lids may allow foreign materials into the tank.

Look at the seat, cover and tank lid. Are they present and intact? Most seats and covers have hard plastic stops that are the bearing points. Cracks often develop around these pads.

The seat and cover are usually secured to the bowl with two bolts. Make sure that the seat is well secured to the toilet. It's not usually a big problem to tighten up these bolts, but you should mention a loose seat to the client as a courtesy.

Tank lids simply rest on the tank.

**IMPLICATIONS****STRATEGY****Surface Defects**

We talked about surface defects such as cracking and staining on other fixtures. These can be issues on toilets as well. Hairline cracks sometimes do not leak, but should be mentioned as a potential leakage spot. It's very common for toilet tank lids to get dropped and broken.

There may be no performance issues related to the surface defects. They may also be the precursor to a leak as a result of a crack.

We do not make a big deal out of surface defects, but as a courtesy to our client, mention them if we see them, and keep them in perspective.

**IMPLICATIONS****STRATEGY****Crowded**

Good practice calls for toilets to have 15 inches on both sides of the center line of the toilet as clear space.

The toilet will still operate, but it is uncomfortable for users. It may also be difficult to install a toilet paper holder.

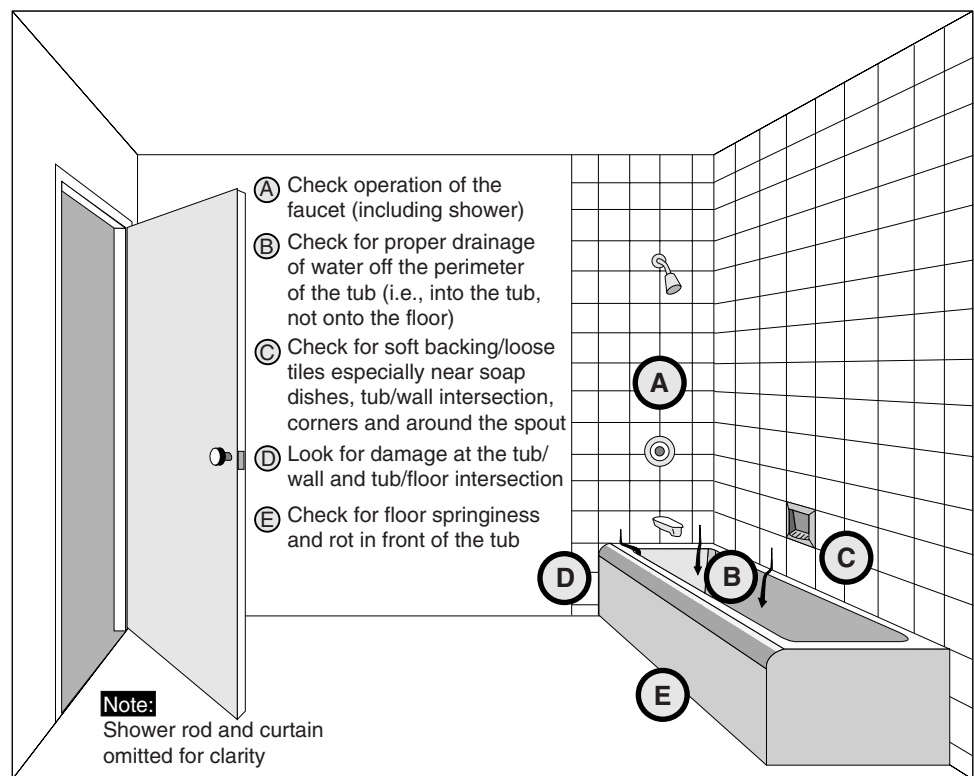
Where the toilet has less than 30 inches of clear space in total (with the center of the toilet side to side falling at the 15-inch line), note this less than ideal arrangement. You can leave it to your client's discretion to decide whether rearrangement is necessary.

**IMPLICATIONS****STRATEGY**

## 13.4.5 Bathtubs

Common bathtub problems include the following:

1. Leaks
2. Rust
3. Surface defects

**FIGURE 13.31** Inspecting a Tub

4. Cross connections
5. Not level
6. Overflow leaking, disconnected or rusted
7. Loose or unstable
8. Slow drains

Figure 13.31 summarizes the inspection issues for a bathtub; let's look at the conditions more closely.

### Leaks

It's unusual for the bathtub itself to leak. It's possible that steel or cast iron will rust through or that acrylic and/or fiberglass may crack, although these are not terribly common problems.

The most common leak issues around bathtubs are those related to—

- the tub-tile intersection around the perimeter of the tub
- the drain connection
- the overflow

Enclosure leakage is covered later in this chapter. Water splashing or running out of the bathtub is not a leak but will cause damage to the tub-floor and tub-wall intersection.

### IMPLICATIONS

The leaks may cause water damage. In this sense the implications are the same as for any other leak. Leaks on drain systems are more serious in one sense than supply plumbing leaks. A supply plumbing leak usually results in a flood that will

be noticed and corrected immediately. A leak in a drain system may be intermittent, and the amount of water that leaks out may not be great. As a result, the problem may not be diagnosed quickly, leading to concealed damage to the structure from the persistent, intermittent wetting and no opportunity for wood to dry out. When this is coupled with little or no damage to interior finishes below, we can have significant concealed damage.

#### STRATEGY

We talked earlier about looking below plumbing fixtures before operating them and looking again after they have been tested. If there is access to the piping for the bathtub in a panel on a wall backing onto the tub, you should look at that both before and after testing the fixture.

#### *Tub-tile Gaps and Mold*

Look for evidence of gaps between the tub and tile. Deteriorated caulking or caulking heavily discolored by mold should also be pointed out. If you can sense the difference, mold that is under the caulking is more serious than surface mold on the caulking. The mildew that's underneath the caulking indicates that the caulking seal has failed and that water is getting in behind it.

#### *Don't Test Overflows*

We recommend not filling the tub up to the overflow level because most overflows leak. You should let your client know that the overflows on bathtubs frequently leak, but we don't feel the need to prove our point.

### Rust

We talked about rust on basins, sinks and tubs. This is a similar issue on cast iron and steel bathtubs.

### Surface Defects

Again, chips, rust, cracks, and scratches are common various bathtub defects. In most cases, they are just cosmetic, but they can lead to performance problems. Rusting metal will eventually fail, for example. Cracks may be only on the surface of materials but may eventually go all the way through materials.

Again, we talked about this when we looked at basins, sinks, and laundry tubs.

#### STRATEGY

### Cross Connections

Older cast iron bathtubs often have the fill spout below the flood rim of the tub. Watch for this common situation. Also, let your clients know that if there are extendable faucets that can be left below the flood rim of the tub, this is another possible cross connection.

### Not Level

This is a fairly common problem with tubs and leads to damage outside the tub at the tub-floor intersection and sometimes at the walls at either end of the tub exterior.

#### IMPLICATIONS

The apron around the perimeter of bathtubs is usually sloped slightly so that water will drain back into the tub. If the tub is not installed level, the water will tend to drain either out of the front of the tub, or back in along one side or corner of the tub. If the water comes out to the front of the tub, it's going to escape and accumulate, usually at the tub-floor intersection. Sometimes the water gets hung up at the tub-wall intersection at either end of the tub and causes damage here.

If the tub slopes back away from the opening, water will stand on the shelf and be more likely to find its way through the tub enclosure finish (tile, acrylic, marble, etc.) and get into the wall system behind.

#### STRATEGY

As you test the shower, you'll usually end up with water on the tub perimeter shelf. Watch for evidence of water running in any direction, other than toward the

Damage at Tub Floor and  
Tub Wall

tub. Sometimes you can visually detect a tub that’s out of level, although since quarters are usually cramped in a bathroom, you often can’t get far enough away to get a good perspective.

Also watch for evidence of damage at the tub-floor intersection and at the tub-wall intersections. Stand on the floor very close to the tub and bounce. The flooring material in a bathroom is usually water resistant (resilient flooring or ceramic tile, for example), but damage to the subfloor below is a concern, especially if it is a wood flooring assembly.

Check the bottom section of the tub enclosure carefully where you see standing water accumulating on the tub shelf.

**Overflow Leaking, Disconnected, or Rusting**

Problems with overflows are common.

The overflow isn’t used on a regular basis. When the overflow is needed, it will cause water damage below if it’s not working properly, even though it is designed to prevent water damage.

In most cases, you won’t be able to get a look at the overflow, so you won’t know how well it’s connected. Sometimes you can grab the interior fitting in the tub, pull on it and feel that it is not tightly connected. If so, you can criticize the overflow. If you can get a look into an access panel behind the tub, you may be able to see the overflow. In these cases, you may be able to visually identify problems without flowing water. If you choose to test the overflow by filling the tub, be prepared for leakage below!

**Loose or Unstable**

Bathtubs are typically installed with support from the floor and the perimeter walls. While it’s not a common problem, sometimes the tub is not well secured in place.

The tub is likely to be out of level, and may be more prone to leakage as a result of strained fittings on both the supply and waste plumbing.

Many home inspectors step into the tub during the course of their inspection. You should be either stepping into the tub or grabbing the edges of the tub and making sure that it won’t move. We recommend that you check this out before doing your plumbing tests, otherwise you are stepping into a wet bathtub!

**Slow Drains**

We consider a slow bathtub drain as one that takes more than 6 minutes to drain water 8 inches deep from a standard 5-foot tub.

**13.4.6 Tub and Shower Stall Enclosures**

Vulnerable to Leakage

Enclosures around tubs and shower stalls are extremely vulnerable to leakage. This leakage is often concealed for some time and is a difficult issue for home inspectors. Shower stall bases have their own set of problems which we’ll talk about shortly.

Types of Enclosures

Enclosures for bathtubs and shower stalls may be ceramic tile, marble or acrylic; any type of durable, water resistant material should work. Ideally, it should also be relatively smooth and easy to clean, as well as being non-porous.

The Easy Inspections

One-piece bathtub or shower enclosures are the easiest to inspect. These are typically acrylic, fiberglass or other laminated plastic-type materials. They have very few seams and, therefore, very few places to leak.

*Ceramic Tile*

A large percentage of enclosures are ceramic tile. Marble, slate, and limestone are less common, but are treated the same as ceramic tile. We will focus our discussion on ceramic tile shower stall and bathtub enclosures.

*Watch Those Exterior Walls*

In cold climates, it's common to find that shower and bathtub enclosures on an exterior wall are most troublesome. The vapor barrier and insulation details are often poorly executed around tubs and shower stalls on exterior walls. Wall board is not typically installed below the level of a bathtub, for example, and so there is not the traditional support for vapor barriers and insulation in this area. Leakage of warm, moist air into this wall cavity can cause condensation and structural problems resulting in rotted wood. It's often difficult to tell the difference between leakage through the enclosure and condensation problems on exterior cold walls around tub and shower enclosures. Watch for damage here and pay particular attention to these areas and the parts of the house immediately below.

*Inspection Tip**Penetrations and Terminations*

How are bathtub and shower enclosures like roofs? With roofs, most of the problems occur at roof penetrations or changes in direction and material. The same is true of bathtub and shower enclosures. Most of the problems occur at penetrations like faucets, soap dishes, grab bars and windows and at changes in direction or material like tub-tile intersections, inside wall corners and outside wall corners where the tile ends. Let's take a look at the problems that we normally find on these very troublesome details. There are two common conditions in bathtub and shower enclosures, and they are usually related. A third problem is less common but is a safety issue.

1. Leaks
2. Loose, broken, or missing tile
3. Electrical problems at the enclosure

**Leaks**

Leaks in bathtub and shower enclosures are both common and difficult to detect during a home inspection. Leakage can be through grout joints in tile or at any of the penetrations and joints we've talked about.

**IMPLICATIONS**

The implications of leakage are cosmetic damage to interior finishes below and structural damage to wood members exposed to the moisture. Since this damage is often concealed, the extent of the damage is often not appreciated until the damage is quite advanced.

**STRATEGY**

With bathtub and shower enclosures, we recommend that you get inside and test these wall systems before flowing any water. Firstly, it's easier on you if you are checking dry surfaces. You don't want to track wet footprints through the house. Secondly, if you see an obvious problem, you may want to avoid running water so you don't cause more water damage.

*Look, Tap, and Press*

Your testing procedure normally includes a visual inspection of all the surfaces. Focus on the weak areas that we've mentioned. The second part of the inspection is to tap or press on the tiles themselves (or whatever material forms the enclosure). You are looking for sponginess or an unusual amount of give. Sometimes you can see tiles starting to fall off. You may only find a problem when you tap on the surface and hear a hollow ringing sound or push on the tiles and find that they give easily. The tiles around penetrations and just above the tub or just above the shower base are important ones to watch for.

*Penetrations*

You should also check how well penetrations through the walls are sealed. Some soap dishes and other fixtures are installed through the tile and secured to the wall system behind. Many of the modern soap dishes and other accessories are

*Look Behind Escutcheon Plates*

simply bonded to the tile with a silicone sealant. These are preferable since there is no wall penetration. Obviously, leakage is not nearly so likely here.

**Escutcheon plates** (the decorative trim rings around pipe penetrations through walls) can sometimes be pulled back easily to determine whether things are well sealed.

Even if there is no leakage, you should note vulnerable areas for water penetration.

*Testing the Fixtures*

Many home inspectors test bathtubs and showers vigorously. You should make sure that water is not spilling out of the fixture onto the floor during your test. Shower curtains or doors should be in place before operating showers. Some inspectors are also quite diligent about bouncing as much water off the shower walls as they can by moving the shower nozzle. There is a slight risk of creating a leak as you manipulate the shower nozzle, especially if it hasn't been moved recently.

*Intermittent Leaks*

We have had a number of situations where, despite our best efforts to wet the walls and flow lots of water at a shower head, we have had clients find a leak. Sometimes the leaks will only occur with people standing in the shower. Obviously the water spray is going to bounce and run down the walls in a different pattern if there is someone standing in the shower. Let your clients know that this possibility exists despite your best efforts during the inspection. You should also let them know that shower and tub enclosures are high leakage, high maintenance areas.

*Check Adjacent Rooms*

After you have flowed water in a tub enclosure or shower stall, you should look closely at adjacent walls in other rooms. Sometimes evidence of leakage will be visible in the walls and/or wall-floor area of adjacent rooms before it shows up either in the bathroom or in the room below, if there is any. Damage to baseboards and quarter round is a common clue that there has been leakage at the bathtub or shower enclosure.

*Check the Floor Outside the Tub or Shower*

One of the big problems with bathtubs and shower stalls is water spilling out over the front of the tub or through the shower door opening. Damage to the wall below the shower threshold or to the tub facade or tub-floor intersection is common. Make sure you look for this. Many inspectors tap on the wall surface just below a shower threshold and similarly check any area on the face of the tub that is subject to water damage. We've already talked about bouncing on the floor close to the tub to check for sponginess or rotted subflooring. The same kind of test should be done on a shower enclosure, particularly in front of the threshold at either corner of the opening.

**IMPLICATIONS****Loose, Broken, or Missing Tile**

The tiles will not be watertight if they are not secure. If the backer material starts to deteriorate, tiles will eventually fall off the wall. The more immediate concern is that water will get in behind the backer material and attack the wall and floor systems behind and below. Considerable water damage can be done to the structure. If we are lucky, interior finishes that are visible below the tub or shower will give us ample warning of this leakage. Unfortunately, this isn't always the case.

**STRATEGY**

We talked about pushing and tapping on tiles, concentrating on areas around and below penetrations and at the top of the tub or shower base.

In many areas you'll be able to detect differences in grout, color and texture, and in some cases the tiles will be different colors, textures and sizes. This usually indicates a repair. Areas that have been repaired are likely to leak again.

*Scanning with a Flashlight*

You can hold a flashlight parallel to the wall and scan along the surface looking for irregularities. Sometimes defects can be seen that are not visible looking straight at the wall.

*Watch for Windows*

Generally speaking, windows in bathtub and shower enclosures are a bad idea. Since virtually all windows contain sills, water can collect here. Also, the presence of a window creates more joints between materials and more changes in direction. An additional problem is the condensation that forms on windows in cold weather when people are showering. Even with double- or triple-glazed windows, you can expect condensation here. This condensation runs down the glass and may get into the window sash or frame or into the sill system and wall below. Look carefully around and below the window for loose tile and soft, spongy backer material. In some cases, people have added a small curtain to cover the window. In effect, the window gets its own shower curtain. While this helps, it may or may not be a positive solution.

*Caulk Those Corner Joints*

In most new work, the ceramic tiles at interior corners in bathtubs are grouted. We recommend caulking this vertical joint since the grout inevitably cracks as framing members shrink and the building settles very slightly.

*Caulking Replacement*

The tub-tile or stall enclosure-base connection is particularly vulnerable. Advise your clients that they should plan on replacing the caulking on a regular basis. This means digging out the old caulking and replacing it, rather than adding new caulking over old.

*Light Fixtures in Enclosures***Electrical Problems at the Enclosure**

Where lighting fixtures are provided inside tub and shower enclosures, they should be a vapor-proof fixture designed for this application. You can usually identify these by the gasket material around the lens.

*Receptacles and Switches Near Enclosures*

Electric light switches and receptacles should be at least 3 feet away (out of the reach of) shower stalls and bathtub enclosures. We don't want soaking wet people operating electrical equipment.

These are life safety issues.

**IMPLICATIONS****STRATEGY**

Look for vapor-proof light fixtures and a safe distance from enclosures to switches and receptacles.

## 13.4.7 Shower Stalls

All shower stalls leak! They don't necessarily leak when you are doing your inspection, but eventually they all leak. Worse than that, shower stalls are very expensive to replace. Worse yet, the leakage can be slow and cause considerable damage without being particularly visible. Be very, very careful with shower stalls. The base or enclosure may be the culprit, but the result is similar.

They leak because they are complicated. The type least likely to leak is a one-piece fiberglass or acrylic shower stall because the floor and walls are made of a single unit (Figure 13.32).

*Floors Should Slope*

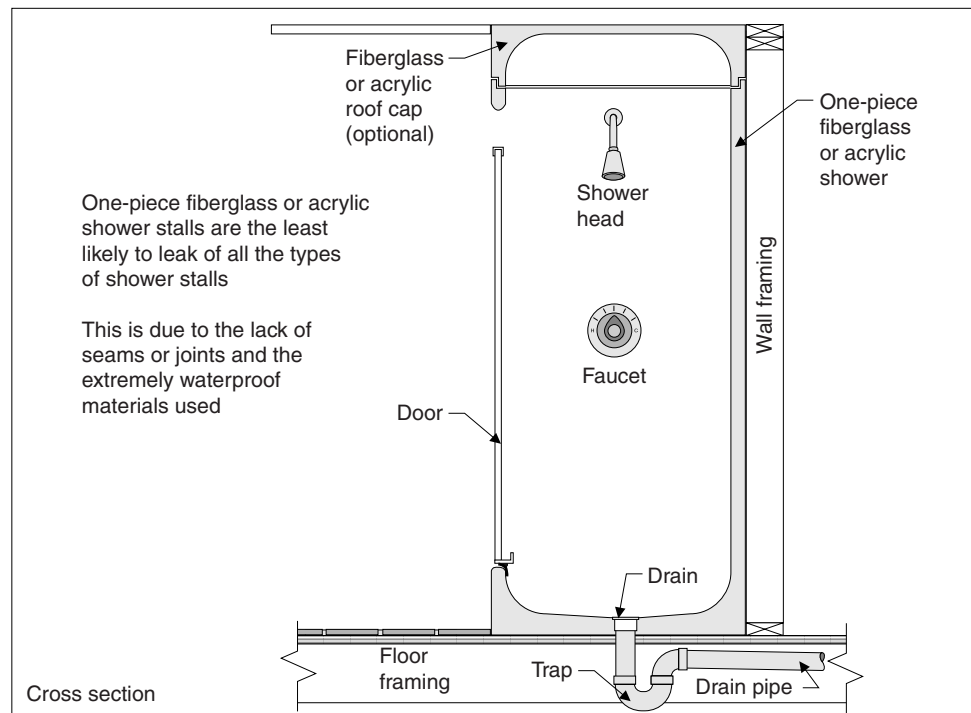
All shower stalls should have a slope of at least  $\frac{1}{4}$  inch per foot in toward the drain. The drain obviously should not be at a high point. We don't want water ponding on the floor of the shower stall.

*Sills and Thresholds*

All shower stalls need to have some way for people to get in to them. They will have a curb or sill at the entrance and it may form a door threshold if there is a shower door rather than a curtain. Typically, the sill or threshold is 1 inch below the top of the liner.

Let's look at some of the common problems with showers themselves. These are in addition to the supply and drain issues and enclosure issues that we've already talked about.



**FIGURE 13.32** One-Piece Shower Stalls

1. Leaks
2. Rust
3. Entrance problems
4. Sill and threshold problems
5. Pooling water on the floor
6. Too small
7. Slow drains

### Leaks

As always, leaks are the big issue.

#### STRATEGY

##### *Flood Testing the Stall*

One of the tests that many home inspectors perform is a flood test of the shower base itself. Some inspectors plug the drain of the shower stall and flood the base with an inch or two of water. Obviously, we don't want to get the water up over the sill or door threshold. This is a fairly rigorous test. The longer the water stays, the more rigorous the test. Some specialists say the water should stay for 24 hours!

##### *The Risk*

This does not necessarily simulate normal usage and has the added disadvantage of a delayed reaction. In some cases, it will take several hours for the leaks to appear. Obviously a home inspection is not going to last several hours in most cases. We'll leave it to you to decide whether you do this type of test. It does go beyond the Standards and can certainly be revealing.

##### *Looking Outside the Shower Stall*

In one sense, you're not going to see many leaks related to the shower stall by looking inside the stall. You need to be looking around and below for leaks, rather than inside, for the most part. Remember to check the walls and floors in adjacent rooms that back onto the stall.

**Rust**

It's fairly common to find that metal shower stalls rust. This isn't surprising given their environment. We also mentioned earlier that many metal shower stalls are relatively low quality systems.

It's tough to keep a rusty shower stall sanitary. Eventually the rust will go through the metal and the shower stall will lose its integrity.

With metal shower stalls, look carefully in all corners and seams for evidence of rust. In some cases, rust has been painted over. Watch for rough textured surfaces that are usually smooth.

**Entrance Problems**

The minimum entrance width is 22 inches for shower stalls. Where they are narrower than this, they should be reported. Also watch for doors that are not tempered glass. In most cases, you'll find an indication in the corner of the door that indicates that it is tempered. This won't always be obvious, however. Where you can't verify that the door is tempered, you may want to raise the question, although this goes beyond the Standards.

Another problem is a shower door that only opens in. There are two disadvantages to this. If the water temperature suddenly gets very hot, people can't get out of the shower quickly. A bigger problem is if someone collapses in a shower stall and falls against the door, it can be very difficult to get the person out. Shower stall doors should open outward.

Narrow shower entrances, non-tempered glass doors and inward opening doors are all life safety issues.

Watch for these situations when looking at shower stalls.

**Sill and Threshold Problems**

Sills and thresholds at shower stalls should be watertight and should be tall enough to control the shower water.

In both cases, leakage is the implication.

Check sills or thresholds carefully. Where these are tile or marble, make sure they are well secured and intact. Ideally, the sills should have a slight slope back into the shower stall so any water that lands on the sill or threshold will drain back into the stall, rather than out onto the floor. With every shower stall, part of your normal routine should be to check outside the sill or threshold carefully for evidence of water damage. It may have been freshly painted, so you may have to be persistent. It is unusual to find no evidence of leakage around and below shower stall sills and thresholds.

**Pooling Water on the Floor**

Stagnant water sitting in a shower stall is unpleasant and may be unhealthy.

Before operating the fixture, make sure that there is no water on the floor. Check that the drain is at a low point on the floor.

When operating the fixture, make sure water does not back up and pool on the floor.

**IMPLICATIONS****STRATEGY**

*Narrow Doors, Non-Tempered Glass*

*Door Opens Inward*

**IMPLICATIONS****STRATEGY****IMPLICATION****STRATEGY****IMPLICATIONS****STRATEGY**

**Too Small**

We need to watch for shower stalls that are too small. Generally, 30 inches by 30 inches is a minimum size.

Shower stalls that are too small may trap people who collapse in the shower.

While it's not necessary to measure each shower stall with a tape measure, you should have some sense of a 30 inch by 30 inch shower stall and note any that are significantly smaller.

**IMPLICATIONS****STRATEGY**

## REVIEW QUESTIONS

1. What are three things that go wrong with water supply?
2. Evaluating the quality of the water available at a private source is beyond the scope of a standard home inspection.  
True False
3. List six materials commonly used for water service piping.
4. What is the typical minimum diameter of water service piping in modern construction?
5. What is the function of a stop and waste valve?
6. List five materials used as water supply piping.
7. When was copper supply piping first used?
  - a. 1850s
  - b. 1900
  - c. 1930s
  - d. 1960s
  - e. 1990
8. At what interval is vertical support required for copper piping?
9. What is the recommended horizontal spacing of supports for copper piping?
10. What is the recommended horizontal spacing of supports for galvanized steel piping?
11. Plastic piping needs less support than copper piping because it tends to expand more.  
True False
12. What is the implication of improper support on piping?
13. Briefly describe a cross connection.
14. List five locations in a house where there may be a cross connection.
15. Is a vacuum necessary to have backflow at a cross connection?  
Yes No
16. Briefly describe water hammer, and why it occurs.
17. Vertical pipes at ends of runs are one method of preventing water hammer.  
True False
18. Why can plastic pipe not typically be used between units in shared dwellings?
19. Polybutylene piping failures seem to be concentrated most at
  - a. the crimp rings
  - b. elbows in the pipes
  - c. horizontal splits in the piping
  - d. at the connection to fixtures
  - e. in vertical pipe runs only
20. Polybutylene supply piping should not be closer than 18 inches to a water heater.  
True False
21. Can polybutylene piping be used on a hot water recirculating loop?  
Yes No

22. A house with several different styles of crimp rings should be a cause for concern.  
True False
23. What is the approximate temperature rise in residential water heaters?
24. A water heater is termed a **single-pass** appliance because \_\_\_\_\_.
25. What is the biggest impact of the single-pass feature on a water heater?
26. Where is the burner located on gas- and oil-fired water heaters?
27. What is the component that introduces cold water near the bottom of a gas-fired water heater tank?
- a. the drip tube
  - b. the relief valve
  - c. the dip tube
  - d. the baffle
  - e. the vent
28. What are the two methods of rating a water heater?
29. How do these two relate?
30. Which type of water heater has the slowest recovery rate?
31. Which type of water heater has the fastest recovery rate?
32. Water running out of the burner compartment on start-up of a gas water heater suggests \_\_\_\_\_.
33. Gas and oil-fired water heaters are not allowed in sleeping areas.  
True False
34. How high above the floor should a gas or oil-fired water heater be in a garage?
- a. 0 inches
  - b. 6 inches
  - c. 12 inches
  - d. 18 inches
  - e. 24 inches
35. Where does the probe for the temperature/pressure relief valve sit in the tank?
36. Where does the discharge tube from the temperature/pressure relief valve have to dump water in a cold climate?
37. Where does the temperature/pressure relief valve have to discharge water in a warm climate?
38. If we only need a pressure relief valve because the water heater has some internal temperature control, where can this pressure relief valve be?
39. What is the purpose of a baffle or turbulator in a gas fired water heater?
40. Why is an explosion with propane more likely than with natural gas?
41. When would a fan-assisted gas water heater typically be used?
42. What is the exhaust pipe material for a fan assisted water heater typically?
43. List the common problems found with fan-assisted gas water heaters.
44. Are fan-assisted water heaters condensing?
45. Describe the function of a trap.
46. Explain briefly why a toilet does not need a trap.

47. A toilet does not need a vent.  
True False
48. A waste pipe is intended to carry solids only.  
True False
49. Why in a cold climate is the trap portion of the drainage system most vulnerable to freezing?
50. Briefly describe the purpose of a trap primer.
51. Explain briefly what is meant by a wet vent.
52. List seven different materials used for piping in drain, waste, and vent systems.
53. Why are leaks more difficult to find in drain, waste and vent systems than in supply systems?
54. Copper pipe does not rust.  
True False
55. List two methods of reducing the freezing risk of pipes in a cold environment.
56. List six locations for possible cross connections in a residence.
57. What is the implication of a cross connection?
58. What is the required slope for nearly horizontal pipes in a drain, waste, and vent system?
59. Pipe size should always increase as you move downstream and more fixtures join in.  
True False
60. Why is PVC or ABS plastic piping usually allowed only in single-family homes?
61. What is the implication of poor manifolding?
62. What is the purpose of a trap?
63. Every trap should be provided with some method of removal or cleaning.  
True False
64. A running trap is a good idea because it provides extra protection.  
True False
65. List four types of traps that are not permitted in residential construction.
66. What is the typical maximum distance between the fixture and the trap?
67. What is the minimum trap arm length?
68. What is the implication of a trap arm that is too long?
69. It's impossible for a trap to be too big.  
True False
70. Where should the floor drain be located?
71. Explain the purpose of a backwater valve.
72. What is the purpose of the venting system?
73. List five common materials used in venting systems.

74. Explain briefly the simple test required to check the venting system when you test functional drainage from a fixture.
75. List seven problems found with vents.
76. The vent is typically located upstream of the trap.  
True False
77. Nearly horizontal vent pipes should slope slightly down away from the trap they are protecting.  
True False
78. The vent diameter must be as large as the trap diameter.  
True False
79. Explain briefly the problems with venting kitchen island sinks.
80. What is the minimum height of vent pipe above the roofline?
81. At least how far should an outdoor vent termination be above an operable door?
- a. 1 foot
  - b. 2 feet
  - c. 3 feet
  - d. 4 feet
  - e. 5 feet
82. What are the three components required on the discharge line from a sewage ejector pump?
83. What is the typical vent size on a sewage ejector pump?
- a. 1<sup>1</sup>/<sub>4</sub> inches
  - b. 1<sup>1</sup>/<sub>2</sub> inches
  - c. 2 inches
  - d. 2<sup>1</sup>/<sub>2</sub> inches
  - e. 3 inches
84. What is the purpose of the sewage ejector pump?
85. Why is a check valve necessary when a sewage ejector is connected to municipal sewers?
86. What are the functions of the three pipes connected to a sewage ejector pump?
87. Sewage ejector pumps should not vibrate significantly when running.  
True False
88. What is the purpose of a high level alarm with a sewage ejector?
89. Surface cracking in an acrylic kitchen sink is not usually a concern because there is no leakage.  
True False
90. Testing for loose wall hung basins is best done by putting all your weight on the basin.  
True False
91. List two possible cross connections at basins and sinks.
92. Why is water hammer more common in a house with cartridge faucets than with compression faucets?
93. List eight common faucet problems.
94. List three types of faucet-related noise.

95. How is a cross connection possible with a hose bibb?
96. How is it prevented?
97. List seven common problems with hose bibbs.
98. Describe in two sentences each, two types of hose bibbs.
99. Describe in one sentence a stop and waste valve.
100. What can cause a toilet to sweat?
101. How is a mixing valve used to stop a toilet sweating?
102. Running toilets are less prone to sweating because the water in the tank is constantly replenished.  
True False
103. List three common leak locations in a two-piece toilet.
104. Describe a test used to determine if the toilet connection to the drain piping is loose.
105. Does a toilet need to have a shutoff valve on the supply line?  
Yes No
106. What is the minimum width of space needed for a toilet installation?
  - a. 18 inches
  - b. 24 inches
  - c. 30 inches
  - d. 36 inches
  - e. there is no minimum
107. List seven common bathtub problems.
108. Why does it matter if the tub is slightly out of level?
109. Why are tile enclosures on exterior walls in cold climates more vulnerable than ones on interior walls?
110. List five specific areas where tile enclosures are vulnerable to leakage.
111. Briefly describe a test used to determine if the tiles in an enclosure are bonded well to the backer material.
112. Because there is a watertight liner underneath the tiles in a shower stall base, the floor is not required to slope to drain.  
True False
113. What is the minimum entrance width for shower stalls?
  - a. 18 inches
  - b. 22 inches
  - c. 26 inches
  - d. 30 inches
  - e. there is no minimum requirement
114. Which way should a hinged shower door open in a shower stall?



